

Access line, 16 MHz STM8S 8-bit MCU, up to 32 Kbyte Flash, integrated EEPROM, 10-bit ADC, timers, UART, SPI, I²C

Datasheet - production data

Features

Core

- 16 MHz advanced STM8 core with Harvard architecture and 3-stage pipeline
- Extended instruction set

Memories

- Program memory: up to 32 Kbyte Flash; data retention 20 years at 55 °C after 10 kcycle
- Data memory: up to 1 Kbyte true data EEPROM; endurance 300 kcycle
- RAM: up to 2 Kbyte

Clock, reset and supply management

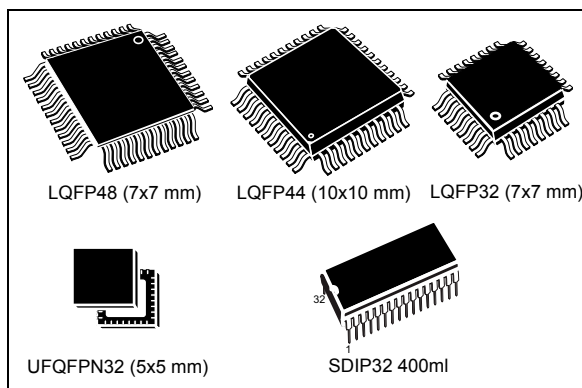
- 2.95 to 5.5 V operating voltage
- Flexible clock control, 4 master clock sources
 - Low power crystal resonator oscillator
 - External clock input
 - Internal, user-trimmable 16 MHz RC
 - Internal low-power 128 kHz RC
- Clock security system with clock monitor
- Power management:
 - Low-power modes (wait, active-halt, halt)
 - Switch-off peripheral clocks individually
- Permanently active, low-consumption power-on and power-down reset

Interrupt management

- Nested interrupt controller with 32 interrupts
- Up to 37 external interrupts on 6 vectors

Timers

- Advanced control timer: 16-bit, 4 CAPCOM channels, 3 complementary outputs, dead-time insertion and flexible synchronization



- 2x16-bit general purpose timer, with 2+3 CAPCOM channels (IC, OC or PWM)
- 8-bit basic timer with 8-bit prescaler
- Auto wake-up timer
- Window watchdog and independent watchdog timers

Communication interfaces

- UART with clock output for synchronous operation, SmartCard, IrDA, LIN master mode
- SPI interface up to 8 Mbit/s
- I2C interface up to 400 kbit/s

Analog to digital converter (ADC)

- 10-bit, ± 1 LSB ADC with up to 10 multiplexed channels, scan mode and analog watchdog

I/Os

- Up to 38 I/Os on a 48-pin package including 16 high sink outputs
- Highly robust I/O design, immune against current injection

Unique ID

- 96-bit unique key for each device

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1 Introduction

This datasheet contains the description of the device features, pinout, electrical characteristics, mechanical data and ordering information.

- For complete information on the STM8S microcontroller memory, registers and peripherals, please refer to the STM8S microcontroller family reference manual (RM0016).
- For information on programming, erasing and protection of the internal Flash memory please refer to the STM8S Flash programming manual (PM0051).
- For information on the debug and SWIM (single wire interface module) refer to the STM8 SWIM communication protocol and debug module user manual (UM0470).
- For information on the STM8 core, please refer to the STM8 CPU programming manual (PM0044).

2 Description

The STM8S105x4/6 access line 8-bit microcontrollers offer from 16 to 32 Kbyte Flash program memory, plus integrated true data EEPROM. The STM8S microcontroller family reference manual (RM0016) refers to devices in this family as medium-density. All devices of the STM8S105x4/6 access line provide the following benefits: reduced system cost, performance and robustness, short development cycles, and product longevity.

The system cost is reduced thanks to an integrated true data EEPROM for up to 300 k write/erase cycles and a high system integration level with internal clock oscillators, watchdog and brown-out reset.

Device performance is ensured by a 16 MHz CPU clock frequency and enhanced characteristics which include robust I/O, independent watchdogs (with a separate clock source), and a clock security system.

Short development cycles are guaranteed due to application scalability across common family product architecture with compatible pinout, memory map and modular peripherals.

Product longevity is ensured in the STM8S family thanks to their advanced core which is made in a state-of-the-art technology for applications with 2.95 V to 5.5 V operating supply.

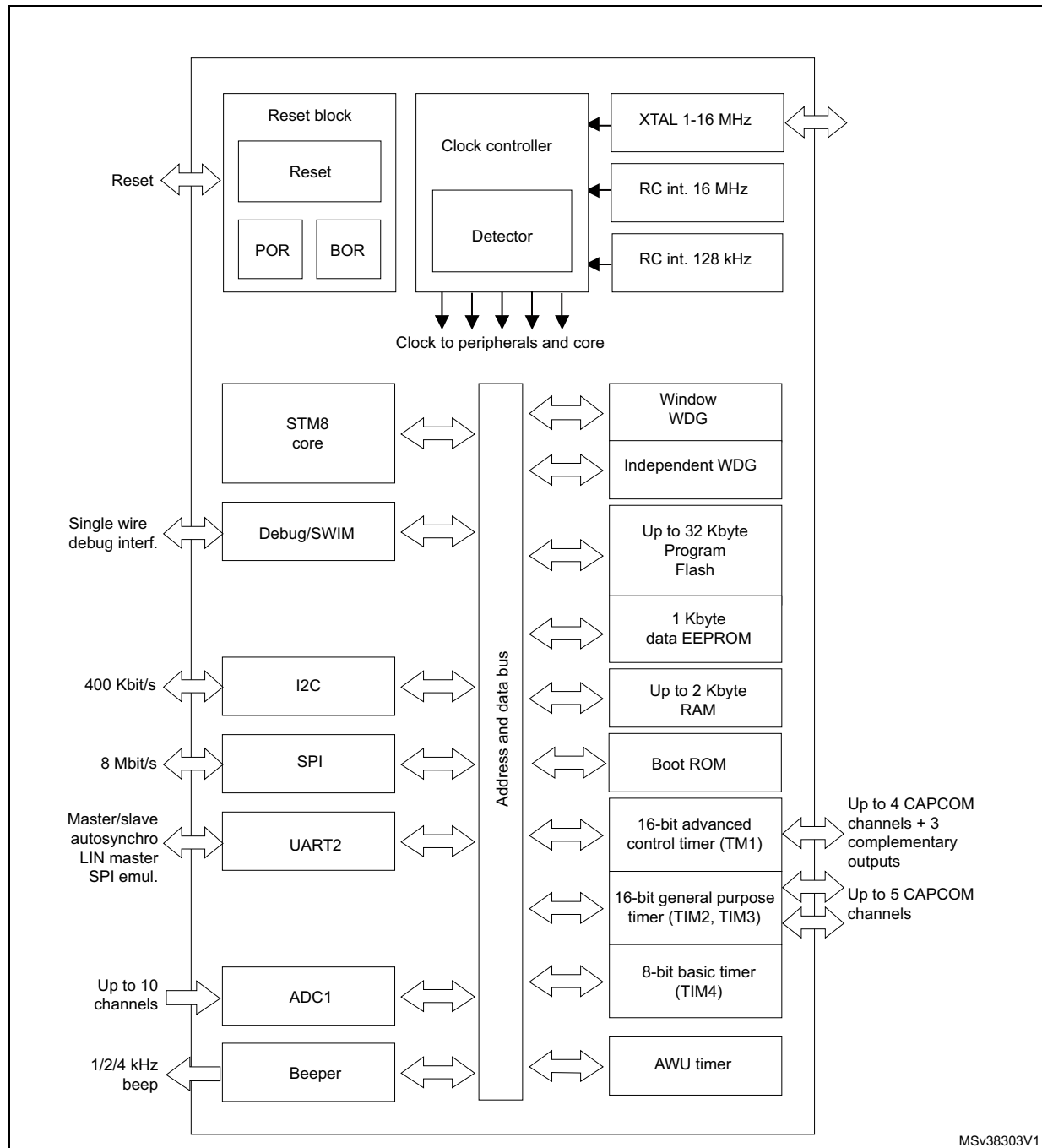
Full documentation is offered as well as a wide choice of development tools.

Table 1. STM8S105x4/6 access line features

Device	STM8S105C6	STM8S105C4	STM8S105S6	STM8S105S4	STM8S105K6	STM8S105K4
Pin count	48	48	44	44	32	32
Maximum number of GPIOs	38	38	34	34	25	25
Ext. Interrupt pins	35	35	31	31	23	23
Timer CAPCOM channels	9	9	8	8	8	8
Timer complementary outputs	3	3	3	3	3	3
A/D Converter channels	10	10	9	9	7	7
High sink I/Os	16	16	15	15	12	12
Medium density Flash Program memory (byte)	32K	16K	32K	16K	32K	16K
Data EEPROM (bytes)	1024	1024	1024	1024	1024	1024
RAM (bytes)	2K	2K	2K	2K	2K	2K
Peripheral set	Advanced control timer (TIM1), General-purpose timers (TIM2 and TIM3), Basic timer (TIM4) SPI, I2C, UART, Window WDG, Independent WDG, ADC					

3 Block diagram

Figure 1. STM8S105x4/6 block diagram



4 Product overview

The following section provides an overview of the basic features of the device functional modules and peripherals.

For more detailed information please refer to the corresponding family reference manual (RM0016).

4.1 Central processing unit STM8

The 8-bit STM8 core is designed for code efficiency and performance.

It contains 6 internal registers which are directly addressable in each execution context, 20 addressing modes including indexed indirect and relative addressing and 80 instructions.

Architecture and registers

- Harvard architecture,
- 3-stage pipeline,
- 32-bit wide program memory bus - single cycle fetching for most instructions,
- X and Y 16-bit index registers - enabling indexed addressing modes with or without offset and read-modify-write type data manipulations,
- 8-bit accumulator,
- 24-bit program counter - 16-Mbyte linear memory space,
- 16-bit stack pointer - access to a 64 K-level stack,
- 8-bit condition code register - 7 condition flags for the result of the last instruction.

Addressing

- 20 addressing modes,
- Indexed indirect addressing mode for look-up tables located anywhere in the address space,
- Stack pointer relative addressing mode for local variables and parameter passing.

Instruction set

- 80 instructions with 2-byte average instruction size,
- Standard data movement and logic/arithmetic functions,
- 8-bit by 8-bit multiplication,
- 16-bit by 8-bit and 16-bit by 16-bit division,
- Bit manipulation,
- Data transfer between stack and accumulator (push/pop) with direct stack access,
- Data transfer using the X and Y registers or direct memory-to-memory transfers.

4.2 Single wire interface module (SWIM) and debug module (DM)

The single wire interface module and debug module permits non-intrusive, real-time in-circuit debugging and fast memory programming.

SWIM

Single wire interface module for direct access to the debug module and memory programming. The interface can be activated in all device operation modes. The maximum data transmission speed is 145 bytes/ms.

Debug module

The non-intrusive debugging module features a performance close to a full-featured emulator. Beside memory and peripherals, also CPU operation can be monitored in real-time by means of shadow registers.

- R/W to RAM and peripheral registers in real-time
- R/W access to all resources by stalling the CPU
- Breakpoints on all program-memory instructions (software breakpoints)
- Two advanced breakpoints, 23 predefined configurations

4.3 Interrupt controller

- Nested interrupts with three software priority levels,
- 32 interrupt vectors with hardware priority,
- Up to 37 external interrupts on 6 vectors including TLI,
- Trap and reset interrupts

4.4 Flash program and data EEPROM memory

- Up to 32 Kbyte of Flash program single voltage Flash memory,
- Up to 1 Kbyte true data EEPROM,
- Read while write: writing in data memory possible while executing code in program memory,
- User option byte area.

Write protection (WP)

Write protection of Flash program memory and data EEPROM is provided to avoid unintentional overwriting of memory that could result from a user software malfunction.

There are two levels of write protection. The first level is known as MASS (memory access security system). MASS is always enabled and protects the main Flash program memory, data EEPROM and option bytes.

To perform in-application programming (IAP), this write protection can be removed by writing a MASS key sequence in a control register. This allows the application to write to data EEPROM, modify the contents of main program memory or the device option bytes.

A second level of write protection, can be enabled to further protect a specific area of memory known as UBC (user boot code). Refer to the figure below.

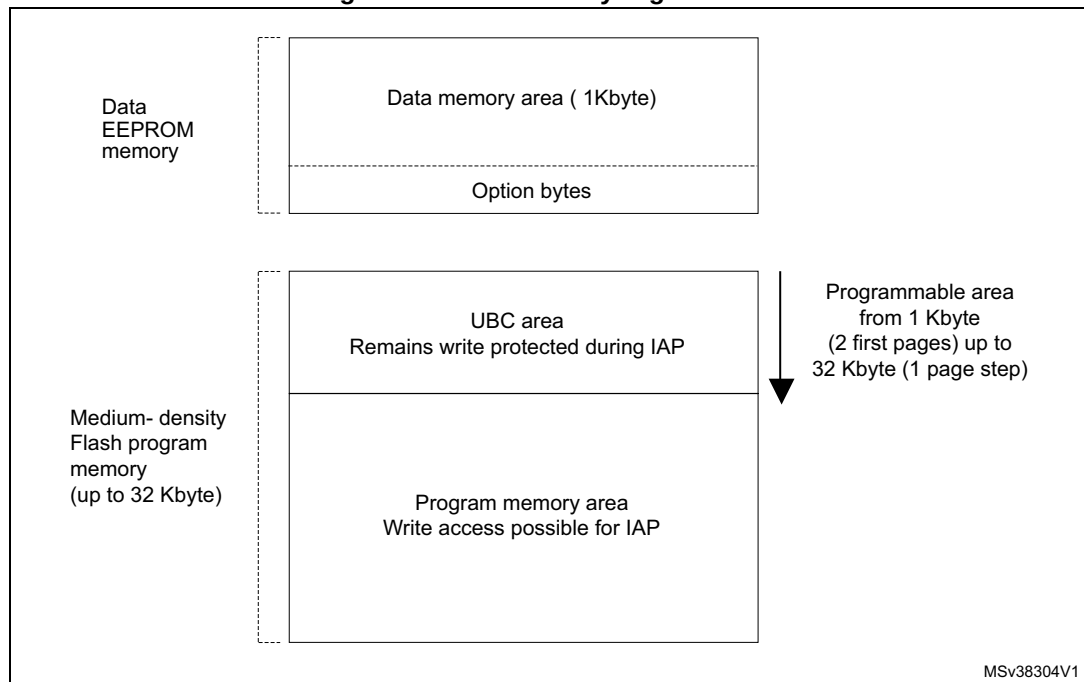
The size of the UBC is programmable through the UBC option byte, in increments of 1 page (512 byte) by programming the UBC option byte in ICP mode.

This divides the program memory into two areas:

- Main program memory: up to 32 Kbyte minus UBC
- User-specific boot code (UBC): Configurable up to 32 Kbyte

The UBC area remains write-protected during in-application programming. This means that the MASS keys do not unlock the UBC area. It protects the memory used to store the boot program, specific code libraries, reset and interrupt vectors, the reset routine and usually the IAP and communication routines.

Figure 2. Flash memory organization



Read-out protection (ROP)

The read-out protection blocks reading and writing the Flash program memory and data EEPROM memory in ICP mode (and debug mode). Once the read-out protection is activated, any attempt to toggle its status triggers a global erase of the program and data memory. Even if no protection can be considered as totally unbreakable, the feature provides a very high level of protection for a general purpose microcontroller.

4.5 Clock controller

The clock controller distributes the system clock (fMASTER) coming from different oscillators to the core and the peripherals. It also manages clock gating for low power modes and ensures clock robustness.

Features

- **Clock prescaler:** to get the best compromise between speed and current consumption the clock frequency to the CPU and peripherals can be adjusted by a programmable prescaler.
- **Safe clock switching:** clock sources can be changed safely on the fly in run mode through a configuration register. The clock signal is not switched until the new clock source is ready. The design guarantees glitch-free switching.
- **Clock management:** to reduce power consumption, the clock controller can stop the clock to the core, individual peripherals or memory.
- **Master clock sources:** four different clock sources can be used to drive the master clock:
 - 1-16 MHz high-speed external crystal (HSE)
 - Up to 16 MHz high-speed user-external clock (HSE user-ext)
 - 16 MHz high-speed internal RC oscillator (HSI)
 - 128 kHz low-speed internal RC (LSI)
- **Startup clock:** After reset, the microcontroller restarts by default with an internal 2 MHz clock (HSI/8). The prescaler ratio and clock source can be changed by the application program as soon as the code execution starts.
- **Clock security system (CSS):** This feature can be enabled by software. If an HSE clock failure occurs, the internal RC (16 MHz/8) is automatically selected by the CSS and an interrupt can optionally be generated.
- **Configurable main clock output (CCO):** This outputs an external clock for use by the application.

Table 2. Peripheral clock gating bit assignments in CLK_PCKENR1/2 registers

Bit	Peripheral clock	Bit	Peripheral clock	Bit	Peripheral clock	Bit	Peripheral clock
PCKEN17	TIM1	PCKEN13	UART2	PCKEN27	Reserved	PCKEN23	ADC
PCKEN16	TIM3	PCKEN12	Reserved	PCKEN26	Reserved	PCKEN22	AWU
PCKEN15	TIM2	PCKEN11	SPI	PCKEN25	Reserved	PCKEN21	Reserved
PCKEN14	TIM4	PCKEN10	I2C	PCKEN24	Reserved	PCKEN20	Reserved

4.6 Power management

For efficient power management, the application can be put in one of four different low-power modes. You can configure each mode to obtain the best compromise between lowest power consumption, fastest start-up time and available wakeup sources.

- **Wait mode:** In this mode, the CPU is stopped, but peripherals are kept running. The wakeup is performed by an internal or external interrupt or reset.
- **Active halt mode with regulator on:** In this mode, the CPU and peripheral clocks are stopped. An internal wakeup is generated at programmable intervals by the auto wake up unit (AWU). The main voltage regulator is kept powered on, so current consumption is higher than in active halt mode with regulator off, but the wakeup time is faster. Wakeup is triggered by the internal AWU interrupt, external interrupt or reset.
- **Active halt mode with regulator off:** This mode is the same as active halt with regulator on, except that the main voltage regulator is powered off, so the wake up time is slower.
- **Halt mode:** In this mode the microcontroller uses the least power. The CPU and peripheral clocks are stopped, the main voltage regulator is powered off. Wakeup is triggered by external event or reset.

4.7 Watchdog timers

The watchdog system is based on two independent timers providing maximum security to the applications.

Activation of the watchdog timers is controlled by option bytes or by software. Once activated, the watchdogs cannot be disabled by the user program without performing a reset.

Window watchdog timer

The window watchdog is used to detect the occurrence of a software fault, usually generated by external interferences or by unexpected logical conditions, which cause the application program to abandon its normal sequence.

The window function can be used to trim the watchdog behavior to match the application perfectly.

The application software must refresh the counter before time-out and during a limited time window.

A reset is generated in two situations:

1. Timeout: At 16 MHz CPU clock the time-out period can be adjusted between 75 μ s up to 64 ms.
2. Refresh out of window: The downcounter is refreshed before its value is lower than the one stored in the window register.

Independent watchdog timer

The independent watchdog peripheral can be used to resolve processor malfunctions due to hardware or software failures.

It is clocked by the 128 kHz LSI internal RC clock source, and thus stays active even in case of a CPU clock failure

The IWDG time base spans from 60 μ s to 1 s.

4.8 Auto wakeup counter

- Used for auto wakeup from active halt mode,
- Clock source: Internal 128 kHz internal low frequency RC oscillator or external clock,
- LSI clock can be internally connected to TIM1 input capture channel 1 for calibration.

4.9 Beeper

The beeper function outputs a signal on the BEEP pin for sound generation. The signal is in the range of 1, 2 or 4 kHz.

The beeper output port is only available through the alternate function remap option bit AFR7.

4.10 TIM1 - 16-bit advanced control timer

This is a high-end timer designed for a wide range of control applications. With its complementary outputs, dead-time control and center-aligned PWM capability, the field of applications is extended to motor control, lighting and half-bridge driver

- 16-bit up, down and up/down autoreload counter with 16-bit prescaler
- Four independent capture/compare channels (CAPCOM) configurable as input capture, output compare, PWM generation (edge and center aligned mode) and single pulse mode output
- Synchronization module to control the timer with external signals
- Break input to force the timer outputs into a defined state
- Three complementary outputs with adjustable dead time
- Encoder mode
- Interrupt sources: 3 x input capture/output compare, 1 x overflow/update, 1 x break

4.11 TIM2, TIM3 - 16-bit general purpose timers

- 16-bit auto reload (AR) up-counter
- 15-bit prescaler adjustable to fixed power of 2 ratios 1...32768
- Timers with 3 or 2 individually configurable capture/compare channels
- PWM mode
- Interrupt sources: 2 or 3 x input capture/output compare, 1 x overflow/update

4.12 TIM4 - 8-bit basic timer

- 8-bit auto reload, adjustable prescaler ratio to any power of 2 from 1 to 128
- Clock source: CPU clock
- Interrupt source: 1 x overflow/update

Table 3. TIM timer features

Timer	Counter size (bits)	Prescaler	Counting mode	CAPCOM channels	Complementary outputs	Ext. trigger	Timer synchronization/chaining
TIM1	16	Any integer from 1 to 65536	Up/down	4	3	Yes	No
TIM2	16	Any power of 2 from 1 to 32768	Up	3	0	No	
TIM3	16	Any power of 2 from 1 to 32768	Up	2	0	No	
TIM4	8	Any power of 2 from 1 to 128	Up	0	0	No	

4.13 Analog-to-digital converter (ADC1)

The STM8S105x4/6 products contain a 10-bit successive approximation A/D converter (ADC1) with up to 10 multiplexed input channels and the following main features:

- Input voltage range: 0 to VDD
- Conversion time: 14 clock cycles
- Single and continuous and buffered continuous conversion modes
- Buffer size (n x 10 bits) where n = number of input channels
- Scan mode for single and continuous conversion of a sequence of channels
- Analog watchdog capability with programmable upper and lower thresholds
- Analog watchdog interrupt
- External trigger input
- Trigger from TIM1 TRGO
- End of conversion (EOC) interrupt

Note: Additional AIN12 analog input is not selectable in ADC scan mode or with analog watchdog. Values converted from AIN12 are stored only into the ADC_DRH/ADC_DRL registers.

4.14 Communication interfaces

The following communication interfaces are implemented:

- UART1: Full feature UART, synchronous mode, SPI master mode, Smartcard mode, IrDA mode, single wire mode, LIN2.1 master capability
- SPI: Full and half-duplex, 8 Mbit/s
- I²C: Up to 400 kbit/s

4.14.1 UART2

Main features

- 1 Mbit/s full duplex SCI
- SPI emulation
- High precision baud rate generator
- Smartcard emulation
- IrDA SIR encoder decoder
- LIN master mode
- LIN slave mode

Asynchronous communication (UART mode)

- Full duplex communication - NRZ standard format (mark/space)
- Programmable transmit and receive baud rates up to 1 Mbit/s (fCPU/16) and capable of following any standard baud rate regardless of the input frequency
- Separate enable bits for transmitter and receiver
- Two receiver wakeup modes:
 - Address bit (MSB)
 - Idle line (interrupt)
- Transmission error detection with interrupt generation
- Parity control

Synchronous communication

- Full duplex synchronous transfers
- SPI master operation
- 8-bit data communication
- Maximum speed: 1 Mbit/s at 16 MHz (fCPU/16)

LIN master mode

- Emission: Generates 13-bit synch. break frame
- Reception: Detects 11-bit break frame

LIN slave mode

- Autonomous header handling - one single interrupt per valid message header
- Automatic baud rate synchronization - maximum tolerated initial clock deviation $\pm 15\%$
- Synch delimiter checking
- 11-bit LIN synch break detection - break detection always active
- Parity check on the LIN identifier field
- LIN error management
- Hot plugging support

4.14.2 SPI

- Maximum speed: 8 Mbit/s ($f_{MASTER}/2$) both for master and slave
- Full duplex synchronous transfers
- Simplex synchronous transfers on two lines with a possible bidirectional data line
- Master or slave operation - selectable by hardware or software
- CRC calculation
- 1 byte Tx and Rx buffer
- Slave/master selection input pin

4.14.3 I²C

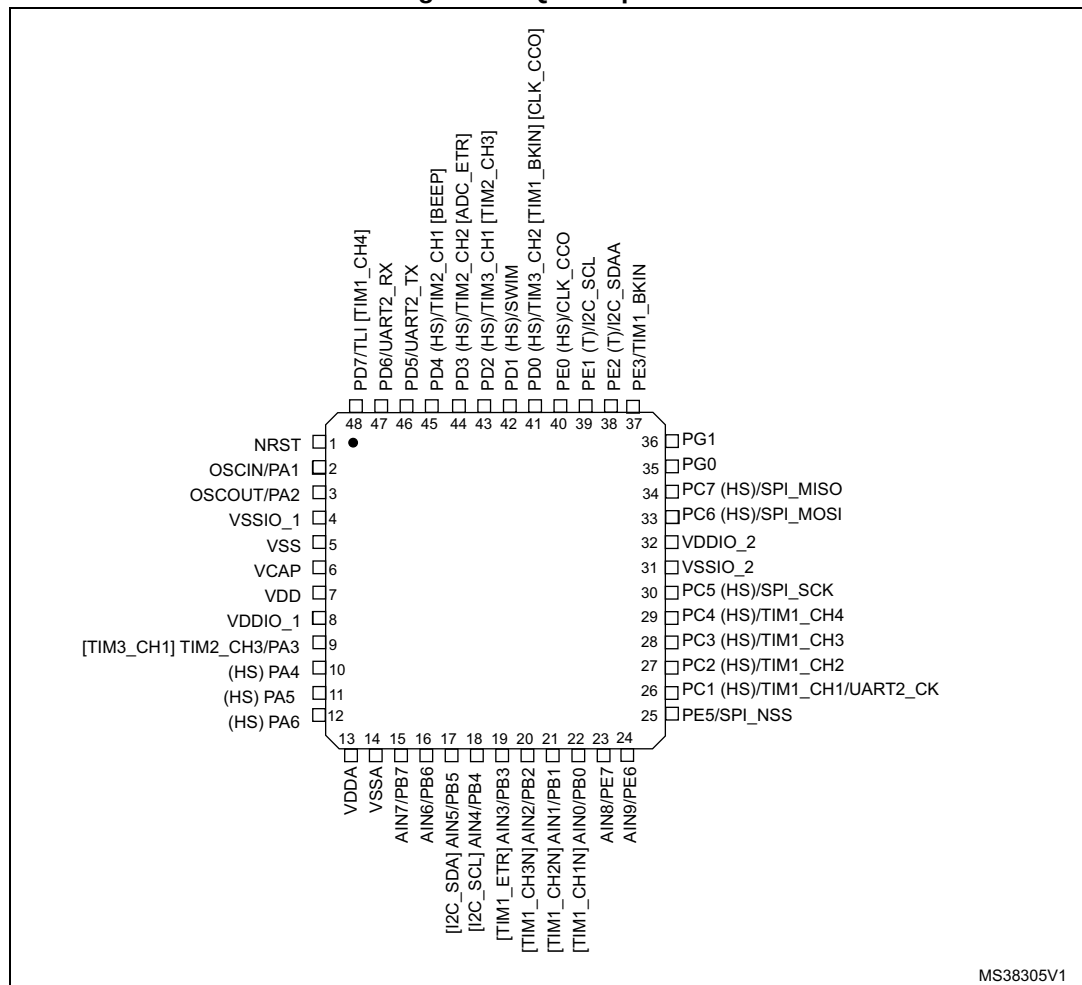
- I²C master features:
 - Clock generation
 - Start and stop generation
- I²C slave features:
 - Programmable I2C address detection
 - Stop bit detection
- Generation and detection of 7-bit/10-bit addressing and general call
- Supports different communication speeds:
 - Standard speed (up to 100 kHz)
 - Fast speed (up to 400 kHz)

5 Pinout and pin description

Table 4. Legend/abbreviations for pin description tables

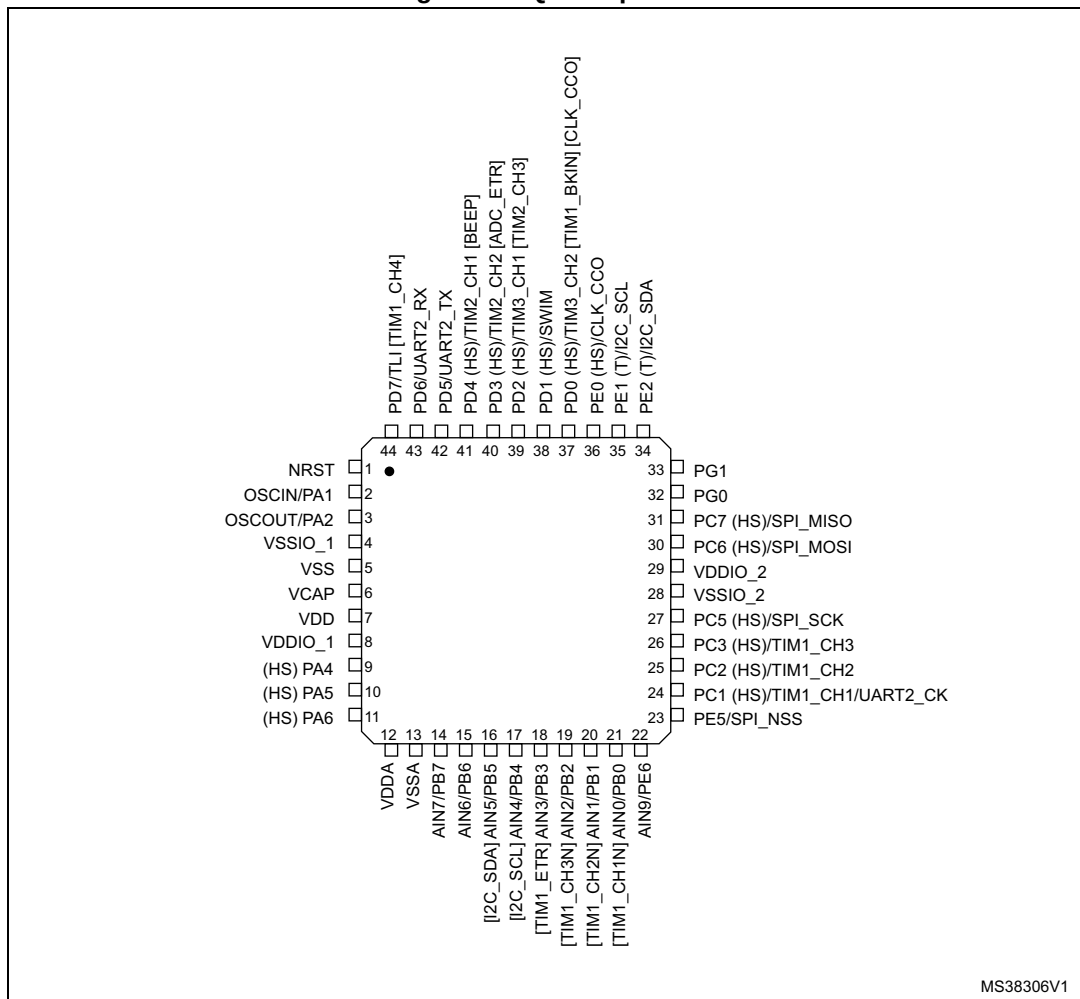
Type	I= Input, O = Output, S = Power supply	
Level	Input	CM = CMOS
	Output	HS = High sink
Output speed	O1 = Slow (up to 2 MHz) O2 = Fast (up to 10 MHz) O3 = Fast/slow programmability with slow as default state after reset O4 = Fast/slow programmability with fast as default state after reset	
Port and control configuration	Input	float = floating, wpu = weak pull-up
	Output	T = True open drain, OD = Open drain, PP = Push pull
Reset state	Bold X (pin state after internal reset release). Unless otherwise specified, the pin state is the same during the reset phase and after the internal reset release.	

Figure 3. LQFP48 pinout



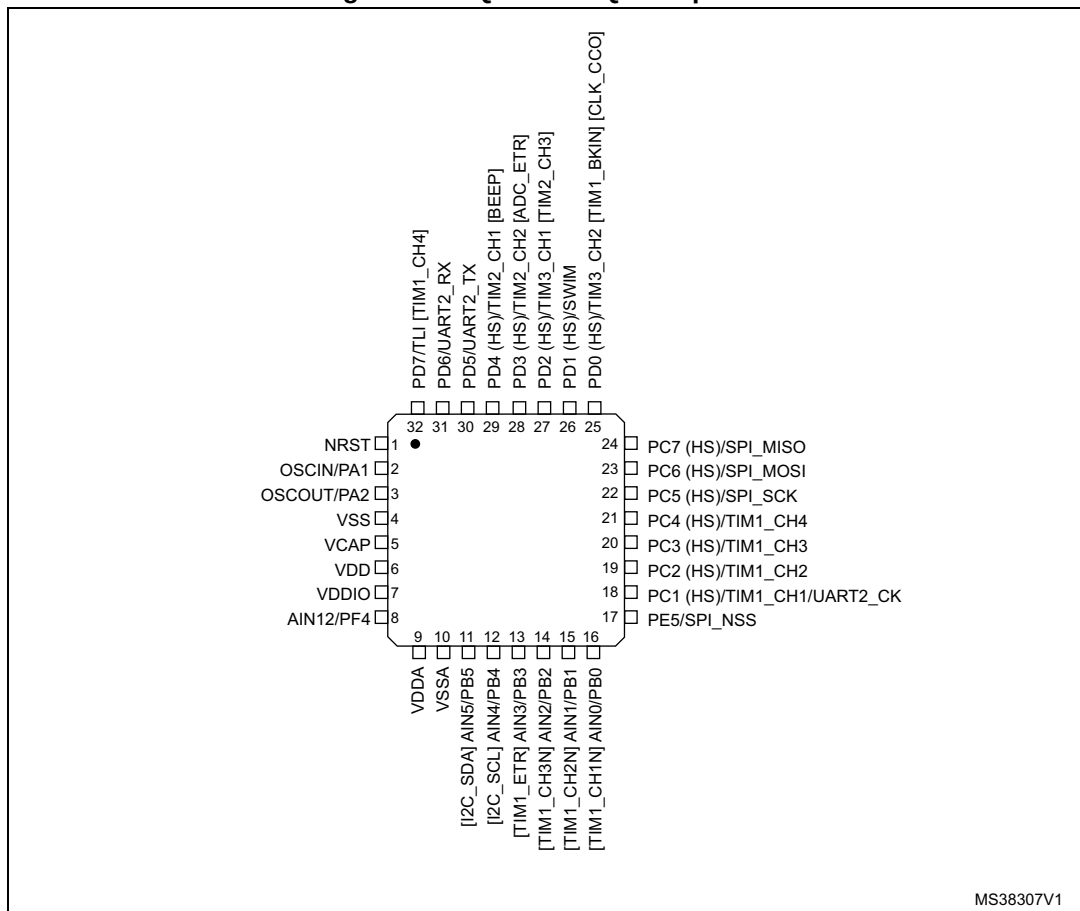
1. (HS) high sink capability.
2. (T) True open drain (P-buffer and protection diode to V_{DD} not implemented).
3. [] alternate function remapping option (if the same alternate function is shown twice, it indicates an exclusive choice not a duplication of the function).

Figure 4. LQFP44 pinout



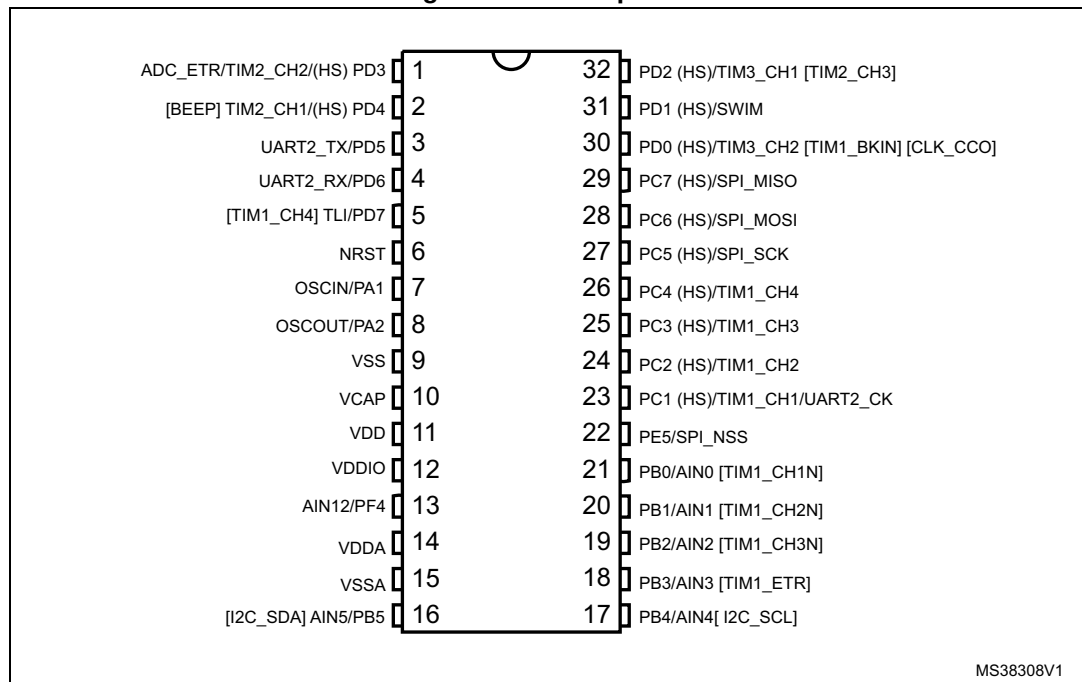
1. (HS) high sink capability.
2. (T) True open drain (P-buffer and protection diode to V_{DD} not implemented).
3. [] alternate function remapping option (if the same alternate function is shown twice, it indicates an exclusive choice not a duplication of the function).

Figure 5. UFQFPN32/LQFP32 pinout



1. (HS) high sink capability.
2. [] alternate function remapping option (if the same alternate function is shown twice, it indicates an exclusive choice not a duplication of the function).

Figure 6. SDIP32 pinout



1. (HS) high sink capability.
2. (T) True open drain (P-buffer and protection diode to V_{DD} not implemented).
3. [] alternate function remapping option (if the same alternate function is shown twice, it indicates an exclusive choice not a duplication of the function).

Table 5. STM8S105x4/6 pin description

Pin number				Pin name	Type	Input			Output				Main function (after reset)	Default alternate function	Alternate function after remap [option bit]
LQFP48	LQFP44	LQFP32/UFQFPN32	SDIP32			Floating	wpu	Ext. interrupt	High sink	Speed	OD	PP			
1	1	1	6	NRST	I/O	-	X	-	-	-	-	-	Reset		-
2	2	2	7	PA1/ OSC IN	I/O	X	X	-	-	O1	X	X	Port A1	Resonator/ crystal in	
3	3	3	8	PA2/ OSC OUT	I/O	X	X	-	-	O1	X	X	Port A1	Resonator/ crystal in	
4	4	-	-	VSSIO_1	S	-	-	-	-	-	-	-	I/O ground		-
5	5	4	9	VSS	S	-	-	-	-	-	-	-	Digital ground		-
6	6	5	10	VCAP	S	-	-	-	-	-	-	-	1.8 V regulator capacitor		-

Table 5. STM8S105x4/6 pin description (continued)

Pin number				Pin name	Type	Input			Output				Main function (after reset)	Default alternate function	Alternate function after remap [option bit]
LQFP48	LQFP44	LQFP32/UFQFPN32	SDIP32			Floating	wpu	Ext. interrupt	High sink	Speed	OD	PP			
7	7	6	11	VDD	S	-	-	-	-	-	-	-	Digital power supply		-
8	8	7	12	VDDIO_1	S	-	-	-	-	-	-	-	I/O power supply		-
9	-	-	-	PA3/TIM2_CH3 [TIM3_CH1]	I/O	<u>X</u>	X	X	-	O1	X	X	Port A3	Timer 2 - channel 3	TIM3_CH1 [AFR1]
10	9	-	-	PA4	I/O	<u>X</u>	X	X	HS	O3	X	X	Port A4	-	-
11	10	-	-	PA5	I/O	<u>X</u>	X	X	HS	O3	X	X	Port A5	-	-
12	11	-	-	PA6	I/O	<u>X</u>	X	X	HS	O3	X	X	Port A6	-	-
-	-	8	13	PF4/ AIN12 ⁽¹⁾	I/O	<u>X</u>	X	-	-	O1	X	X	Port F4	Analog input 12 ⁽²⁾	-
13	12	9	14	VDDA	S	-	-	-	-	-	-	-	Analog power supply		-
14	13	10	15	VSSA	S	-	-	-	-	-	-	-	Analog ground		-
15	14	-	-	PB7/ AIN7	I/O	<u>X</u>	X	X	-	O1	X	X	Port B7	Analog input 7	-
16	15	-	-	PB6/ AIN6	I/O	<u>X</u>	X	X	-	O1	X	X	Port B6	Analog input 6	-
17	16	11	16	PB5/ AIN5 [I2C_SDA]	I/O	<u>X</u>	X	X	-	O1	X	X	Port B5	Analog input 5	I2C_SDA [AFR6]
18	17	12	17	PB4/ AIN4 [I2C_SCL]	I/O	<u>X</u>	X	X	-	O1	X	X	Port B4	Analog input 4	I2C_SCL [AFR6]
19	18	13	18	PB3/ AIN3 [TIM1_ETR]	I/O	<u>X</u>	X	X	-	O1	X	X	Port B3	Analog input 3	TIM1_ETR [AFR5]
20	19	14	19	PB2/ AIN2 [TIM1_CH3N]	I/O	<u>X</u>	X	X	-	O1	X	X	Port B2	Analog input 2	TIM1_CH3N [AFR5]
21	20	15	20	PB1/ AIN1 [TIM1_CH2N]	I/O	<u>X</u>	X	X	-	O1	X	X	Port B1	Analog input 1	TIM1_CH2N [AFR5]
22	21	16	21	PB0/ AIN0 [TIM1_CH1N]	I/O	<u>X</u>	X	X	-	O1	X	X	Port B0	Analog input 0	TIM1_CH1N [AFR5]
23	-	-	-	PE7/ AIN8	I/O	<u>X</u>	X	X	-	O1	X	X	Port E7	Analog input 8	-

Table 5. STM8S105x4/6 pin description (continued)

Pin number				Pin name	Type	Input			Output				Main function (after reset)	Default alternate function	Alternate function after remap [option bit]
LQFP48	LQFP44	LQFP32/UFQFPN32	SDIP32			Floating	wpu	Ext. interrupt	High sink	Speed	OD	PP			
24	22	-	-	PE6/ AIN9	I/O	<u>X</u>	X	X	-	O1	X	X	Port E6	Analog input 9 ⁽³⁾	-
25	23	17	22	PE5/ SPI_NSS	I/O	<u>X</u>	X	X	-	O1	X	X	Port E5	SPI master/ slave select	-
26	24	18	23	PC1/ TIM1_CH1/ UART2_CK	I/O	<u>X</u>	X	X	HS	O3	X	X	Port C1	Timer 1 - channel 1/UART2 synchronous clock	-
27	25	19	24	PC2/ TIM1_CH2	I/O	<u>X</u>	X	X	HS	O3	X	X	Port C2	Timer 1 - channel 2	-
28	26	20	25	PC3/ TIM1_CH3	I/O	<u>X</u>	X	X	HS	O3	X	X	Port C3	Timer 1 - channel 3	-
29	-	21	26	PC4/ TIM1_CH4	I/O	<u>X</u>	X	X	HS	O3	X	X	Port C4	Timer 1 - channel 4	-
30	27	22	27	PC5/ SPI_SCK	I/O	<u>X</u>		X	HS	O3	X	X	Port C5	SPI clock	-
31	28	-	-	VSSIO_2	S	-	-	-	-	-	-	-	I/O ground		-
32	29	-	-	VDDIO_2	S	-	-	-	-	-	-	-	I/O power supply		-
33	30	23	28	PC6/ SPI_MOSI	I/O	<u>X</u>	X	X	HS	O3	X	X	Port C6	SPI master out/slave in	-
34	31	24	29	PC7/ SPI_MISO	I/O	<u>X</u>	X	X	HS	O3	X	X	Port C7	SPI master in/ slave out	-
35	32	-	-	PG0	I/O	<u>X</u>	X	-	-	O1	X	X	Port G0	-	-
36	33	-	-	PG1	I/O	<u>X</u>	X	-	-	O1	X	X	Port G1	-	-
37	-	-	-	PE3/ TIM1_BKIN	I/O	<u>X</u>	X	X	-	O1	X	X	Port E3	Timer 1 - break input	-
38	34	-	-	PE2/ I2C_SDA	I/O	<u>X</u>	-	X	-	O1	T ₍₄₎	-	Port E2	I2C data	-
39	35	-	-	PE1/ I2C_SCL	I/O	<u>X</u>	-	X	-	O1	T ₍₄₎	-	Port E1	I2C clock	-

Table 5. STM8S105x4/6 pin description (continued)

Pin number				Pin name	Type	Input			Output				Main function (after reset)	Default alternate function	Alternate function after remap [option bit]
LQFP48	LQFP44	LQFP32/UFQFPN32	SDIP32			Floating	wpu	Ext. interrupt	High sink	Speed	OD	PP			
40	36	-	-	PE0/ CLK_CCO	I/O	<u>X</u>	X	X	HS	O3	X	X	Port E0	Configurable clock output	-
41	37	25	30	PD0/ TIM3_CH2 [TIM1_BKIN] [CLK_CCO]	I/O	<u>X</u>	X	X	HS	O3	X	X	Port D0	Timer 3 - channel 2	TIM1_BKIN [AFR3]/CLK_CCO [AFR2]
42	38	26	31	PD1/ SWIM ⁽⁵⁾	I/O	<u>X</u>	X	X	X	HS	O4	X	Port D1	SWIM data interface	-
43	39	27	32	PD2/ TIM3_CH1 [TIM2_CH3]	I/O	<u>X</u>	X	X	HS	O3	X	X	Port D2	Timer 3 - channel 1	TIM2_CH3 [AFR1]
44	40	28	1	PD3/ TIM2_CH2 [ADC_ETR]	I/O	<u>X</u>	X	X	HS	O3	X	X	Port D3	Timer 2 - channel 2	ADC_ETR [AFR0]
45	41	29	2	PD4/ TIM2_CH1 [BEEP]	I/O	<u>X</u>	X	X	HS	O3	X	X	Port D4	Timer 2 - channel 1	BEEP output [AFR7]
46	42	30	3	PD5/ UART2_TX	I/O	<u>X</u>	X	X	-	O1	X	X	Port D5	UART2 data transmit	-
47	43	31	4	PD6/ UART2_RX	I/O	<u>X</u>	X	X	-	O1	X	X	Port D6	UART2 data receive	-
48	44	32	5	PD7/ TLI [TIM1_CH4]	I/O	<u>X</u>	X	X	-	O1	X	X	Port D7	Top level interrupt	TIM1_CH4 [AFR4]

1. A pull-up is applied to PF4 during the reset phase. This pin is input floating after reset release.
2. AIN12 is not selectable in ADC scan mode or with analog watchdog.
3. In 44-pin package, AIN9 cannot be used by ADC scan mode.
4. In the open-drain output column, 'T' defines a true open-drain I/O (P-buffer, weak pull-up and protection diode to V_{DD} are not implemented).
5. The PD1 pin is in input pull-up during the reset phase and after internal reset release.

5.1 Alternate function remapping

As shown in the rightmost column of the pin description table, some alternate functions can be remapped at different I/O ports by programming one of eight AFR (alternate function remap) option bits. When the remapping option is active, the default alternate function is no longer available.

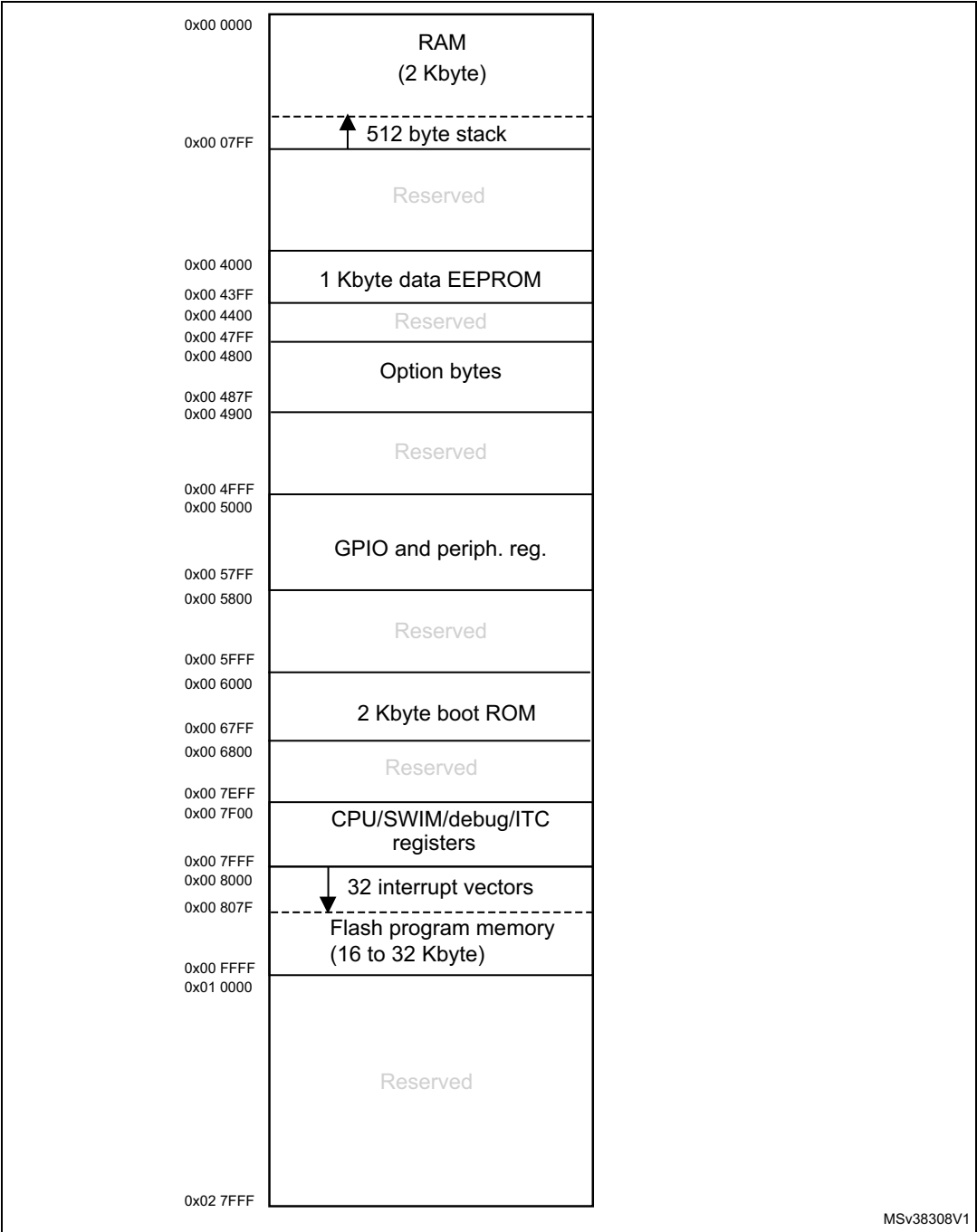
To use an alternate function, the corresponding peripheral must be enabled in the peripheral registers.

Alternate function remapping does not effect GPIO capabilities of the I/O ports (see the GPIO section of the family reference manual, RM0016).

6 Memory and register map

6.1 Memory map

Figure 7. Memory map



The following table lists the boundary addresses for each memory size. The top of the stack is at the RAM end address in each case.

Table 6. Flash, data EEPROM and RAM boundary address

Memory area	Size (byte)	Start address	End address
Flash program memory	32 K	0x00 8000	0x00 FFFF
	16 K	0x00 8000	0x00 BFFF
RAM	2 K	0x00 0000	0x00 07FF
Data EEPROM	1024	0x00 4000	0x00 43FF

6.2 Register map

6.2.1 I/O port hardware register map

Table 7. I/O port hardware register map

Address	Block	Register label	Register name	Reset status
0x00 5000	Port A	PA_ODR	Port A data output latch register	0x00
0x00 5001		PA_IDR	Port A input pin value register	0xXX ⁽¹⁾
0x00 5002		PA_DDR	Port A data direction register	0x00
0x00 5003		PA_CR1	Port A control register 1	0x00
0x00 5004		PA_CR2	Port A control register 2	0x00
0x00 5005	Port B	PB_ODR	Port B data output latch register	0x00
0x00 5006		PB_IDR	Port B input pin value register	0xXX ⁽¹⁾
0x00 5007		PB_DDR	Port B data direction register	0x00
0x00 5008		PB_CR1	Port B control register 1	0x00
0x00 5009		PB_CR2	Port B control register 2	0x00
0x00 500A	Port C	PC_ODR	Port C data output latch register	0x00
0x00 500B		PC_IDR	Port C input pin value register	0xXX ⁽¹⁾
0x00 500C		PC_DDR	Port C data direction register	0x00
0x00 500D		PC_CR1	Port C control register 1	0x00
0x00 500E		PC_CR2	Port C control register 2	0x00
0x00 500F	Port D	PD_ODR	Port D data output latch register	0x00
0x00 5010		PD_IDR	Port D input pin value register	0xXX ⁽¹⁾
0x00 5011		PD_DDR	Port D data direction register	0x00
0x00 5012		PD_CR1	Port D control register 1	0x02
0x00 5013		PD_CR2	Port D control register 2	0x00

Table 7. I/O port hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 5014	Port E	PE_ODR	Port E data output latch register	0x00
0x00 5015		PE_IDR	Port E input pin value register	0xFF ⁽¹⁾
0x00 5016		PE_DDR	Port E data direction register	0x00
0x00 5017		PE_CR1	Port E control register 1	0x00
0x00 5018		PE_CR2	Port E control register 2	0x00
0x00 5019	Port F	PF_ODR	Port F data output latch register	0x00
0x00 501A		PF_IDR	Port F input pin value register	0xFF ⁽¹⁾
0x00 501B		PF_DDR	Port F data direction register	0x00
0x00 501C		PF_CR1	Port F control register 1	0x00
0x00 501D		PF_CR2	Port F control register 2	0x00
0x00 501E	Port G	PG_ODR	Port G data output latch register	0x00
0x00 501F		PG_IDR	Port G input pin value register	0xFF ⁽¹⁾
0x00 5020		PG_DDR	Port G data direction register	0x00
0x00 5021		PG_CR1	Port G control register 1	0x00
0x00 5022		PG_CR2	Port G control register 2	0x00
0x00 5023	Port H	PH_ODR	Port H data output latch register	0x00
0x00 5024		PH_IDR	Port H input pin value register	0xFF ⁽¹⁾
0x00 5025		PH_DDR	Port H data direction register	0x00
0x00 5026		PH_CR1	Port H control register 1	0x00
0x00 5027		PH_CR2	Port H control register 2	0x00
0x00 5028	Port I	PI_ODR	Port I data output latch register	0x00
0x00 5029		PI_IDR	Port I input pin value register	0xFF ⁽¹⁾
0x00 502A		PI_DDR	Port I data direction register	0x00
0x00 502B		PI_CR1	Port I control register 1	0x00
0x00 502C		PI_CR2	Port I control register 2	0x00

1. Depends on the external circuitry.

6.2.2 General hardware register map

Table 8. General hardware register map

Address	Block	Register label	Register name	Reset status
0x00 5050 to 0x00 5059	Reserved area (10 byte)			
0x00 505A	Flash	FLASH_CR1	Flash control register 1	0x00
0x00 505B		FLASH_CR2	Flash control register 2	0x00
0x00 505C		FLASH_NCR2	Flash complementary control register 2	0xFF
0x00 505D		FLASH_FPR	Flash protection register	0x00
0x00 505E		FLASH_NFPR	Flash complementary protection register	0xFF
0x00 505F		FLASH_IAPSR	Flash in-application programming status register	0x00
0x00 5060 to 0x00 5061	Reserved area (2 byte)			
0x00 5062	Flash	FLASH_PUKR	Flash program memory unprotection register	0x00
0x00 5063	Reserved area (1 byte)			
0x00 5064	Flash	FLASH_DUKR	Data EEPROM unprotection register	0x00
0x00 5065 to 0x00 509F	Reserved area (59 byte)			
0x00 50A0	ITC	EXTI_CR1	External interrupt control register 1	0x00
0x00 50A1		EXTI_CR2	External interrupt control register 2	0x00
0x00 50A2 to 0x00 50B2	Reserved area (17 byte)			
0x00 50B3	RST	RST_SR	Reset status register	0xFF ⁽¹⁾
0x00 50B4 to 0x00 50BF	Reserved area (12 byte)			
0x00 50C0	CLK	CLK_IICKR	Internal clock control register	0x01
0x00 50C1		CLK_EICKR	External clock control register	0x00
0x00 50C2	Reserved area (1 byte)			

Table 8. General hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 50C3	CLK	CLK_CMSR	Clock master status register	0xE1
0x00 50C4		CLK_SWR	Clock master switch register	0xE1
0x00 50C5		CLK_SWCR	Clock switch control register	0xFF
0x00 50C6		CLK_CKDIVR	Clock divider register	0x18
0x00 50C7		CLK_PCKENR1	Peripheral clock gating register 1	0xFF
0x00 50C8		CLK_CSSR	Clock security system register	0x00
0x00 50C9		CLK_CCOR	Configurable clock control register	0x00
0x00 50CA		CLK_PCKENR2	Peripheral clock gating register 2	0xFF
0x00 50CC		CLK_HSI TRIMR	HSI clock calibration trimming register	0x00
0x00 50CD		CLK_SWIMCCR	SWIM clock control register	0bXXXX XXX0
0x00 50CE to 0x00 50D0	Reserved area (3 byte)			
0x00 50D1	WWDG	WWDG_CR	WWDG control register	0x7F
0x00 50D2		WWDG_WR	WWDG window register	0x7F
0x00 50D3 to 00 50DF	Reserved area (13 byte)			
0x00 50E0	IWDG	IWDG_KR	IWDG key register	0xFF ⁽²⁾
0x00 50E1		IWDG_PR	IWDG prescaler register	0x00
0x00 50E2		IWDG_RLR	IWDG reload register	0xFF
0x00 50E3 to 0x00 50EF	Reserved area (13 byte)			
0x00 50F0	AWU	AWU_CSR1	AWU control/status register 1	0x00
0x00 50F1		AWU_APR	AWU asynchronous prescaler buffer register	0x3F
0x00 50F2		AWU_TBR	AWU timebase selection register	0x00
0x00 50F3	BEEP	BEEP_CSR	BEEP control/status register	0x1F
0x00 50F4 to 0x00 50FF	Reserved area (12 byte)			
0x00 5200	SPI	SPI_CR1	SPI control register 1	0x00
0x00 5201		SPI_CR2	SPI control register 2	0x00
0x00 5202		SPI_ICR	SPI interrupt control register	0x00
0x00 5203		SPI_SR	SPI status register	0x02
0x00 5204		SPI_DR	SPI data register	0x00
0x00 5205		SPI_CRCPR	SPI CRC polynomial register	0x07
0x00 5206		SPI_RXCRCR	SPI Rx CRC register	0xFF
0x00 5207		SPI_TXCRCR	SPI Tx CRC register	0xFF

Table 8. General hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 5208 to 0x00 520F	Reserved area (8 byte)			
0x00 5210	I2C	I2C_CR1	I2C control register 1	0x00
0x00 5211		I2C_CR2	I2C control register 2	0x00
0x00 5212		I2C_FREQR	I2C frequency register	0x00
0x00 5213		I2C_OARL	I2C Own address register low	0x00
0x00 5214		I2C_OARH	I2C Own address register high	0x00
0x00 5215		Reserved		
0x00 5216		I2C_DR	I2C data register	0x00
0x00 5217		I2C_SR1	I2C status register 1	0x00
0x00 5218		I2C_SR2	I2C status register 2	0x00
0x00 5219		I2C_SR3	I2C status register 3	0x0X
0x00 521A		I2C_ITR	I2C interrupt control register	0x00
0x00 521B		I2C_CCRL	I2C Clock control register low	0x00
0x00 521C		I2C_CCRH	I2C Clock control register high	0x00
0x00 521D		I2C_TRISER	I2C TRISE register	0x02
0x00 521E		I2C_PECR	I2C packet error checking register	0x00
0x00 521F to 0x00 522F	Reserved area (17 byte)			
0x00 5230 to 0x00 523F	Reserved area (6 byte)			
0x00 5240	UART2	UART2_SR	UART2 status register	0xC0
0x00 5241		UART2_DR	UART2 data register	0xXX
0x00 5242		UART2_BRR1	UART2 baud rate register 1	0x00
0x00 5243		UART2_BRR2	UART2 baud rate register 2	0x00
0x00 5244		UART2_CR1	UART2 control register 1	0x00
0x00 5245		UART2_CR2	UART2 control register 2	0x00
0x00 5246		UART2_CR3	UART2 control register 3	0x00
0x00 5247		UART2_CR4	UART2 control register 4	0x00
0x00 5248		UART2_CR5	UART2 control register 5	0x00
0x00 5249		UART2_CR6	UART2 control register 6	0x00
0x00 524A		UART2_GTR	UART2 guard time register	0x00
0x00 524B		UART2_PSCR	UART2 prescaler register	0x00
0x00 524C to 0x00 524F	Reserved area (4 byte)			

Table 8. General hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 5250	TIM1	TIM1_CR1	TIM1 control register 1	0x00
0x00 5251		TIM1_CR2	TIM1 control register 2	0x00
0x00 5252		TIM1_SMCR	TIM1 slave mode control register	0x00
0x00 5253		TIM1_ETR	TIM1 external trigger register	0x00
0x00 5254		TIM1_IER	TIM1 interrupt enable register	0x00
0x00 5255		TIM1_SR1	TIM1 status register 1	0x00
0x00 5256		TIM1_SR2	TIM1 status register 2	0x00
0x00 5257		TIM1_EGR	TIM1 event generation register	0x00
0x00 5258		TIM1_CCMR1	TIM1 capture/compare mode register 1	0x00
0x00 5259		TIM1_CCMR2	TIM1 capture/compare mode register 2	0x00
0x00 525A		TIM1_CCMR3	TIM1 capture/compare mode register 3	0x00
0x00 525B		TIM1_CCMR4	TIM1 capture/compare mode register 4	0x00
0x00 525C		TIM1_CCER1	TIM1 capture/compare enable register 1	0x00
0x00 525D		TIM1_CCER2	TIM1 capture/compare enable register 2	0x00
0x00 525E		TIM1_CNTRH	TIM1 counter high	0x00
0x00 525F		TIM1_CNTRL	TIM1 counter low	0x00
0x00 5260		TIM1_PSCRH	TIM1 prescaler register high	0x00
0x00 5261		TIM1_PSCRL	TIM1 prescaler register low	0x00
0x00 5262		TIM1_ARRH	TIM1 auto-reload register high	0xFF
0x00 5263		TIM1_ARRL	TIM1 auto-reload register low	0xFF
0x00 5264		TIM1_RCR	TIM1 repetition counter register	0x00
0x00 5265		TIM1_CCR1H	TIM1 capture/compare register 1 high	0x00
0x00 5266		TIM1_CCR1L	TIM1 capture/compare register 1 low	0x00
0x00 5267		TIM1_CCR2H	TIM1 capture/compare register 2 high	0x00
0x00 5268		TIM1_CCR2L	TIM1 capture/compare register 2 low	0x00
0x00 5269		TIM1_CCR3H	TIM1 capture/compare register 3 high	0x00

Table 8. General hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 526A	TIM1	TIM1_CCR3L	TIM1 capture/compare register 3 low	0x00
0x00 526B		TIM1_CCR4H	TIM1 capture/compare register 4 high	0x00
0x00 526C		TIM1_CCR4L	TIM1 capture/compare register 4 low	0x00
0x00 526D		TIM1_BKR	TIM1 break register	0x00
0x00 526E		TIM1_DTR	TIM1 dead-time register	0x00
0x00 526F		TIM1_OISR	TIM1 output idle state register	0x00
0x00 5270 to 0x00 52FF	Reserved area (147 byte)			

Table 8. General hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 5300	TIM2	TIM2_CR1	TIM2 control register 1	0x00
0x00 5301		TIM2_IER	TIM2 Interrupt enable register	0x00
0x00 5302		TIM2_SR1	TIM2 status register 1	0x00
0x00 5303		TIM2_SR2	TIM2 status register 2	0x00
0x00 5304		TIM2_EGR	TIM2 event generation register	0x00
0x00 5305		TIM2_CCMR1	TIM2 capture/compare mode register 1	0x00
0x00 5306		TIM2_CCMR2	TIM2 capture/compare mode register 2	0x00
0x00 5307		TIM2_CCMR3	TIM2 capture/compare mode register 3	0x00
0x00 5308		TIM2_CCER1	TIM2 capture/compare enable register 1	0x00
0x00 5309		TIM2_CCER2	TIM2 capture/compare enable register 2	0x00
0x00 530A		TIM2_CNTRH	TIM2 counter high	0x00
0x00 530B		TIM2_CNTRL	TIM2 counter low	0x00
0x00 530C		TIM2_PSCR	IM2 prescaler register	0x00
0x00 530D		TIM2_ARRH	TIM2 auto-reload register high	0xFF
0x00 530E		TIM2_ARRL	TIM2 auto-reload register low	0xFF
0x00 530F		TIM2_CCR1H	TIM2 capture/compare register 1 high	0x00
0x00 5310		TIM2_CCR1L	TIM2 capture/compare register 1 low	0x00
0x00 5311		TIM2_CCR2H	TIM2 capture/compare reg. 2 high	0x00
0x00 5312		TIM2_CCR2L	TIM2 capture/compare register 2 low	0x00
0x00 5313		TIM2_CCR3H	TIM2 capture/compare register 3 high	0x00
0x00 5314		TIM2_CCR3L	TIM2 capture/compare register 3 low	0x00
0x00 5315 to 0x00 531F	Reserved area (11 byte)			

Table 8. General hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 5320	TIM3	TIM3_CR1	TIM3 control register 1	0x00
0x00 5321		TIM3_IER	TIM3 Interrupt enable register	0x00
0x00 5322		TIM3_SR1	TIM3 status register 1	0x00
0x00 5323		TIM3_SR2	TIM3 status register 2	0x00
0x00 5324		TIM3_EGR	TIM3 event generation register	0x00
0x00 5325		TIM3_CCMR1	TIM3 capture/compare mode register 1	0x00
0x00 5326		TIM3_CCMR2	TIM3 capture/compare mode register 2	0x00
0x00 5327		TIM3_CCER1	TIM3 capture/compare enable register 1	0x00
0x00 5328		TIM3_CNTRH	TIM3 counter high	0x00
0x00 5329		TIM3_CNTRL	TIM3 counter low	0x00
0x00 532A		TIM3_PSCR	TIM3 prescaler register	0x00
0x00 532B		TIM3_ARRH	TIM3 auto-reload register high	0xFF
0x00 532C		TIM3_ARRL	TIM3 auto-reload register low	0xFF
0x00 532D		TIM3_CCR1H	TIM3 capture/compare register 1 high	0x00
0x00 532E		TIM3_CCR1L	TIM3 capture/compare register 1 low	0x00
0x00 532F		TIM3_CCR2H	TIM3 capture/compare reg. 2 high	0x00
0x00 5330		TIM3_CCR2L	TIM3 capture/compare register 2 low	0x00
0x00 5331 to 0x00 533F	Reserved area (15 byte)			
0x00 5340	TIM4	TIM4_CR1	TIM4 control register 1	0x00
0x00 5341		TIM4_IER	TIM4 interrupt enable register	0x00
0x00 5342		TIM4_SR	TIM4 status register	0x00
0x00 5343		TIM4_EGR	TIM4 event generation register	0x00
0x00 5344		TIM4_CNTR	TIM4 counter	0x00
0x00 5345		TIM4_PSCR	TIM4 prescaler register	0x00
0x00 5346		TIM4_ARR	TIM4 auto-reload register	0xFF
0x00 5347 to 0x00 53DF	Reserved area (153 byte)			
0x00 53E0 to 0x00 53F3	ADC1	ADC_DBxR	ADC data buffer registers	0x00
0x00 53F4 to 0x00 53FF	Reserved area (12 byte)			

Table 8. General hardware register map (continued)

Address	Block	Register label	Register name	Reset status
0x00 5400	ADC1 cont'd	ADC_CSR	ADC control/status register	0x00
0x00 5401		ADC_CR1	ADC configuration register 1	0x00
0x00 5402		ADC_CR2	ADC configuration register 2	0x00
0x00 5403		ADC_CR3	ADC configuration register 3	0x00
0x00 5404		ADC_DRH	ADC data register high	0xXX
0x00 5405		ADC_DRL	ADC data register low	0xXX
0x00 5406		ADC_TDRH	ADC Schmitt trigger disable register high	0x00
0x00 5407		ADC_TDRL	ADC Schmitt trigger disable register low	0x00
0x00 5408		ADC_HTRH	ADC high threshold register high	0x03
0x00 5409		ADC_HTRL	ADC high threshold register low	0xFF
0x00 540A		ADC_LTRH	ADC low threshold register high	0x00
0x00 540B		ADC_LTRL	ADC low threshold register low	0x00
0x00 540C		ADC_AWSRH	ADC analog watchdog status register high	0x00
0x00 540D		ADC_AWSRL	ADC analog watchdog status register low	0x00
0x00 540E		ADC_AWCRH	ADC analog watchdog control register high	0x00
0x00 540F		ADC_AWCRL	ADC analog watchdog control register low	0x00
0x00 5410 to 0x00 57FF	Reserved area (1008 byte)			

1. Depends on the previous reset source.
2. Write-only register.

6.2.3 CPU/SWIM/debug module/interrupt controller registers

Table 9. CPU/SWIM/debug module/interrupt controller registers

Address	Block	Register label	Register name	Reset status
0x00 7F00	CPU ⁽¹⁾	A	Accumulator	0x00
0x00 7F01		PCE	Program counter extended	0x00
0x00 7F02		PCH	Program counter high	0x00
0x00 7F03		PCL	Program counter low	0x00
0x00 7F04		XH	X index register high	0x00
0x00 7F05		XL	X index register low	0x00
0x00 7F06		YH	Y index register high	0x00
0x00 7F07		YL	Y index register low	0x00
0x00 7F08		SPH	Stack pointer high	0x03
0x00 7F09		SPL	Stack pointer low	0xFF
0x00 7F0A		CCR	Condition code register	0x28
0x00 7F0B to 0x00 7F5F	Reserved area (85 byte)			
0x00 7F60	CPU	CFG_GCR	Global configuration register	0x00
0x00 7F70	ITC	ITC_SPR1	Interrupt software priority register 1	0xFF
0x00 7F71		ITC_SPR2	Interrupt software priority register 2	0xFF
0x00 7F72		ITC_SPR3	Interrupt software priority register 3	0xFF
0x00 7F73		ITC_SPR4	Interrupt software priority register 4	0xFF
0x00 7F74		ITC_SPR5	Interrupt software priority register 5	0xFF
0x00 7F75		ITC_SPR6	Interrupt software priority register 6	0xFF
0x00 7F76		ITC_SPR7	Interrupt software priority register 7	0xFF
0x00 7F77		ITC_SPR8	Interrupt software priority register 8	0xFF
0x00 7F78 to 0x00 7F79	Reserved area (2 byte)			
0x00 7F80	SWIM	SWIM_CSR	SWIM control status register	0x00
0x00 7F81 to 0x00 7F8F	Reserved area (15 byte)			

Table 9. CPU/SWIM/debug module/interrupt controller registers (continued)

Address	Block	Register label	Register name	Reset status
0x00 7F90	DM	DM_BK1RE	DM breakpoint 1 register extended byte	0xFF
0x00 7F91		DM_BK1RH	DM breakpoint 1 register high byte	0xFF
0x00 7F92		DM_BK1RL	DM breakpoint 1 register low byte	0xFF
0x00 7F93		DM_BK2RE	DM breakpoint 2 register extended byte	0xFF
0x00 7F94		DM_BK2RH	DM breakpoint 2 register high byte	0xFF
0x00 7F95		DM_BK2RL	DM breakpoint 2 register low byte	0xFF
0x00 7F96		DM_CR1	DM debug module control register 1	0x00
0x00 7F97		DM_CR2	DM debug module control register 2	0x00
0x00 7F98		DM_CSR1	DM debug module control/status register 1	0x10
0x00 7F99		DM_CSR2	DM debug module control/status register 2	0x00
0x00 7F9A		DM_ENFCTR	DM enable function register	0xFF
0x00 7F9B to 0x00 7F9F	Reserved area (5 byte)			

1. Accessible by debug module only.

7 Interrupt vector mapping

Table 10. Interrupt mapping

IRQ no.	Source block	Description	Wakeup from halt mode	Wakeup from active-halt mode	Vector address
-	RESET	Reset	Yes	Yes	0x00 8000
-	TRAP	Software interrupt	-	-	0x00 8004
0	TLI	External top level interrupt	-	-	0x00 8008
1	AWU	Auto wake up from halt	-	Yes	0x00 800C
2	CLK	Clock controller	-	-	0x00 8010
3	EXTI0	Port A external interrupts	Yes ⁽¹⁾	Yes ⁽¹⁾	0x00 8014
4	EXTI1	Port B external interrupts	Yes	Yes	0x00 8018
5	EXTI2	Port C external interrupts	Yes	Yes	0x00 801C
6	EXTI3	Port D external interrupts	Yes	Yes	0x00 8020
7	EXTI4	Port E external interrupts	Yes	Yes	0x00 8024
8	Reserved	-	-	-	0x00 8028
9	Reserved	-	-	-	0x00 802C
10	SPI	End of transfer	Yes	Yes	0x00 8030
11	TIM1	TIM1 update/ overflow/ underflow/ trigger/ break	-	-	0x00 8034
12	TIM1	TIM1 capture/ compare	-	-	0x00 8038
13	TIM2	TIM2 update/ overflow	-	-	0x00 803C
14	TIM2	TIM2 capture/ compare	-	-	0x00 8040
15	TIM3	TIM3 update/ overflow	-	-	0x00 8044
16	TIM3	TIM3 capture/ compare	-	-	0x00 8048
17	Reserved	-	-	-	0x00 804C
18	Reserved	-	-	-	0x00 8050
19	I2C	I2C interrupt	Yes	Yes	0x00 8054

Table 10. Interrupt mapping (continued)

IRQ no.	Source block	Description	Wakeup from halt mode	Wakeup from active-halt mode	Vector address
20	UART2	Tx complete	-	-	0x00 8058
21	UART2	Receive register DATA FULL	-	-	0x00 805C
22	ADC1	ADC1 end of conversion/ analog watchdog interrupt	-	-	0x00 8060
23	TIM4	TIM4 update/ overflow	-	-	0x00 8064
24	Flash	EOP/WR_PG_DIS	-	-	0x00 8068
Reserved					0x00 806C to 0x00 807C

1. Except PA1.

8 Option byte

Option byte contain configurations for device hardware features as well as the memory protection of the device. They are stored in a dedicated block of the memory. Except for the ROP (read-out protection) byte, each option byte has to be stored twice, in a regular form (OPTx) and a complemented one (NOPTx) for redundancy.

Option byte can be modified in ICP mode (via SWIM) by accessing the EEPROM address shown in the table below.

Option byte can also be modified 'on the fly' by the application in IAP mode, except the ROP option that can only be modified in ICP mode (via SWIM).

Refer to the STM8S Flash programming manual (PM0051) and STM8 SWIM communication protocol and debug module user manual (UM0470) for information on SWIM programming procedures.

Table 11. Option byte

Addr.	Option name	Option byte no.	Option bits								Factory default setting
			7	6	5	4	3	2	1	0	
0x4800	Read-out protection (ROP)	OPT0	ROP [7:0]								0x00
0x4801	User boot code (UBC)	OPT1	UBC [7:0]								0x00
0x4802		NOPT1	NUBC [7:0]								0xFF
0x4803	Alternate function remapping (AFR)	OPT2	AFR7	AFR6	AFR5	AFR4	AFR3	AFR2	AFR1	AFR0	0x00
0x4804		NOPT2	NAFR7	NAFR6	NAFR5	NAFR4	NAFR3	NAFR2	NAFR1	NAFR0	0xFF
0x4805h	Misc. option	OPT3	Reserved			HSI TRIM	LSI _ EN	IWDG _HW	WWDG _HW	WWDG _HALT	0x00
0x4806		NOPT3	Reserved			NHSI TRIM	NLSI _ EN	NIWDG _HW	NWWDG _HW	NWWG _HALT	0xFF
0x4807	Clock option	OPT4	Reserved				EXT CLK	CKAWU SEL	PRS C1	PRS C0	0x00
0x4808		NOPT4	Reserved				NEXT CLK	NCKA WUSEL	NPRSC1	NPR SC0	0xFF
0x4809	HSE clock startup	OPT5	HSECNT [7:0]								0x00
0x480A		NOPT5	NHSECNT [7:0]								0xFF
0x480B	Reserved	OPT6	Reserved								0x00
0x480C		NOPT6	Reserved								0xFF
0x480D	Reserved	OPT7	Reserved								0x00
0x480E		NOPT7	Reserved								0xFF
0x480F	Reserved	-	Reserved								-
0x48FD		-	Reserved								-

Table 11. Option byte (continued)

Addr.	Option name	Option byte no.	Option bits								Factory default setting
			7	6	5	4	3	2	1	0	
0x487E	Bootloader	OPTBL	BL[7:0]								0x00
0x487F		NOPTBL	NBL[7:0]								0xFF

Table 12. Option byte description

Option byte no.	Description
OPT0	ROP[7:0] Memory readout protection (ROP) 0xAA: Enable readout protection (write access via SWIM protocol) <i>Note: Refer to the family reference manual (RM0016) section on Flash/EEPROM memory readout protection for details.</i>
OPT1	UBC[7:0] User boot code area 0x00: no UBC, no write-protection 0x01: Page 0 to 1 defined as UBC, memory write-protected 0x02: Page 0 to 3 defined as UBC, memory write-protected 0x03: Page 0 to 4 defined as UBC, memory write-protected ... 0x3E: Pages 0 to 63 defined as UBC, memory write-protected Other values: Reserved <i>Note: Refer to the family reference manual (RM0016) section on Flash write protection for more details.</i>
OPT2	AFR[7:0] Refer to the following table for the description of the alternate function remapping description of bits [7:2].
OPT3	HSITRIM: High speed internal clock trimming register size 0: 3-bit trimming supported in CLK_HSITRIMR register 1: 4-bit trimming supported in CLK_HSITRIMR register
	LSI_EN: Low speed internal clock enable 0: LSI clock is not available as CPU clock source 1: LSI clock is available as CPU clock source
	IWDG_HW: Independent watchdog 0: IWDG Independent watchdog activated by software 1: IWDG Independent watchdog activated by hardware
	WWDG_HW: Window watchdog activation 0: WWDG window watchdog activated by software 1: WWDG window watchdog activated by hardware
	WWDG_HALT: Window watchdog reset on halt 0: No reset generated on halt if WWDG active 1: Reset generated on halt if WWDG active

Table 12. Option byte description (continued)

Option byte no.	Description
OPT4	EXTCLK: External clock selection 0: External crystal connected to OSCIN/OSCOU 1: External clock signal on OSCIN
	CKAWUSEL: Auto wake-up unit/clock 0: LSI clock source selected for AWU 1: HSE clock with prescaler selected as clock source for AWU
	PRSC[1:0] AWU clock prescaler 0x: 16 MHz to 128 kHz prescaler 10: 8 MHz to 128 kHz prescaler 11: 4 MHz to 128 kHz prescaler
OPT5	HSECNT[7:0]: HSE crystal oscillator stabilization time 0x00: 2048 HSE cycles 0xB4: 128 HSE cycles 0xD2: 8 HSE cycles 0xE1: 0.5 HSE cycles
OPT6	Reserved
OPT7	Reserved
OPTBL	BL[7:0]: Bootloader option byte For STM8S products, this option is checked by the boot ROM code after reset. Depending on the content of addresses 0x487E, 0x487F, and 0x8000 (reset vector), the CPU jumps to the bootloader or to the reset vector. Refer to the UM0560 (STM8L/S bootloader manual) for more details. For STM8L products, the bootloader option bytes are on addresses 0xFFFF and 0xFFFF+1 (2 byte). These option bytes control whether the bootloader is active or not. For more details, refer to the UM0560 (STM8L/S bootloader manual) for more details.

8.1 Alternate function remapping bits

Table 13. Alternate function remapping bits [7:0] of OPT2

Option byte no.	Description ⁽¹⁾
OPT2	AFR7 Alternate function remapping option 7 0: AFR7 remapping option inactive: Default alternate functions.⁽²⁾ 1: Port D4 alternate function = BEEP. AFR6 Alternate function remapping option 6 0: AFR6 remapping option inactive: Default alternate function.⁽²⁾ 1: Port B5 alternate function = I2C_SDA; port B4 alternate function = I2C_SCL. AFR5 Alternate function remapping option 5 0: AFR5 remapping option inactive: Default alternate function.⁽²⁾ 1: Port B3 alternate function = TIM1_ETR; port B2 alternate function = TIM1_NCC3; port B1 alternate function = TIM1_CH2N; port B0 alternate function = TIM1_CH1N. AFR4 Alternate function remapping option 4 0: AFR4 remapping option inactive: Default alternate functions.⁽²⁾ 1: Port D7 alternate function = TIM1_CH4. AFR3 Alternate function remapping option 3 0: AFR3 remapping option inactive: Default alternate function.⁽²⁾ 1: Port D0 alternate function = TIM1_BKIN. AFR2 Alternate function remapping option 2 0: AFR2 remapping option inactive: Default alternate functions.⁽²⁾ 1: Port D0 alternate function = CLK_CCO. Note: AFR2 option has priority over AFR3 if both are activated. AFR1 Alternate function remapping option 1 0: AFR1 remapping option inactive: Default alternate functions.⁽²⁾ 1: Port A3 alternate function = TIM3_CH1; port D2 alternate function TIM2_CH3 AFR0 Alternate function remapping option 0 0: AFR0 remapping option inactive: Default alternate functions.⁽²⁾ 1: Port D3 alternate function = ADC_ETR.

1. Do not use more than one remapping option in the same port.

2. Refer to STM8S105x4/6 pin descriptions.

9 Unique ID

The devices feature a 96-bit unique device identifier which provides a reference number that is unique for any device and in any context. The 96 bits of the identifier can never be altered by the user.

The unique device identifier can be read in single byte and may then be concatenated using a custom algorithm.

The unique device identifier is ideally suited:

- For use as serial numbers
- For use as security keys to increase the code security in the program memory while using and combining this unique ID with software cryptographic primitives and protocols before programming the internal memory.
- To activate secure boot processes

Table 14. Unique ID registers (96 bits)

Address	Content description	Unique ID bits							
		7	6	5	4	3	2	1	0
0x48CD	X co-ordinate on the wafer	U_ID[7:0]							
0x48CE		U_ID[15:8]							
0x48CF	Y co-ordinate on the wafer	U_ID[23:16]							
0x48D0		U_ID[31:24]							
0x48D1	Wafer number	U_ID[39:32]							
0x48D2	Lot number	U_ID[47:40]							
0x48D3		U_ID[55:48]							
0x48D4		U_ID[63:56]							
0x48D5		U_ID[71:64]							
0x48D6		U_ID[79:72]							
0x48D7		U_ID[87:80]							
0x48D8		U_ID[95:88]							

10 Electrical characteristics

10.1 Parameter conditions

Unless otherwise specified, all voltages are referred to V_{SS} .

10.1.1 Minimum and maximum values

Unless otherwise specified the minimum and maximum values are guaranteed in the worst conditions of ambient temperature, supply voltage and frequencies by tests in production on 100% of the devices with an ambient temperature at $T_A = 25\text{ }^{\circ}\text{C}$, and $T_A = T_{Amax}$ (given by the selected temperature range).

Data based on characterization results, design simulation and/or technology characteristics are indicated in the table footnotes and are not tested in production. Based on characterization, the minimum and maximum values refer to sample tests and represent the mean value plus or minus three times the standard deviation (mean $\pm 3\sigma$).

10.1.2 Typical values

Unless otherwise specified, typical data are based on $T_A = 25\text{ }^{\circ}\text{C}$, $V_{DD} = 5.0\text{ V}$. They are given only as design guidelines and are not tested.

Typical ADC accuracy values are determined by characterization of a batch of samples from a standard diffusion lot over the full temperature range, where 95% of the devices have an error less than or equal to the value indicated (mean $\pm 2\sigma$).

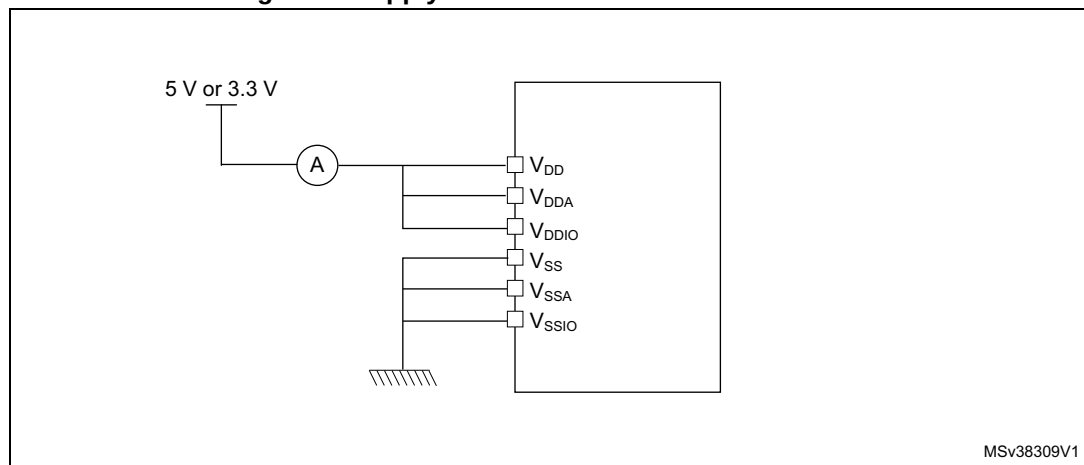
10.1.3 Typical curves

Unless otherwise specified, all typical curves are given only as design guidelines and are not tested.

10.1.4 Typical current consumption

For typical current consumption measurements, V_{DD} , V_{DDIO} and V_{DDA} are connected together in the configuration shown in the following figure.

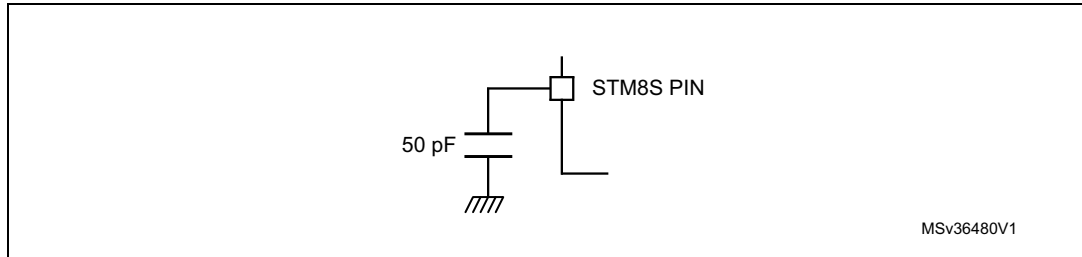
Figure 8. Supply current measurement conditions



10.1.5 Loading capacitor

The loading conditions used for pin parameter measurement are shown in [Figure 9](#).

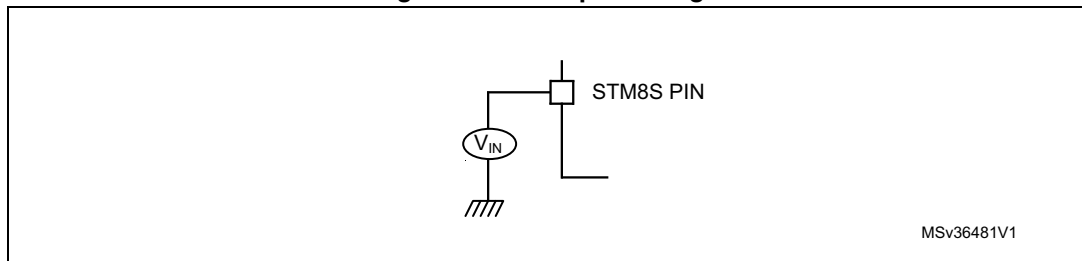
Figure 9. Pin loading conditions



10.1.6 Pin input voltage

The input voltage measurement on a pin of the device is described in [Figure 10](#).

Figure 10. Pin input voltage



10.2 Absolute maximum ratings

Stresses above those listed as 'absolute maximum ratings' may cause permanent damage to the device. This is a stress rating only and functional operation of the device under these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 15. Voltage characteristics

Symbol	Ratings	Min	Max	Unit
V _{DDx} - V _{SS}	Supply voltage (including V _{DDA} and V _{DDIO}) ⁽¹⁾	-0.3	6.5	V
V _{IN}	Input voltage on true open drain pins (PE1, PE2) ⁽²⁾	V _{SS} - 0.3	6.5	V
	Input voltage on any other pin ⁽²⁾	V _{SS} - 0.3	V _{DD} + 0.3	
V _{DDx} - V _{DD}	Variations between different power pins	-	50	mV
V _{SSx} - V _{SS}	Variations between all the different ground pins	-	50	
V _{ESD}	Electrostatic discharge voltage	see <i>Absolute maximum ratings (electrical sensitivity) on page 89</i>		

1. All power (V_{DD}) and ground (V_{SS}) pins must always be connected to the external power supply
2. $I_{INJ(PIN)}$ must never be exceeded. This is implicitly insured if V_{IN} maximum is respected. If V_{IN} maximum cannot be respected, the injection current must be limited externally to the $I_{INJ(PIN)}$ value. A positive injection is induced by $V_{IN} > V_{DD}$ while a negative injection is induced by $V_{IN} < V_{SS}$. For true open-drain pads, there is no positive injection current, and the corresponding V_{IN} maximum must always be respected

Table 16. Current characteristics

Symbol	Ratings	Max. ⁽¹⁾	Unit
I_{VDD}	Total current into V_{DD} power lines (source) ⁽²⁾	100	mA
I_{VSS}	Total current out of V_{SS} ground lines (sink) ⁽¹⁾	80	
I_{IO}	Output current sunk by any I/O and control pin	20	
	Output current source by any I/Os and control pin	-20	
ΣI_{IO}	Total output current sourced (sum of all I/O and control pins) for devices with two V_{DDIO} pins ⁽³⁾	200	
	Total output current sourced (sum of all I/O and control pins) for devices with one V_{DDIO} pin ⁽³⁾	100	
	Total output current sunk (sum of all I/O and control pins) for devices with two V_{SSIO} pins ⁽³⁾	160	
	Total output current sunk (sum of all I/O and control pins) for devices with one V_{SSIO} pin ⁽³⁾	80	
$I_{INJ(PIN)}$ ^{(4) (5)}	Injected current on NRST pin	±4	
	Injected current on OSCIN pin	±4	
	Injected current on any other pin ⁽⁶⁾	±4	
$\Sigma I_{INJ(PIN)}$ ⁽⁴⁾	Total injected current (sum of all I/O and control pins) ⁽⁶⁾	±20	

1. Data based on characterization results, not tested in production.
2. All power (V_{DD} , V_{DDIO} , V_{DDA}) and ground (V_{SS} , V_{SSIO} , V_{SSA}) pins must always be connected to the external supply.

3. I/O pins used simultaneously for high current source/sink must be uniformly spaced around the package between the VDDIO/VSSIO pins.
4. $I_{INJ(PIN)}$ must never be exceeded. This condition is implicitly insured if V_{IN} maximum is respected. If V_{IN} maximum cannot be respected, the injection current must be limited externally to the $I_{INJ(PIN)}$ value. A positive injection is induced by $V_{IN} > V_{DD}$ while a negative injection is induced by $V_{IN} < V_{SS}$. For true open-drain pads, there is no positive injection current allowed and the corresponding V_{IN} maximum must always be respected.
5. Negative injection disturbs the analog performance of the device. See note in *Section: TIM2, TIM3 - 16-bit general purpose timers*.
6. When several inputs are submitted to a current injection, the maximum $\Sigma I_{INJ(PIN)}$ is the absolute sum of the positive and negative injected currents (instantaneous values). These results are based on characterization with $\Sigma I_{INJ(PIN)}$ maximum current injection on four I/O port pins of the device.

Table 17. Thermal characteristics

Symbol	Ratings	Value	Unit
T_{STG}	Storage temperature range	-65 to 150	°C
T_J	Maximum junction temperature	150	

10.3 Operating conditions

The device must be used in operating conditions that respect the parameters described in the table below. In addition, full account must be taken of all physical capacitor characteristics and tolerances.

Table 18. General operating conditions

Symbol	Parameter	Conditions	Min	Max	Unit
f_{CPU}	Internal CPU clock frequency	-	0	16	MHz
V_{DD}/V_{DDIO}	Standard operating voltage	-	2.95	5.5	V
$V_{CAP}^{(1)}$	C_{EXT} : capacitance of external capacitor	-	470	3300	nF
	ESR of external capacitor	at 1 MHz ⁽²⁾	-	0.3	Ω
	ESL of external capacitor		-	15	nH
$P_D^{(3)}$	Power dissipation at $T_A = 85^\circ\text{C}$ for suffix 6 or $T_A = 125^\circ\text{C}$ for suffix 3	44- and 48-pin devices, with output on eight standard ports, two high sink ports and two open drain ports simultaneously ⁽⁴⁾	-	443	mW
		32-pin package, with output on eight standard ports and two high sink ports simultaneously ⁽⁴⁾	-	360	

Table 18. General operating conditions (continued)

Symbol	Parameter	Conditions	Min	Max	Unit
T_A	Ambient temperature for suffix 6 version	Maximum power dissipation	-40	85	°C
T_A	Ambient temperature for suffix 3 version	Maximum power dissipation	-40	125	
T_J	Junction temperature range	Suffix 6 version	-40	105	
		Suffix 3 version	-40	130	

1. Care should be taken when selecting the capacitor, due to its tolerance, as well as the parameter dependency on temperature, DC bias and frequency in addition to other factors. The parameter maximum value must be respected for the full application range.
2. This frequency of 1 MHz as a condition for V_{CAP} parameters is given by design of internal regulator.
3. To calculate $P_{Dmax}(T_A)$, use the formula $P_{Dmax} = (T_{Jmax} - T_A) / \Theta_{JA}$ (see [Section 12: Thermal characteristics](#)) with the value for T_{Jmax} given in the previous table and the value for Θ_{JA} given in [Section 12: Thermal characteristics](#).
4. See [Section 12: Thermal characteristics](#).

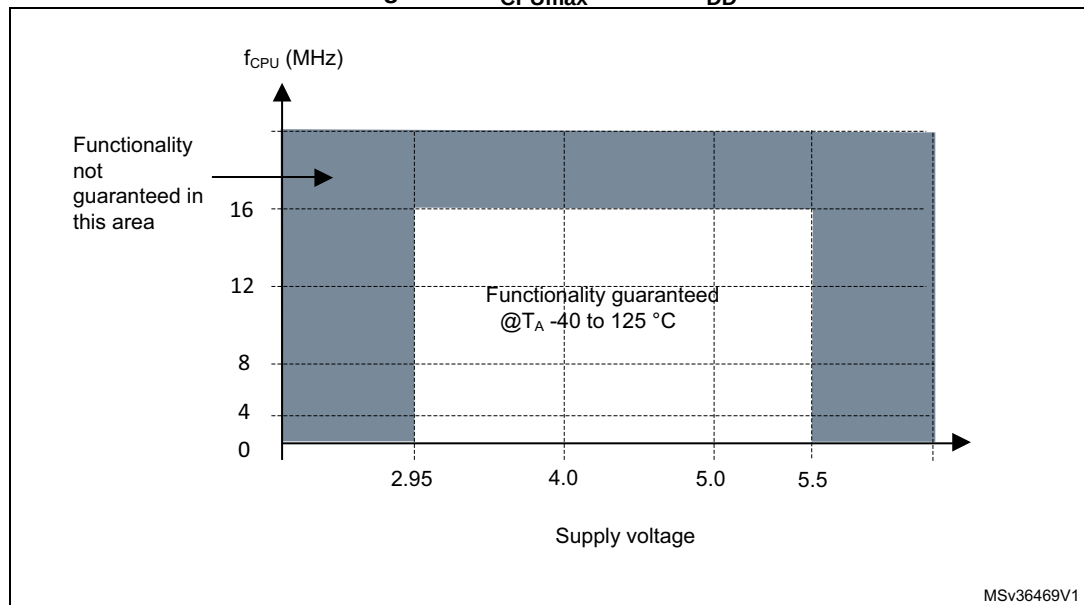
Figure 11. f_{CPUmax} versus V_{DD} 

Table 19. Operating conditions at power-up/power-down

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
t_{VDD}	V_{DD} rise time rate	-	2 ⁽¹⁾	-	∞	$\mu s/V$
	V_{DD} fall time rate	-	2 ⁽¹⁾	-	∞	
t_{TEMP}	Reset release delay	V_{DD} rising	-	-	1.7 ⁽¹⁾	ms

Table 19. Operating conditions at power-up/power-down (continued)

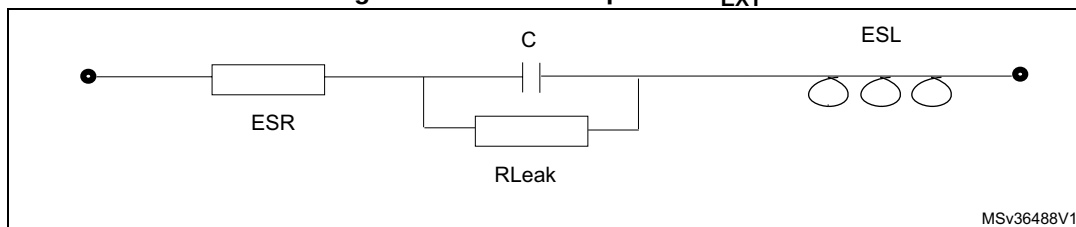
Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{IT+}	Power-on reset threshold	-	2.65	2.8	2.95	V
V_{IT-}	Brown-out reset threshold	-	2.58	2.65	2.88	
$V_{HYS(BOR)}$	Brown-out reset hysteresis	-	-	70	-	mV

1. Guaranteed by design, not tested in production.

10.3.1 VCAP external capacitor

The stabilization for the main regulator is achieved by connecting an external capacitor C_{EXT} to the V_{CAP} pin. C_{EXT} is specified in [Table 18](#). Care should be taken to limit the series inductance to less than 15 nH.

Figure 12. External capacitor C_{EXT}



1. ESR is the equivalent series resistance and ESL is the equivalent inductance.

10.3.2 Supply current characteristics

The current consumption is measured as illustrated in [Figure 10: Pin input voltage](#).

Total current consumption in run mode

Table 20. Total current consumption with code execution in run mode at $V_{DD} = 5\text{ V}$

Symbol	Parameter	Conditions		Typ	Max ⁽¹⁾	Unit
$I_{DD(RUN)}$	Supply current in Run mode, code executed from RAM	$f_{CPU} = f_{MASTER} = 16\text{ MHz}$	HSE crystal osc. (16 MHz)	3.2	-	mA
			HSE user ext. clock (16 MHz)	2.6	3.2	
			HSI RC osc. (16 MHz)	2.5	3.2	
		$f_{CPU} = f_{MASTER} / 128 = 125\text{ kHz}$	HSE user ext. clock (16 MHz)	1.6	2.2	
			HSI RC osc. (16 MHz)	1.3	2.0	
		$f_{CPU} = f_{MASTER} / 128 = 15.625\text{ kHz}$	HSI RC osc. (16 MHz/8)	0.75	-	
		$f_{CPU} = f_{MASTER} = 128\text{ kHz}$	LSI RC osc. (128 kHz)	0.55	-	
$I_{DD(RUN)}$	Supply current in Run mode, code executed from Flash	$f_{CPU} = f_{MASTER} = 16\text{ MHz}$	HSE crystal osc. (16 MHz)	7.7	-	mA
			HSE user ext. clock (16 MHz)	7.0	8.0	
			HSI RC osc. (16 MHz)	7.0	8.0	
		$f_{CPU} = f_{MASTER} = 2\text{ MHz}$	HSI RC osc. (16 MHz/8) ⁽²⁾	1.5	-	
		$f_{CPU} = f_{MASTER} / 128 = 125\text{ kHz}$	HSI RC osc. (16 MHz)	1.35	2.0	
		$f_{CPU} = f_{MASTER} / 128 = 15.625\text{ kHz}$	HSI RC osc. (16 MHz/8)	0.75	-	
		$f_{CPU} = f_{MASTER} = 128\text{ kHz}$	LSI RC osc. (128 kHz)	0.6	-	

1. Data based on characterization results, not tested in production.
 2. Default clock configuration measured with all peripherals off.

Table 21. Total current consumption with code execution in run mode at $V_{DD} = 3.3\text{ V}$

Symbol	Parameter	Conditions		Typ	Max ⁽¹⁾	Unit
$I_{DD(RUN)}$	Supply current in Run mode, code executed from RAM	$f_{CPU} = f_{MASTER} = 16\text{ MHz}$	HSE crystal osc. (16 MHz)	2.8	-	mA
			HSE user ext. clock (16 MHz)	2.6	3.2	
			HSI RC osc. (16 MHz)	2.5	3.2	
		$f_{CPU} = f_{MASTER} / 128 = 125\text{ kHz}$	HSE user ext. clock (16 MHz)	1.6	2.2	
			HSI RC osc. (16 MHz)	1.3	2.0	
		$f_{CPU} = f_{MASTER} / 128 = 15.625\text{ kHz}$	HSI RC osc. (16 MHz/8)	0.75	-	
		$f_{CPU} = f_{MASTER} = 128\text{ kHz}$	LSI RC osc. (128 kHz)	0.55	-	
$I_{DD(RUN)}$	Supply current in Run mode, code executed from Flash	$f_{CPU} = f_{MASTER} = 16\text{ MHz}$	HSE crystal osc. (16 MHz)	7.3	-	mA
			HSE user ext. clock (16 MHz)	7.0	8.0	
			HSI RC osc. (16 MHz)	7.0	8.0	
		$f_{CPU} = f_{MASTER} = 2\text{ MHz}$	HSI RC osc. (16 MHz/8) ⁽²⁾	1.5	-	
		$f_{CPU} = f_{MASTER} / 128 = 125\text{ kHz}$	HSI RC osc. (16 MHz)	1.35	2.0	
		$f_{CPU} = f_{MASTER} / 128 = 15.625\text{ kHz}$	HSI RC osc. (16 MHz/8)	0.75	-	
		$f_{CPU} = f_{MASTER} = 128\text{ kHz}$	LSI RC osc. (128 kHz)	0.6	-	

1. Data based on characterization results, not tested in production.

2. Default clock configuration measured with all peripherals off.

Total current consumption in wait mode

Table 22. Total current consumption in wait mode at $V_{DD} = 5\text{ V}$

Symbol	Parameter	Conditions		Typ	Max ⁽¹⁾	Unit
$I_{DD(WFI)}$	Supply current in wait mode	$f_{CPU} = f_{MASTER} = 16\text{ MHz}$	HSE crystal osc. (16 MHz)	2.15	-	mA
			HSE user ext. clock (16 MHz)	1.55	2.0	
			HSI RC osc. (16 MHz)	1.5	1.9	
		$f_{CPU} = f_{MASTER} / 128 = 125\text{ kHz}$	HSI RC osc. (16 MHz)	1.3	-	
		$f_{CPU} = f_{MASTER} / 128 = 15.625\text{ kHz}$	HSI RC osc. (16 MHz/8) ⁽²⁾	0.7	-	
		$f_{CPU} = f_{MASTER} = 128\text{ kHz}$	LSI RC osc. (128 kHz)	0.5	-	

1. Data based on characterization results, not tested in production.

2. Default clock configuration measured with all peripherals off.

Table 23. Total current consumption in wait mode at $V_{DD} = 3.3\text{ V}$

Symbol	Parameter	Conditions		Typ	Max ⁽¹⁾	Unit
$I_{DD(WFI)}$	Supply current in wait mode	$f_{CPU} = f_{MASTER} = 16\text{ MHz}$	HSE crystal osc. (16 MHz)	1.75	-	mA
			HSE user ext. clock (16 MHz)	1.55	2.0	
			HSI RC osc. (16 MHz)	1.5	1.9	
		$f_{CPU} = f_{MASTER} / 128 = 125\text{ kHz}$	HSI RC osc. (16 MHz)	1.3	-	
		$f_{CPU} = f_{MASTER} / 128 = 15.625\text{ kHz}$	HSI RC osc. (16 MHz/8) ⁽²⁾	0.7	-	
		$f_{CPU} = f_{MASTER} = 128\text{ kHz}$	LSI RC osc. (128 kHz)	0.5	-	

1. Data based on characterization results, not tested in production.

2. Default clock configuration measured with all peripherals off.

Total current consumption in active halt mode

Table 24. Total current consumption in active halt mode at $V_{DD} = 5\text{ V}$

Symbol	Parameter	Conditions			Typ	Max at 85 °C ⁽¹⁾	Max at 85 °C ⁽¹⁾	Unit
		Main voltage regulator (MVR) ⁽²⁾	Flash mode ⁽³⁾	Clock source				
$I_{DD(AH)}$	Supply current in active halt mode	On	Operating mode	HSE crystal osc. (16 MHz)	1080	-	-	μA
			Operating mode	LSI RC osc. (128 kHz)	200	320	400	
			Power down mode	HSE crystal osc. (16 MHz)	1030	-	-	
			Power down mode	LSI RC osc. (128 kHz)	140	270	350	
		Off	Operating mode	LSI RC osc. (128 kHz)	68	120	220	
			Power down mode	LSI RC osc. (128 kHz)	12	60	150	

1. Data based on characterization results, not tested in production.

2. Configured by the REGAH bit in the CLK_ICR register.

3. Configured by the AHALT bit in the FLASH_CR1 register.

Table 25. Total current consumption in active halt mode at $V_{DD} = 3.3\text{ V}$

Symbol	Parameter	Conditions			Typ	Max at 85 °C ⁽¹⁾	Max at 85 °C ⁽¹⁾	Unit
		Main voltage regulator (MVR) ⁽²⁾	Flash mode ⁽³⁾	Clock source				
$I_{DD(AH)}$	Supply current in active halt mode	On	Operating mode	HSE crystal osc. (16 MHz)	680	-	-	μA
			Operating mode	LSI RC osc. (128 kHz)	200	320	400	
			Power down mode	HSE crystal osc. (16 MHz)	630	-	-	
			Power down mode	LSI RC osc. (128 kHz)	140	270	350	
		Off	Operating mode	LSI RC osc. (128 kHz)	66	120	220	
			Power down mode	LSI RC osc. (128 kHz)	10	60	150	

1. Data based on characterization results, not tested in production.

2. Configured by the REGAH bit in the CLK_ICR register.

3. Configured by the AHALT bit in the FLASH_CR1 register.

Total current consumption in halt mode

Table 26. Total current consumption in halt mode at $V_{DD} = 5\text{ V}$

Symbol	Parameter	Conditions	Typ	Max at 85 °C ⁽¹⁾	Max at 85 °C ⁽¹⁾	Unit
$I_{DD(H)}$	Supply current in halt mode	Flash in operating mode, HSI clock after wakeup	62	90	150	μA
		Flash in power-down mode, HSI clock after wakeup	6.5	25	80	

1. Data based on characterization results, not tested in production.

Table 27. Total current consumption in halt mode at $V_{DD} = 3.3\text{ V}$

Symbol	Parameter	Conditions	Typ	Max at 85 °C ⁽¹⁾	Max at 85 °C ⁽¹⁾	Unit
$I_{DD(H)}$	Supply current in halt mode	Flash in operating mode, HSI clock after wakeup	60	90	150	μA
		Flash in power-down mode, HSI clock after wakeup	4.5	20	80	

1. Data based on characterization results, not tested in production.

Low power mode wakeup times

Table 28. Wakeup times

Symbol	Parameter	Conditions			Typ	Max ⁽¹⁾	Unit
$t_{WU(WFI)}$	Wakeup time from wait mode to run mode ⁽²⁾	0 to 16 MHz			-	See note ⁽³⁾	μs
$t_{WU(WFI)}$	Wakeup time from run mode ⁽²⁾	$f_{CPU} = f_{MASTER} = 16 \text{ MHz}$			0.56	-	
$t_{WU(AH)}$	Wakeup time active halt mode to run mode ⁽²⁾	MVR voltage regulator on ⁽⁴⁾	Flash in operating mode ⁽⁵⁾	HSI (after wakeup)	1 ⁽⁶⁾	2 ⁽⁶⁾	
$t_{WU(AH)}$	Wakeup time active halt mode to run mode ⁽²⁾	MVR voltage regulator off ⁽⁴⁾	Flash in operating mode ⁽⁵⁾	HSI (after wakeup)	3 ⁽⁶⁾	-	
$t_{WU(AH)}$	Wakeup time active halt mode to run mode ⁽²⁾	MVR voltage regulator off ⁽⁴⁾	Flash in operating mode ⁽⁵⁾	HSI (after wakeup)	48 ⁽⁶⁾	-	
$t_{WU(AH)}$	Wakeup time active halt mode to run mode ⁽²⁾	MVR voltage regulator off ⁽⁴⁾	Flash in power-down mode ⁽⁵⁾	HSI (after wakeup)	50 ⁽⁶⁾	-	
$t_{WU(H)}$	Wakeup time from halt mode to run mode ⁽²⁾	Flash in operating mode ⁽⁵⁾			52	-	
$t_{WU(H)}$	Wakeup time from halt mode to run mode ⁽²⁾	Flash in power-down mode ⁽⁵⁾			54	-	

1. Data based on characterization results, not tested in production.

2. Measured from interrupt event to interrupt vector fetch

3. $t_{WU(WFI)} = 2 \times 1/f_{master} + 67 \times 1/f_{CPU}$

4. Configured by the REGAH bit in the CLK_ICR register.

5. Configured by the AHALT bit in the FLASH_CR1 register.

6. Plus 1 LSI clock depending on synchronization.

Total current consumption and timing in forced reset state

Table 29. Total current consumption and timing in forced reset state

Symbol	Parameter	Conditions	Typ	Max ⁽¹⁾	Unit
$I_{DD(R)}$	Supply current in reset state ⁽²⁾	$V_{DD} = 5 \text{ V}$	500	-	μA
		$V_{DD} = 3.3 \text{ V}$	400	-	
$t_{RESETBL}$	Reset pin release to vector fetch	-	-	150	μs

1. Data guaranteed by design, not tested in production.

2. Characterized with all I/Os tied to V_{SS} .

Current consumption of on-chip peripherals

Subject to general operating conditions for V_{DD} and T_A .

HSI internal RC/ $f_{\text{CPU}} = f_{\text{MASTER}} = 16 \text{ MHz}$, $V_{\text{DD}} = 5 \text{ V}$

Table 30. Peripheral current consumption

Symbol	Parameter	Typ	Unit
$I_{\text{DD(TIM1)}}$	TIM1 supply current ⁽¹⁾	230	μA
$I_{\text{DD(TIM2)}}$	TIM2 supply current ⁽¹⁾	115	
$I_{\text{DD(TIM3)}}$	TIM3 supply current ⁽¹⁾	90	
$I_{\text{DD(TIM4)}}$	TIM4 supply current ⁽¹⁾	30	
$I_{\text{DD(UART2)}}$	UART2 supply current ⁽²⁾	110	
$I_{\text{DD(SPI)}}$	SPI supply current ⁽²⁾	45	
$I_{\text{DD(I2C)}}$	I2C supply current ⁽²⁾	65	
$I_{\text{DD(ADC1)}}$	ADC1 supply current when converting ⁽³⁾	955	

1. Data based on a differential I_{DD} measurement between reset configuration and timer counter running at 16 MHz. No IC/OC programmed (no I/O pads toggling). Not tested in production.
2. Data based on a differential I_{DD} measurement between the on-chip peripheral when kept under reset and not clocked and the on-chip peripheral when clocked and not kept under reset. No I/O pads toggling. Not tested in production.
3. Data based on a differential I_{DD} measurement between reset configuration and continuous A/D conversions. Not tested in production.

Current consumption curves

The following figures show typical current consumption measured with code executing in RAM.

Figure 13. Typ $I_{\text{DD(RUN)}}$ vs. V_{DD} HSE user external clock, $f_{\text{CPU}} = 16 \text{ MHz}$

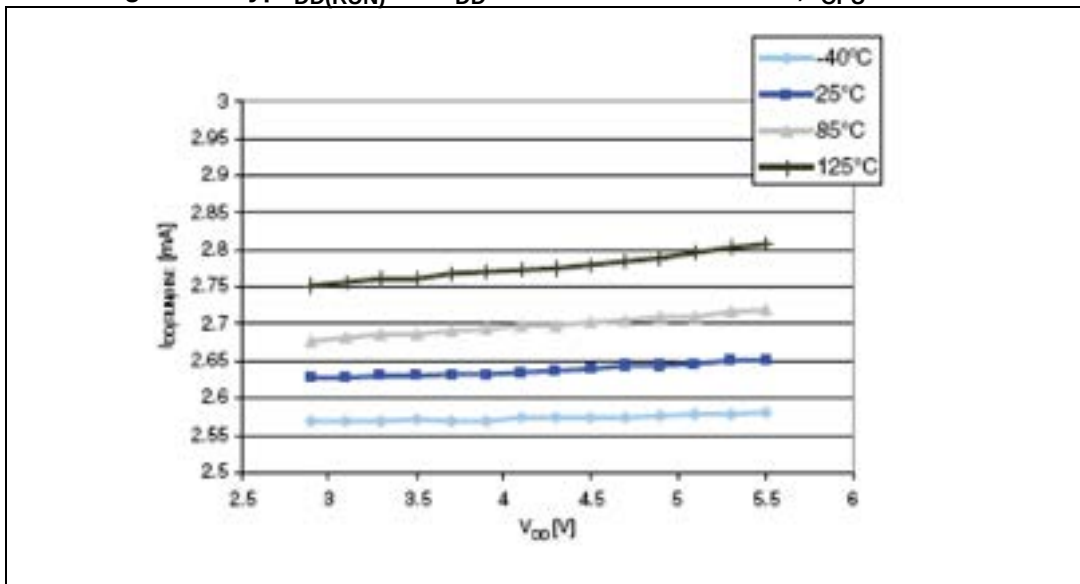


Figure 14. Typ $I_{DD(RUN)}$ vs. f_{CPU} HSE user external clock, $V_{DD} = 5\text{ V}$

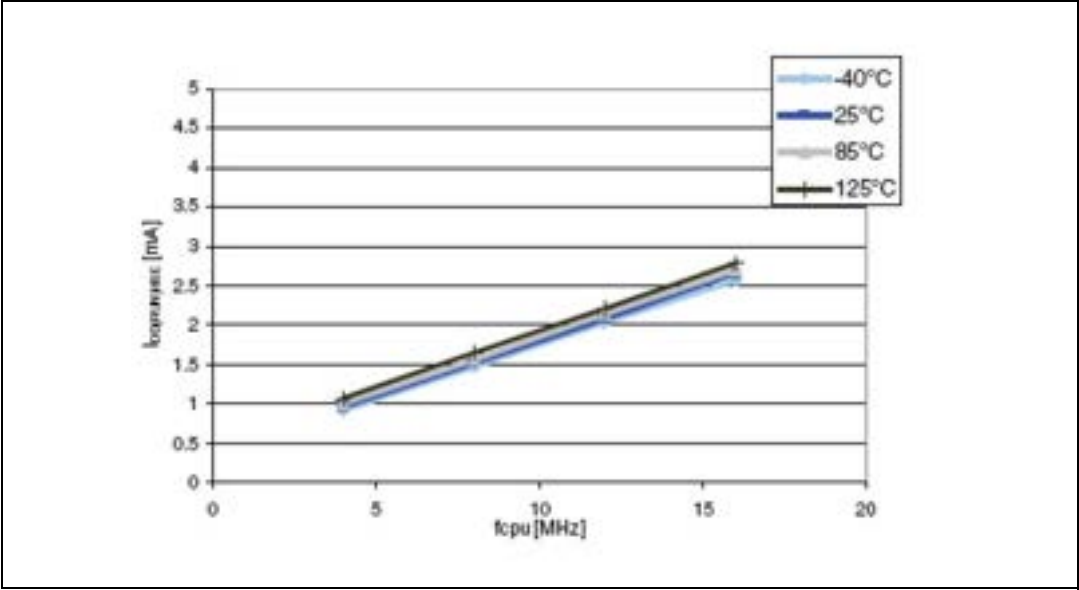


Figure 15. Typ $I_{DD(RUN)}$ vs. V_{DD} HSI RC osc, $f_{CPU} = 16\text{ MHz}$

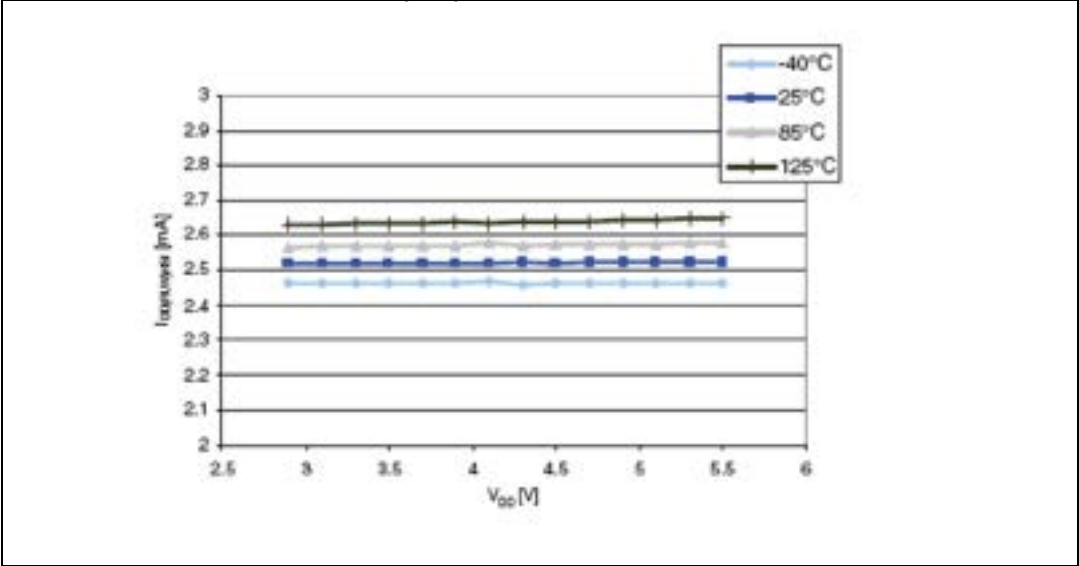


Figure 16. Typ $I_{DD(WFI)}$ vs. V_{DD} HSE external clock, $f_{CPU} = 16\text{ MHz}$

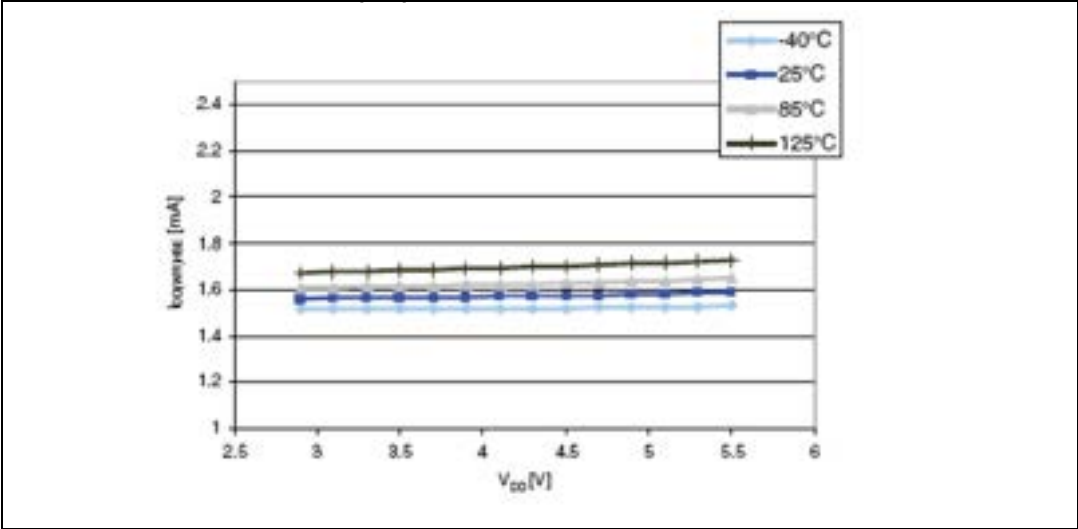


Figure 17. Typ $I_{DD(WFI)}$ vs. f_{CPU} HSE external clock, $V_{DD} = 5\text{ V}$

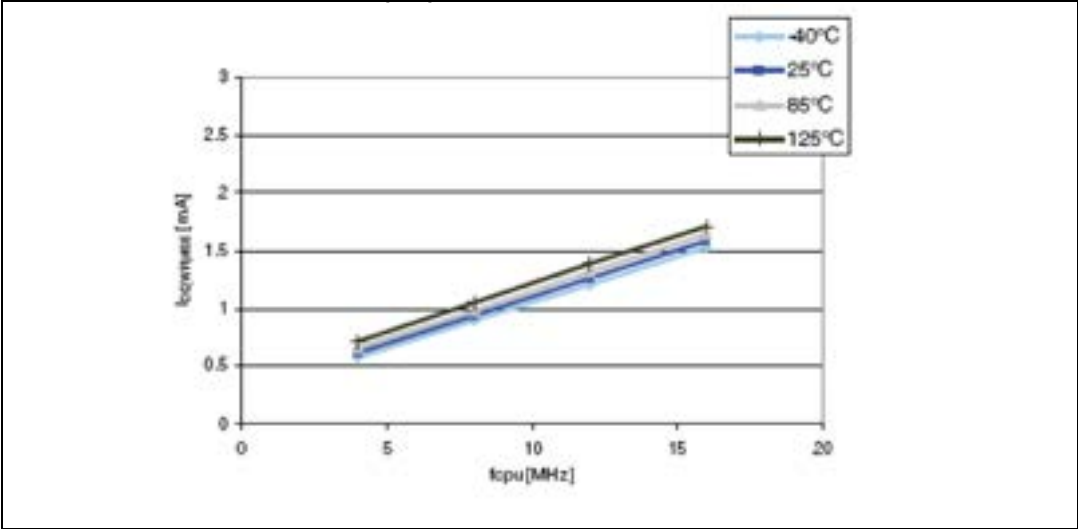
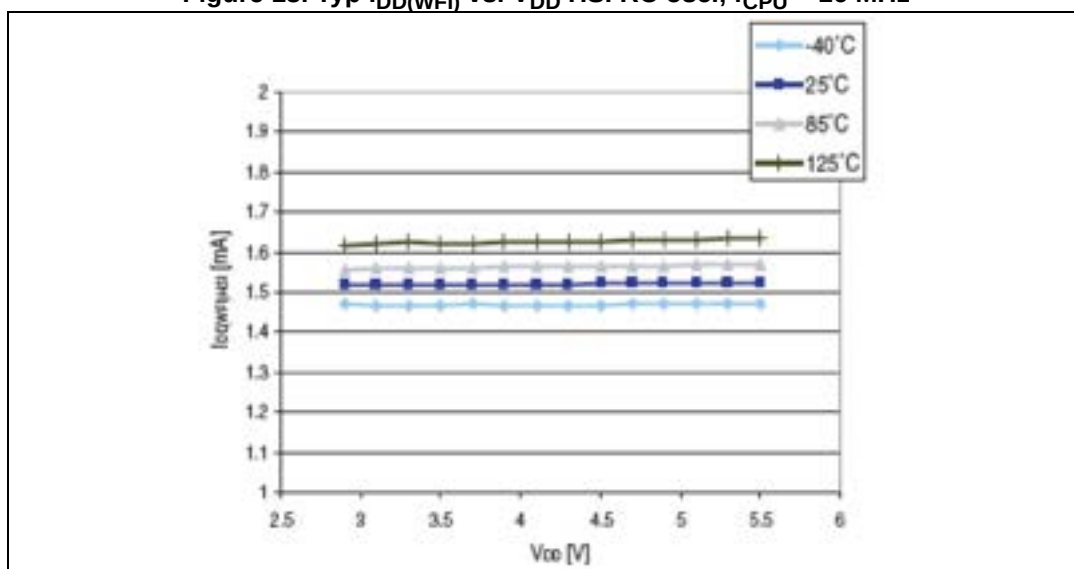


Figure 18. Typ $I_{DD(WFI)}$ vs. V_{DD} HSI RC osc., $f_{CPU} = 16$ MHz

10.3.3 External clock sources and timing characteristics

HSE user external clock

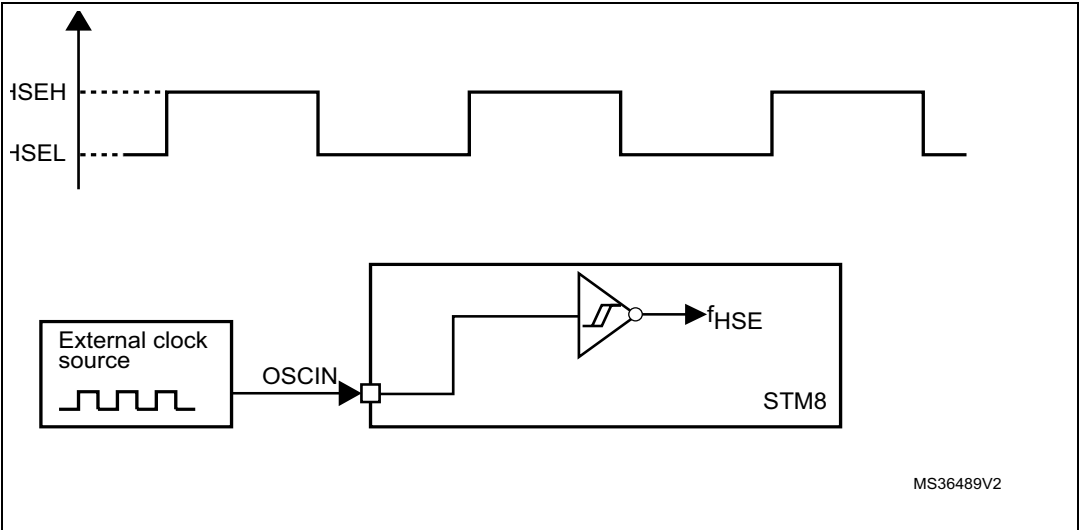
Subject to general operating conditions for V_{DD} and T_A .

Table 31. HSE user external clock characteristics

Symbol	Parameter	Conditions	Min	Max	Unit
f_{HSE_ext}	User external clock source frequency	-	0	16	MHz
$V_{HSEH}^{(1)}$	OSCIN input pin high level voltage	-	$0.7 \times V_{DD}$	$V_{DD} + 0.3\text{ V}$	V
$V_{HSEL}^{(1)}$	OSCIN input pin low level voltage	-	V_{SS}	$0.3 \times V_{DD}$	
I_{LEAK_HSE}	OSCIN input leakage current	$V_{SS} < V_{IN} < V_{DD}$	-1	+1	μA

1. Data based on characterization results, not tested in production.

Figure 19. HSE external clock source



HSE crystal/ceramic resonator oscillator

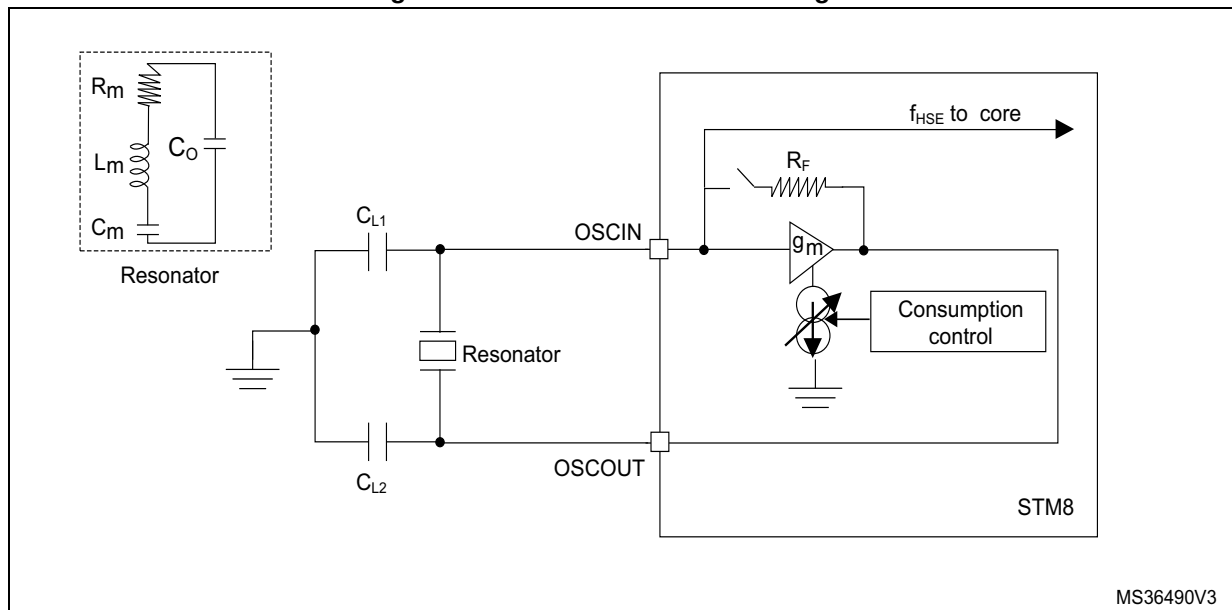
The HSE clock can be supplied with a 1 to 16 MHz crystal/ceramic resonator oscillator. All the information given in this paragraph is based on characterization results with specified typical external components. In the application, the resonator and the load capacitors have to be placed as close as possible to the oscillator pins in order to minimize output distortion and startup stabilization time. Refer to the crystal resonator manufacturer for more details (frequency, package, accuracy...).

Table 32. HSE oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSE}	External high speed oscillator frequency	-	1	-	16	MHz
R_F	Feedback resistor	-	-	220	-	k Ω
$C^{(1)}$	Recommended load capacitance ⁽²⁾	-	-	-	20	pF
$I_{DD(HSE)}$	HSE oscillator power consumption	C = 20 pF $f_{OSC} = 16$ MHz	-	-	6 (start up) 1.6 (stabilized) ⁽³⁾	mA
		C = 10 pF $f_{OSC} = 16$ MHz	-	-	6 (start up) 1.2 (stabilized) ⁽³⁾	
g_m	Oscillator transconductance	-	5	-	-	mA/V
$t_{SU(HSE)}^{(4)}$	Startup time	V_{DD} is stabilized	-	1	-	ms

1. C is approximately equivalent to 2 x crystal Cload.
2. The oscillator selection can be optimized in terms of supply current using a high quality resonator with small Rm value. Refer to crystal manufacturer for more details
3. Data based on characterization results, not tested in production.
4. $t_{SU(HSE)}$ is the start-up time measured from the moment it is enabled (by software) to a stabilized 16 MHz oscillation is reached. This value is measured for a standard crystal resonator and it can vary significantly with the crystal manufacturer.

Figure 20. HSE oscillator circuit diagram



MS36490V3

HSE oscillator critical g_m equation

$$g_{m\text{crit}} = (2 \times \Pi \times f_{\text{HSE}})^2 \times R_m (2C_o + C)^2$$

R_m : Notional resistance (see crystal specification)

L_m : Notional inductance (see crystal specification)

C_m : Notional capacitance (see crystal specification)

C_o : Shunt capacitance (see crystal specification)

$C_{L1} = C_{L2} = C$: Grounded external capacitance

$g_m \gg g_{m\text{crit}}$

10.3.4 Internal clock sources and timing characteristics

Subject to general operating conditions for V_{DD} and T_A .

High speed internal RC oscillator (HSI)

Table 33. HSI oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{HSI}	Frequency	-	-	16	-	MHz
ACC_{HS}	Accuracy of HSI oscillator	User-trimmed with CLK_HSITRIMR register for given V_{DD} and T_A conditions ⁽¹⁾	-	-	1 ⁽²⁾	%
	HSI oscillator accuracy (factory calibrated)	$V_{DD} = 5\text{ V}$, $T_A = 25\text{ °C}$ ⁽³⁾	-1.0	-	1.0	
		$V_{DD} = 5\text{ V}$, $-25\text{ °C} \leq T_A \leq 85\text{ °C}$	-2.0	-	2.0	
		$2.95\text{ V} \leq V_{DD} \leq 5.5\text{ V}$, $-40\text{ °C} \leq T_A \leq 125\text{ °C}$	-3.0 ⁽³⁾	-	3.0 ⁽³⁾	
$t_{su(HSI)}$	HSI oscillator wakeup time including calibration	-	-	-	1.0 ⁽²⁾	μs
$I_{DD(HSI)}$	HSI oscillator power consumption	-	-	170	250 ⁽³⁾	μA

1. Refer to application note.

2. Guaranteed by design, not tested in production.

3. Data based on characterization results, not tested in production.

Figure 21. Typical HSI accuracy @ $V_{DD} = 5\text{ V}$ vs 5 temperatures

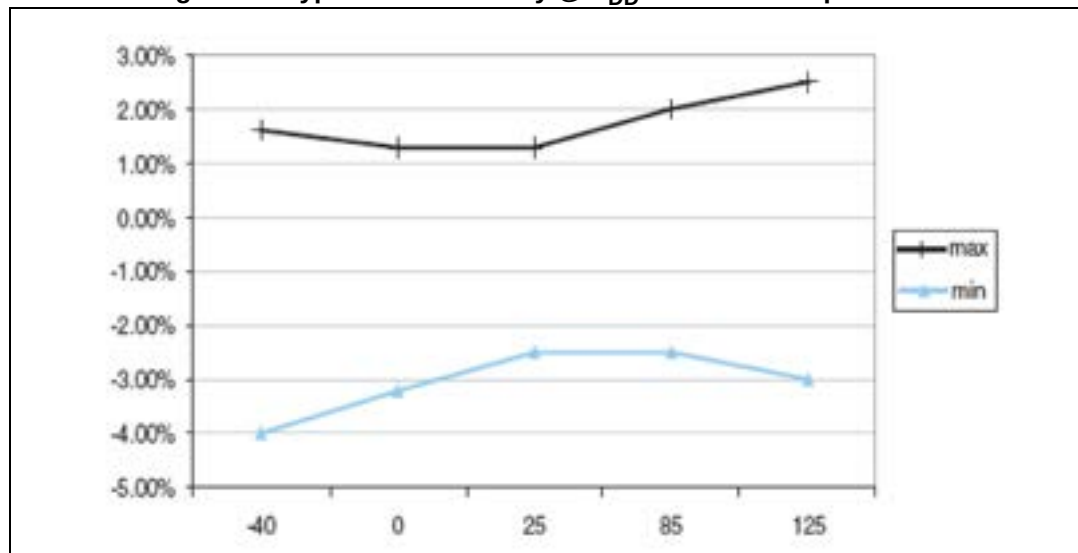
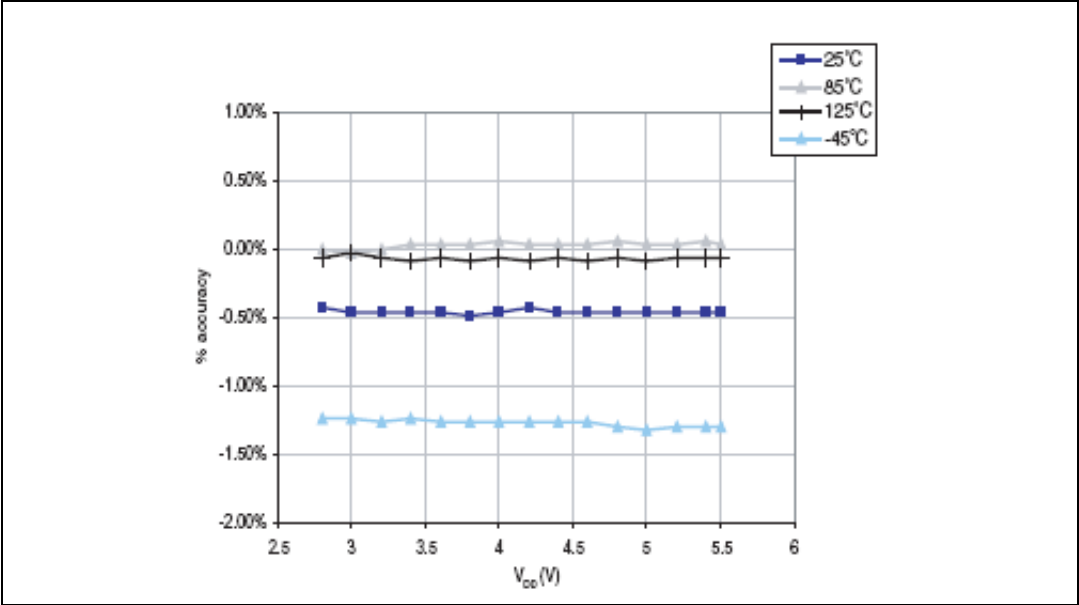


Figure 22. Typical HSI frequency variation vs V_{DD} @ 4 temperatures



Low speed internal RC oscillator (LSI)

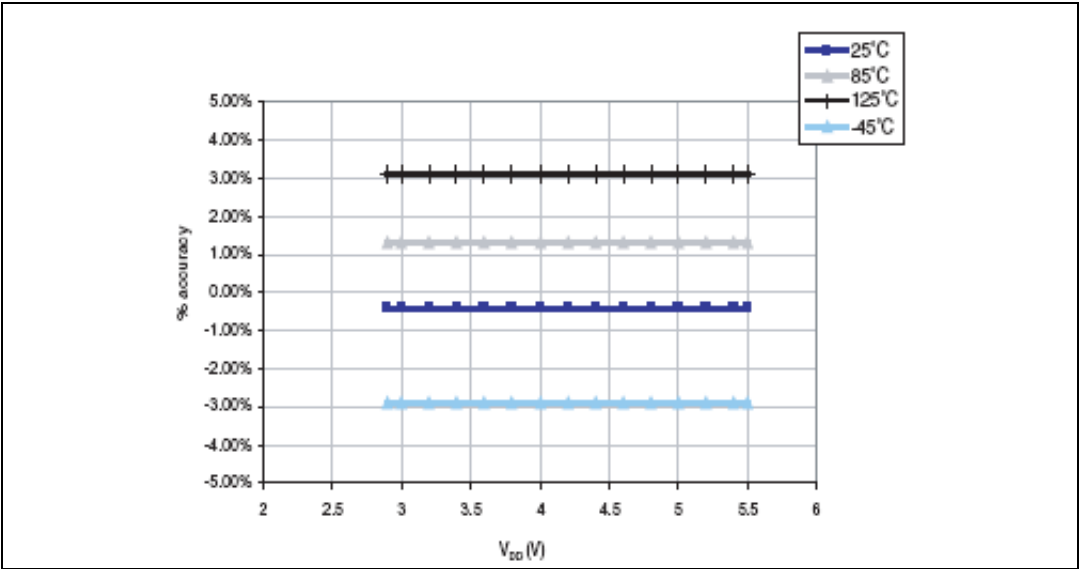
Subject to general operating conditions for V_{DD} and T_A .

Table 34. LSI oscillator characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{LSI}	Frequency	-	110	128	150	kHz
$t_{su(LSI)}$	LSI oscillator wakeup time	-	-	-	7 ⁽¹⁾	μs
$I_{DD(LSI)}$	LSI oscillator power consumption	-	-	5	-	μA

1. Guaranteed by design, not tested in production.

Figure 23. Typical LSI frequency variation vs V_{DD} @ 4 temperatures



10.3.5 Memory characteristics

RAM and hardware registers

Table 35. RAM and hardware registers

Symbol	Parameter	Conditions	Min	Unit
V_{RM}	Data retention mode ⁽¹⁾	Halt mode (or reset)	V_{IT-max} ⁽²⁾	V

1. Minimum supply voltage without losing data stored in RAM (in halt mode or under reset) or in hardware registers (only in halt mode). Guaranteed by design, not tested in production.

2. Refer to [Section 10.3: Operating conditions](#) for the value of V_{IT-max} .

Flash program memory/data EEPROM memory

Table 36. Flash program memory/data EEPROM memory

Symbol	Parameter	Conditions	Min ⁽¹⁾	Typ	Max	Unit
V_{DD}	Operating voltage (all modes, execution/write/erase)	$f_{CPU} \leq 16 \text{ MHz}$	2.95	-	5.5	V
t_{prog}	Standard programming time (including erase) for byte/word/block (1 byte/4 byte/128 byte)	-	-	6	6.6	ms
	Fast programming time for 1 block (128 byte)	-	-	3	3.33	
t_{erase}	Erase time for 1 block (128 byte)	-	-	3	3.33	
N_{RW}	Erase/write cycles (program memory) ⁽²⁾	$T_A = +85 \text{ }^\circ\text{C}$	10k	-	-	cycle
	Erase/write cycles (data memory) ⁽²⁾	$T_A = +125 \text{ }^\circ\text{C}$	300k	1M	-	
t_{RET}	Data retention (program and data memory) after 10k erase/write cycles at $T_A = +55 \text{ }^\circ\text{C}$	$T_{RET} = 55 \text{ }^\circ\text{C}$	20	-	-	year
	Data retention (data memory) after 300k erase/write cycles at $T_A = +125 \text{ }^\circ\text{C}$	$T_{RET} = 85 \text{ }^\circ\text{C}$	1	-	-	
I_{DD}	Supply current (Flash programming or erasing for 1 to 128 byte)	-	-	2	-	mA

1. Data based on characterization results, not tested in production.

2. The physical granularity of the memory is 4 byte, so cycling is performed on 4 byte even when a write/erase operation addresses a single byte.

10.3.6 I/O port pin characteristics

General characteristics

Subject to general operating conditions for V_{DD} and T_A unless otherwise specified. All unused pins must be kept at a fixed voltage, using the output mode of the I/O for example or an external pull-up or pull-down resistor.

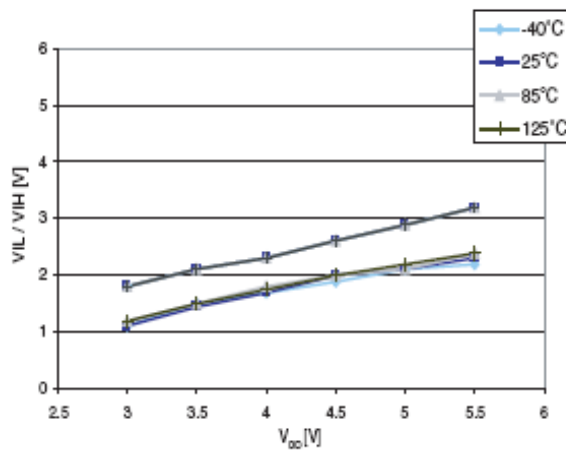
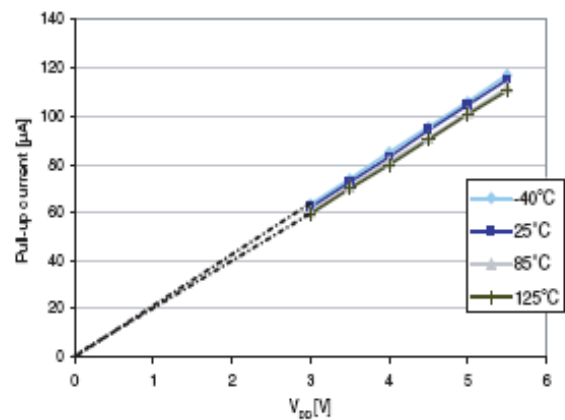
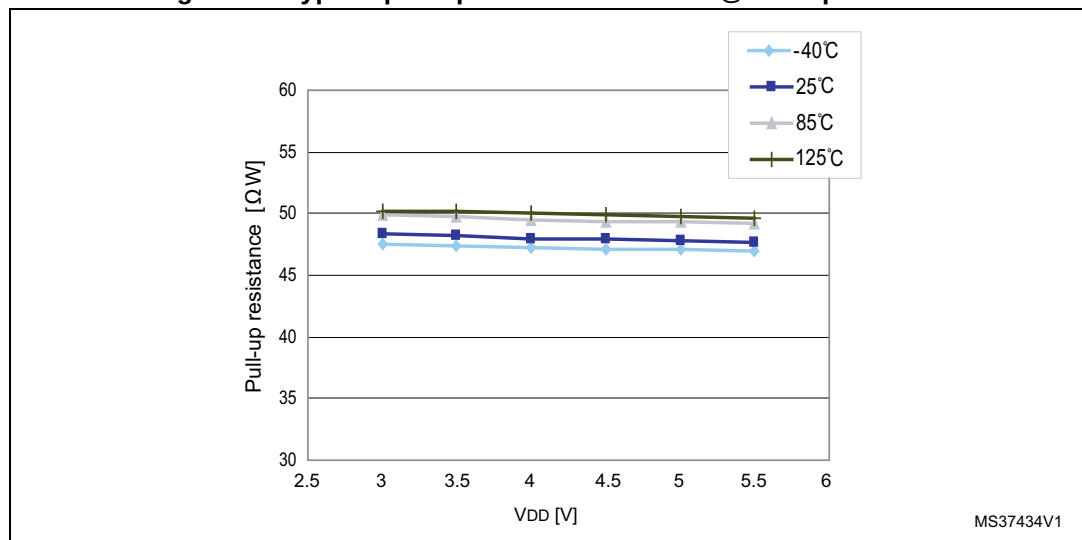
Table 37. I/O static characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{IL}	Input low level voltage	$V_{DD} = 5\text{ V}$	-0.3 V	-	$0.3 \times V_{DD}$	V
V_{IH}	Input high level voltage		$0.7 \times V_{DD}$	-	$V_{DD} + 0.3\text{ V}$	
V_{hys}	Hysteresis ⁽¹⁾		-	700	-	mV
R_{pu}	Pull-up resistor	$V_{DD} = 5\text{ V}, V_{IN} = V_{SS}$	30	55	80	k Ω
t_R, t_F	Rise and fall time (10% - 90%)	Fast I/Os Load = 50 pF	-	-	$35^{(2)}$	ns
		Standard and high sink I/Os Load = 50 pF	-	-	$125^{(2)}$	
t_R, t_F	Rise and fall time (10% - 90%)	Fast I/Os Load = 20 pF	-	-	$20^{(2)}$	ns
		Standard and high sink I/Os Load = 20 pF	-	-	$50^{(2)}$	
I_{lkg}	Digital input leakage current	$V_{SS} \leq V_{IN} \leq V_{DD}$	-	-	$\pm 1^{(3)}$	μA
$I_{lkg\text{ ana}}$	Analog input leakage current	$V_{SS} \leq V_{IN} \leq V_{DD}$	-	-	$\pm 250^{(3)}$	nA
$I_{lkg(inj)}$	Leakage current in adjacent I/O	Injection current $\pm 4\text{ mA}$	-	-	$\pm 1^{(3)}$	μA

1. Hysteresis voltage between Schmitt trigger switching levels. Based on characterization results, not tested in production.

2. Data guaranteed by design.

3. Data based on characterization results, not tested in production

Figure 24. Typical V_{IL} and V_{IH} vs V_{DD} @ 4 temperaturesFigure 25. Typical pull-up current vs V_{DD} @ 4 temperaturesFigure 26. Typical pull-up resistance vs V_{DD} @ 4 temperatures

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Table 38. Output driving current (standard ports)

Symbol	Parameter	Conditions	Min	Max	Unit
V_{OL}	Output low level with 8 pins sunk	$I_{IO} = 10 \text{ mA}$, $V_{DD} = 5 \text{ V}$	-	2.0	V
	Output low level with 4 pins sunk	$I_{IO} = 4 \text{ mA}$, $V_{DD} = 3.3 \text{ V}$	-	1.0 ⁽¹⁾	
V_{OH}	Output high level with 8 pins sourced	$I_{IO} = 10 \text{ mA}$, $V_{DD} = 5 \text{ V}$	2.4	-	
	Output high level with 4 pins sourced	$I_{IO} = 4 \text{ mA}$, $V_{DD} = 3.3 \text{ V}$	2.0 ⁽¹⁾	-	

1. Data based on characterization results, not tested in production

Table 39. Output driving current (true open drain ports)

Symbol	Parameter	Conditions	Min	Max	Unit
V_{OL}	Output low level with 2 pins sunk	$I_{IO} = 10\text{ mA}$, $V_{DD} = 5\text{ V}$	-	1.0	V
	Output low level with 2 pins sunk	$I_{IO} = 10\text{ mA}$, $V_{DD} = 3.3\text{ V}$	-	1.5 ⁽¹⁾	
V_{OH}	Output high level with 2 pins sourced	$I_{IO} = 10\text{ mA}$, $V_{DD} = 5\text{ V}$	-	2.0 ⁽¹⁾	

1. Data based on characterization results, not tested in production

Table 40. Output driving current (high sink ports)

Symbol	Parameter	Conditions	Min	Max	Unit
V_{OL}	Output low level with 8 pins sunk	$I_{IO} = 10\text{ mA}$, $V_{DD} = 5\text{ V}$	-	0.9	V
	Output low level with 4 pins sunk	$I_{IO} = 10\text{ mA}$, $V_{DD} = 3.3\text{ V}$	-	1.1 ⁽¹⁾	
		$I_{IO} = 20\text{ mA}$, $V_{DD} = 5\text{ V}$	-	1.6 ⁽¹⁾	
V_{OH}	Output high level with 8 pins sourced	$I_{IO} = 10\text{ mA}$, $V_{DD} = 5\text{ V}$	3.8	-	
	Output high level with 4 pins sourced	$I_{IO} = 10\text{ mA}$, $V_{DD} = 3.3\text{ V}$	1.9 ⁽¹⁾	-	
		$I_{IO} = 20\text{ mA}$, $V_{DD} = 5\text{ V}$	2.9 ⁽¹⁾	-	

1. Data based on characterization results, not tested in production.

10.3.7 Typical output level curves

The following figures show the typical output level curves measured with the output on a single pin.

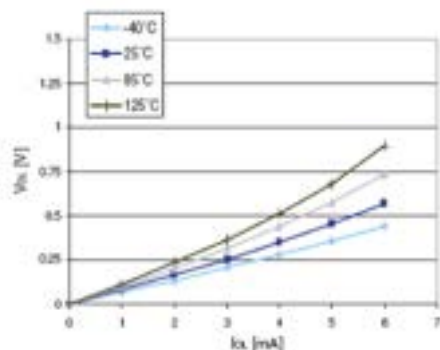
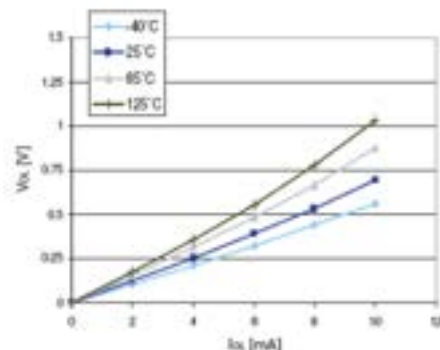
Figure 27. Typ. V_{OL} @ $V_{DD} = 3.3\text{ V}$ (standard ports)Figure 28. Typ. V_{OL} @ $V_{DD} = 5.0\text{ V}$ (standard ports)

Figure 29. Typ. V_{OL} @ $V_{DD} = 3.3\text{ V}$ (true open drain ports)

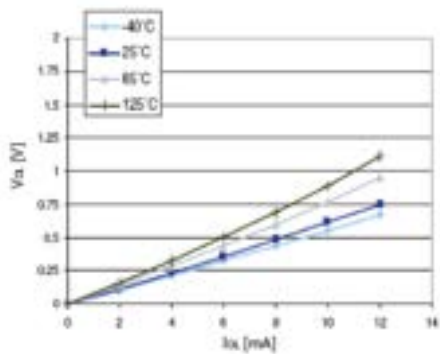


Figure 30. Typ. V_{OL} @ $V_{DD} = 5.0\text{ V}$ (true open drain ports)

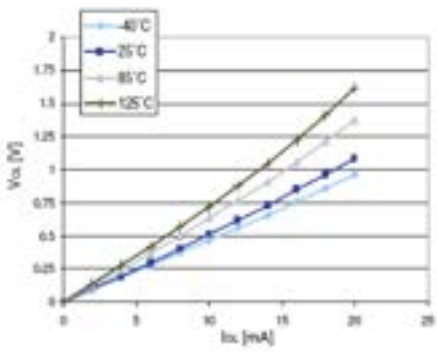


Figure 31. Typ. V_{OL} @ $V_{DD} = 3.3\text{ V}$ (high sink ports)

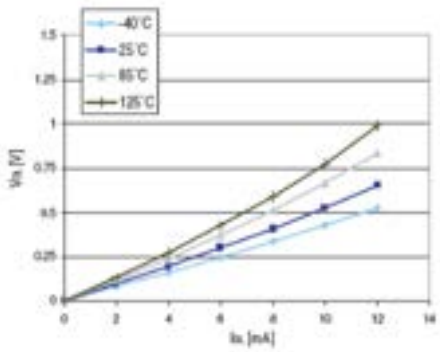


Figure 32. Typ. V_{OL} @ $V_{DD} = 5.0\text{ V}$ (high sink ports)

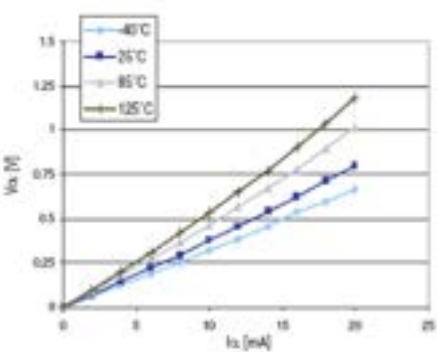


Figure 33. Typ. $V_{DD} - V_{OH}$ @ $V_{DD} = 3.3\text{ V}$
(standard ports)

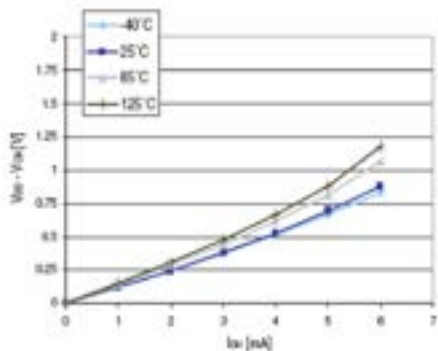


Figure 34. Typ. $V_{DD} - V_{OH}$ @ $V_{DD} = 5.0\text{ V}$
(standard ports)

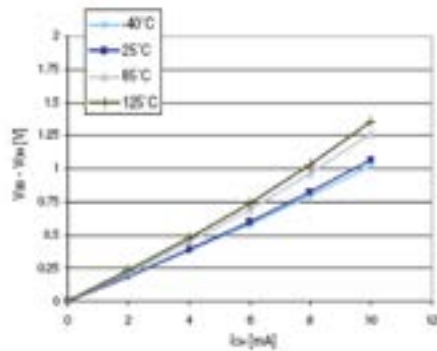


Figure 35. Typ. $V_{DD} - V_{OH}$ @ $V_{DD} = 3.3\text{ V}$ (high
sink ports)

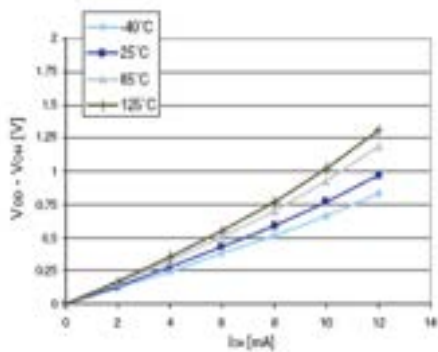
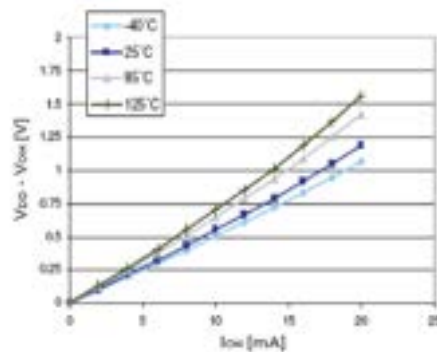


Figure 36. Typ. $V_{DD} - V_{OH}$ @ $V_{DD} = 5.0\text{ V}$ (high
sink ports)



10.3.8 Reset pin characteristics

Subject to general operating conditions for V_{DD} and T_A unless otherwise specified.

Table 41. NRST pin characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{IL(NRST)}$	NRST input low level voltage ⁽¹⁾	-	-0.3	-	$0.3 \times V_{DD}$	V
$V_{IH(NRST)}$	NRST input high level voltage ⁽¹⁾	$I_{OL} = 2 \text{ mA}$	$0.7 \times V_{DD}$	-	$V_{DD} + 0.3$	
$V_{OL(NRST)}$	NRST output low level voltage ⁽¹⁾	$I_{OL} = 3 \text{ mA}$	-	-	0.5	
$R_{PU(NRST)}$	NRST pull-up resistor ⁽²⁾	-	30	55	80	k Ω
$t_{IFP(NRST)}$	NRST input filtered pulse ⁽³⁾	-	-	-	75	ns
$t_{INFP(NRST)}$	NRST Input not filtered pulse ⁽³⁾	-	500	-	-	
$t_{OP(NRST)}$	NRST output pulse ⁽³⁾	-	20	-	-	μs

1. Data based on characterization results, not tested in production.

2. The R_{PU} pull-up equivalent resistor is based on a resistive transistor.

3. Data guaranteed by design, not tested in production.

Figure 37. Typical NRST V_{IL} and V_{IH} vs V_{DD} @ 4 temperatures

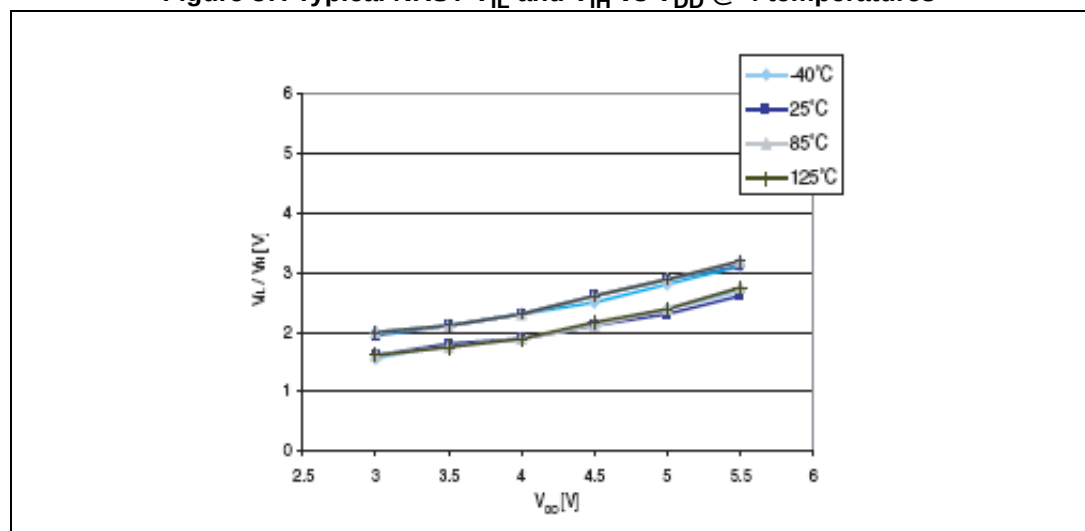
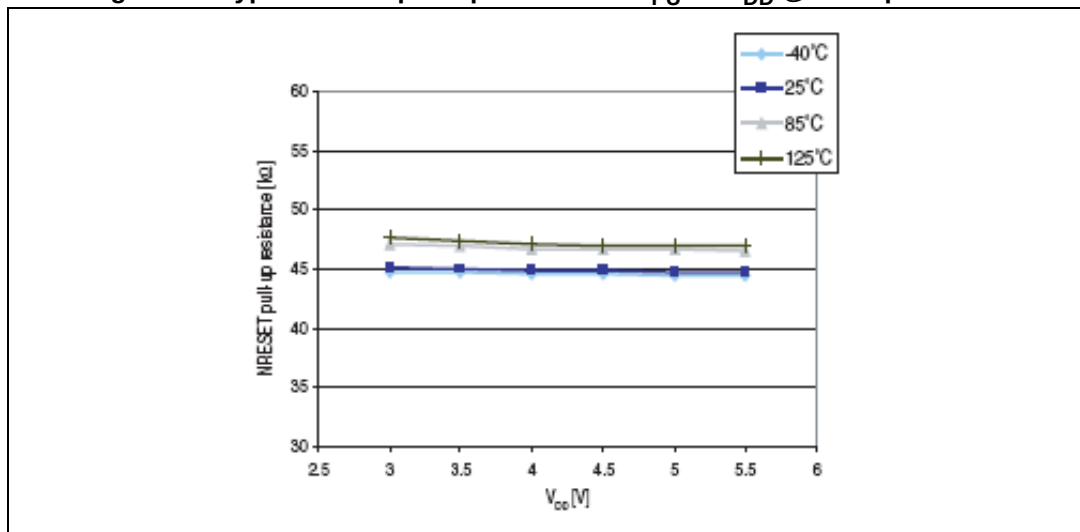
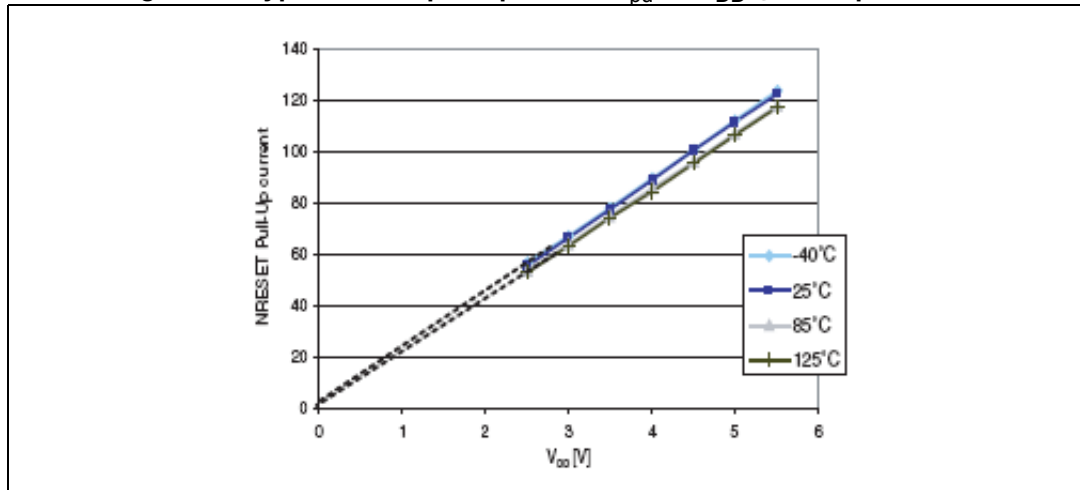
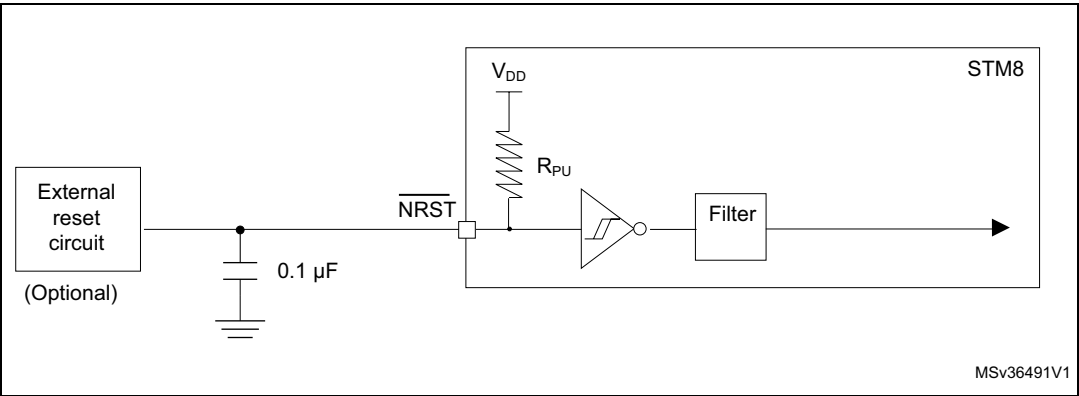


Figure 38. Typical NRST pull-up resistance R_{PU} vs V_{DD} @ 4 temperaturesFigure 39. Typical NRST pull-up current I_{PU} vs V_{DD} @ 4 temperatures

The reset network shown in [Figure 40](#) protects the device against parasitic resets. The user must ensure that the level on the NRST pin can go below $V_{IL(NRST)}$ max (see [Table 41: NRST pin characteristics](#)), otherwise the reset is not taken into account internally.

For power consumption sensitive applications, the external reset capacitor value can be reduced to limit the charge/discharge current. If NRST signal is used to reset external circuitry, attention must be taken to the charge/discharge time of the external capacitor to fulfill the external devices reset timing conditions. Minimum recommended capacity is 100 nF.

Figure 40. Recommended reset pin protection



10.3.9 SPI serial peripheral interface

Unless otherwise specified, the parameters given in [Table 42](#) are derived from tests performed under ambient temperature, f_{MASTER} frequency and V_{DD} supply voltage conditions. $t_{\text{MASTER}} = 1/f_{\text{MASTER}}$.

Refer to I/O port characteristics for more details on the input/output alternate function characteristics (NSS, SCK, MOSI, MISO).

Table 42. SPI characteristics

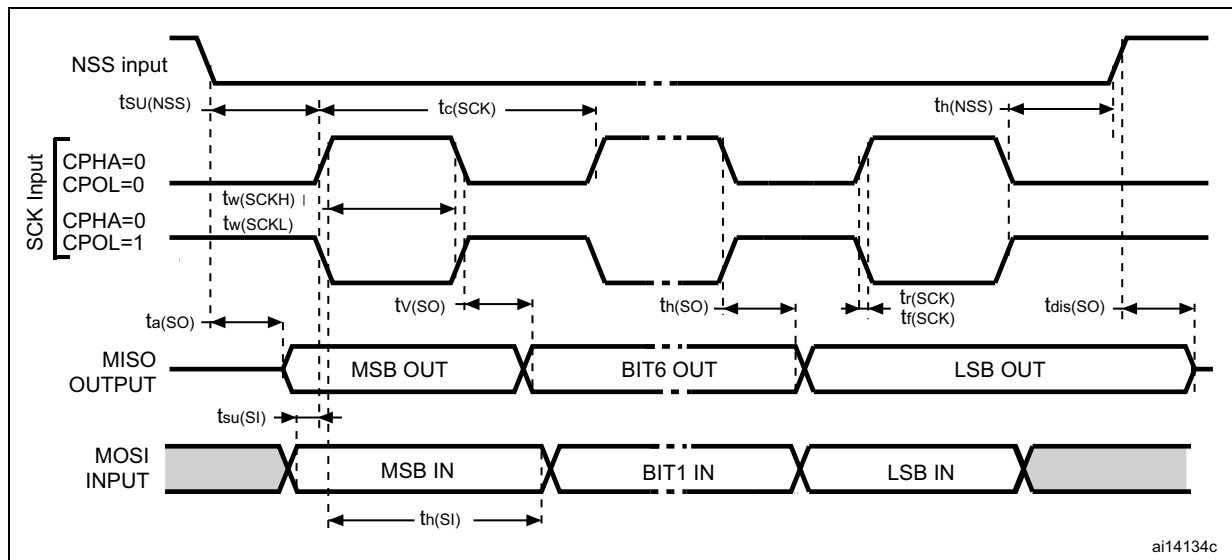
Symbol	Parameter	Conditions ⁽¹⁾	Min	Max	Unit
f_{SCK} $1/t_{\text{c(SCK)}}$	SPI clock frequency	Master mode	0	8	MHz
		Slave mode	0	6	

Table 42. SPI characteristics (continued)

Symbol	Parameter	Conditions ⁽¹⁾	Min	Max	Unit
$t_{r(SCK)}$ $t_{f(SCK)}$	SPI clock rise and fall time	Capacitive load: C = 30 pF	-	25	ns
$t_{su(NSS)}^{(2)}$	NSS setup time	Slave mode	$4 * t_{MASTER}$	-	
$t_{h(NSS)}^{(2)}$	NSS hold time	Slave mode	70	-	
$t_{w(SCKH)}^{(2)}$ $t_{w(SCKL)}^{(2)}$	SCK high and low time	Master mode	$t_{SCK}/2 - 15$	$t_{SCK}/2 + 15$	
$t_{su(MI)}^{(2)}$ $t_{su(SI)}^{(2)}$	Data input setup time	Master mode	5	-	
		Slave mode	5	-	
$t_{h(MI)}^{(2)}$ $t_{h(SI)}^{(2)}$	Data input hold time	Master mode	7	-	
		Slave mode	10	-	
$t_{a(SO)}^{(2)(3)}$	Data output access time	Slave mode	-	$3 * t_{MASTER}$	
$t_{dis(SO)}^{(2)(4)}$	Data output disable time	Slave mode	25	-	
$t_{v(SO)}^{(2)}$	Data output valid time	Slave mode (after enable edge)	-	73	
$t_{v(MO)}^{(2)}$	Data output valid time	Master mode (after enable edge)	-	36	
$t_{h(SO)}^{(2)}$	Data output hold time	Slave mode (after enable edge)	28	-	
$t_{h(MO)}^{(2)}$		Master mode (after enable edge)	12	-	

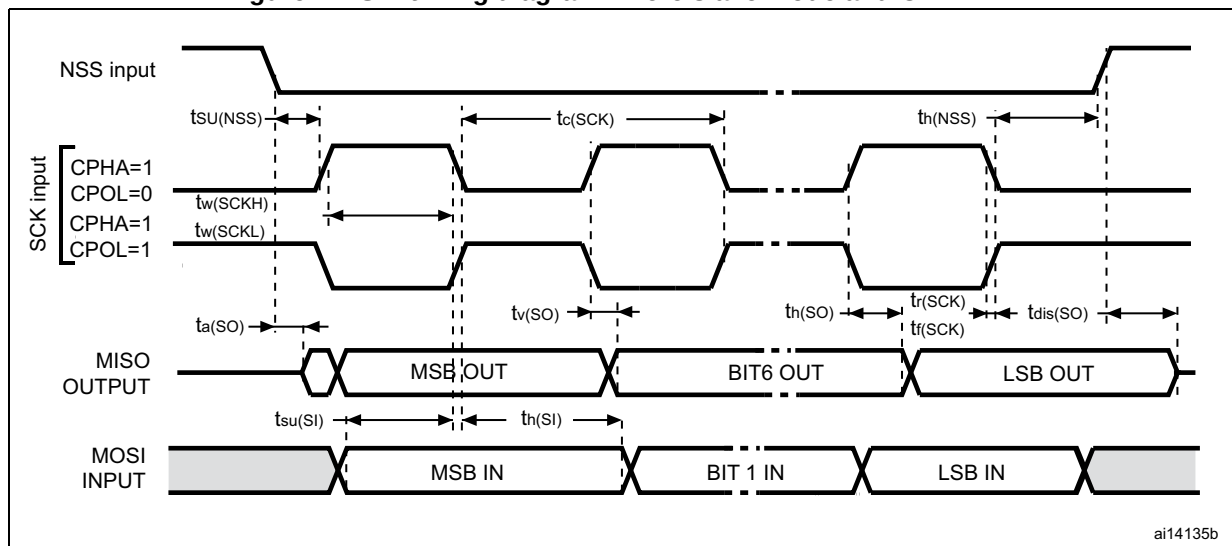
- Parameters are given by selecting 10 MHz I/O output frequency.
- Values based on design simulation and/or characterization results, and not tested in production.
- Min time is for the minimum time to drive the output and the max time is for the maximum time to validate the data.
- Min time is for the minimum time to invalidate the output and the max time is for the maximum time to put the data in Hi-Z.

Figure 41. SPI timing diagram where slave mode and CPHA = 0



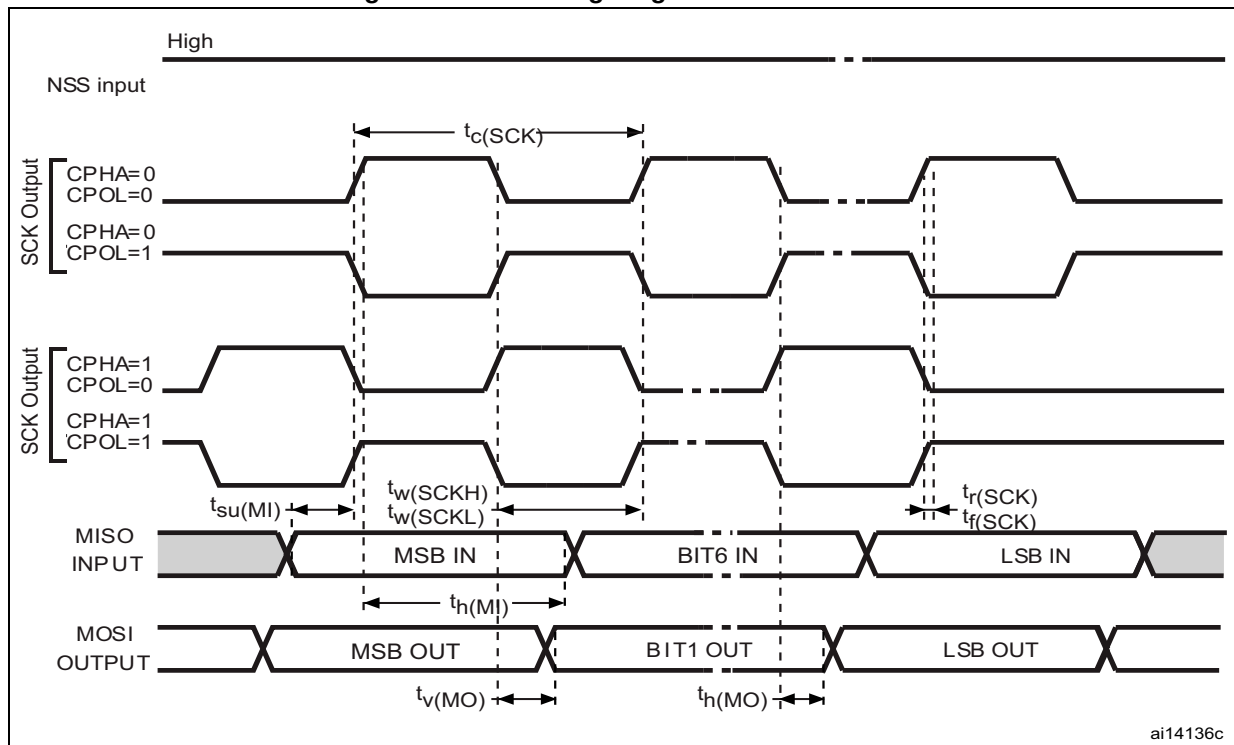
1. Measurement points are at CMOS levels: $0.3 V_{DD}$ and $0.7 V_{DD}$.

Figure 42. SPI timing diagram where slave mode and CPHA = 1



1. Measurement points are at CMOS levels: $0.3 V_{DD}$ and $0.7 V_{DD}$.

Figure 43. SPI timing diagram - master mode

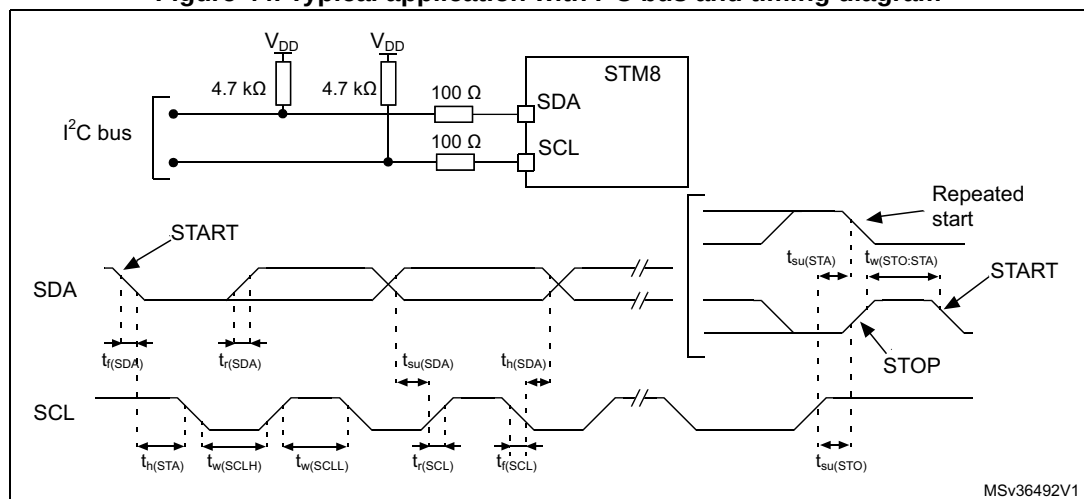


1. Measurement points are at CMOS levels: $0.3 V_{DD}$ and $0.7 V_{DD}$.

10.3.10 I²C interface characteristicsTable 43. I²C characteristics

Symbol	Parameter	Standard mode I ² C		Fast mode I ² C ⁽¹⁾		Unit
		Min ⁽²⁾	Max ⁽²⁾	Min ⁽²⁾	Max ⁽²⁾	
$t_{w(SCLL)}$	SCL clock low time	4.7	-	1.3	-	μs
$t_{w(SCLH)}$	SCL clock high time	4.0	-	0.6	-	
$t_{su(SDA)}$	SDA setup time	250	-	100	-	ns
$t_h(SDA)$	SDA data hold time	0 ⁽³⁾	-	0 ⁽⁴⁾	900 ⁽³⁾	
$t_r(SDA)$ $t_r(SCL)$	SDA and SCL rise time ($V_{DD} = 3$ to 5.5 V)	-	1000	-	300	
$t_f(SDA)$ $t_f(SCL)$	SDA and SCL fall time ($V_{DD} = 3$ to 5.5 V)	-	300	-	300	
$t_h(STA)$	START condition hold time	4.0	-	0.6	-	μs
$t_{su(STA)}$	Repeated START condition setup time	4.7	-	0.6	-	
$t_{su(STO)}$	STOP condition setup time	4.0	-	0.6	-	
$t_w(STO:STA)$	STOP to START condition time (bus free)	4.7	-	1.3	-	
C_b	Capacitive load for each bus line	-	400	-	400	pF

1. f_{MASTER} must be at least 8 MHz to achieve max fast I²C speed (400 kHz)
2. Data based on standard I²C protocol requirement, not tested in production
3. The maximum hold time of the start condition has only to be met if the interface does not stretch the low time
4. The device must internally provide a hold time of at least 300 ns for the SDA signal in order to bridge the undefined region of the falling edge of SCL

Figure 44. Typical application with I²C bus and timing diagram

10.3.11 10-bit ADC characteristics

Subject to general operating conditions for V_{DDA} , f_{MASTER} , and T_A unless otherwise specified.

Table 44. ADC characteristics

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
f_{ADC}	ADC clock frequency	$V_{DD} = 2.95$ to 5.5 V	1	-	4	MHz
		$V_{DD} = 4.5$ to 5.5 V	1	-	6	
V_{DDA}	Analog supply	-	3.0	-	5.5	V
V_{REF+}	Positive reference voltage	-	2.75 ⁽¹⁾	-	V_{DDA}	
V_{REF-}	Negative reference voltage	-	V_{SSA}	-	0.5 ⁽¹⁾	
V_{AIN}	Conversion voltage range ⁽²⁾	-	V_{SSA}	-	V_{DDA}	V
		Devices with external V_{REF+}/V_{REF-}	V_{REF-}	-	V_{REF+}	
C_{ADC}	Internal sample and hold capacitor	-	-	3	-	pF
$t_S^{(2)}$	Minimum sampling time	$f_{ADC} = 4$ MHz	-	0.75	-	μs
		$f_{ADC} = 6$ MHz	-	0.5	-	
t_{STAB}	Wakeup time from standby	-	-	7.0	-	μs
t_{CONV}	Minimum total conversion time (including sampling time, 10-bit resolution)	$f_{ADC} = 4$ MHz	3.5			μs
		$f_{ADC} = 6$ MHz	2.33			μs
		-	14			$1/f_{ADC}$

1. Data guaranteed by design, not tested in production.

2. During the sample time, the sampling capacitance, C_{AIN} (3 pF max), can be charged/discharged by the external source. The internal resistance of the analog source must allow the capacitance to reach its final voltage level within t_S . After the end of the sample time t_S , changes of the analog input voltage have no effect on the conversion result. Values for the sample clock t_S depend on programming.

Table 45. ADC accuracy with $R_{AIN} < 10\text{ k}\Omega$, $V_{DDA} = 5\text{ V}$

Symbol	Parameter	Conditions	Typ	Max ⁽¹⁾	Unit
E _T	Total unadjusted error ⁽²⁾	f _{ADC} = 2 MHz	1.0	2.5	LSB
		f _{ADC} = 4 MHz	1.4	3.0	
		f _{ADC} = 6 MHz	1.6	3.5	
E _O	Offset error ⁽²⁾	f _{ADC} = 2 MHz	0.6	2.0	
		f _{ADC} = 4 MHz	1.1	2.5	
		f _{ADC} = 6 MHz	1.2	2.5	
E _G	Gain error ⁽²⁾	f _{ADC} = 2 MHz	0.2	2.0	
		f _{ADC} = 4 MHz	0.6	2.5	
		f _{ADC} = 6 MHz	0.8	2.5	
E _D	Differential linearity error ⁽²⁾	f _{ADC} = 2 MHz	0.7	1.5	
		f _{ADC} = 4 MHz	0.7	1.5	
		f _{ADC} = 6 MHz	0.8	1.5	
E _L	Integral linearity error ⁽²⁾	f _{ADC} = 2 MHz	0.6	1.5	
		f _{ADC} = 4 MHz	0.6	1.5	
		f _{ADC} = 6 MHz	0.6	1.5	

1. Data based on characterization results, not tested in production.

2. ADC accuracy vs. negative injection current: Injecting negative current on any of the analog input pins should be avoided as this significantly reduces the accuracy of the conversion being performed on another analog input. It is recommended to add a Schottky diode (pin to ground) to standard analog pins which may potentially inject negative current. Any positive injection current within the limits specified for I_{INJ(PIN)} and ΣI_{INJ(PIN)} in [Section 10.3.6](#) does not affect the ADC accuracy.

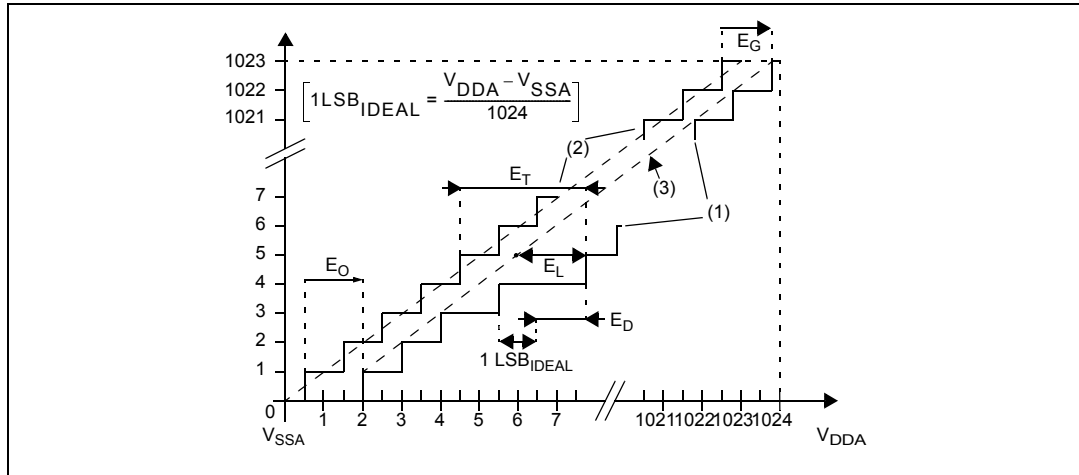
Table 46. ADC accuracy with $R_{AIN} < 10\text{ k}\Omega$, $V_{DDA} = 3.3\text{ V}$

Symbol	Parameter	Conditions	Typ	Max ⁽¹⁾	Unit
E _T	Total unadjusted error ⁽²⁾	f _{ADC} = 2 MHz	1.1	2.0	LSB
		f _{ADC} = 4 MHz	1.6	2.5	
E _O	Offset error ⁽²⁾	f _{ADC} = 2 MHz	0.7	1.5	
		f _{ADC} = 4 MHz	1.3	2.0	
E _G	Gain error ⁽²⁾	f _{ADC} = 2 MHz	0.2	1.5	
		f _{ADC} = 4 MHz	0.5	2.0	
E _D	Differential linearity error ⁽²⁾	f _{ADC} = 2 MHz	0.7	1.0	
		f _{ADC} = 4 MHz	0.7	1.0	
E _L	Integral linearity error ⁽²⁾	f _{ADC} = 2 MHz	0.6	1.5	
		f _{ADC} = 4 MHz	0.6	1.5	

1. Data based on characterization results, not tested in production.

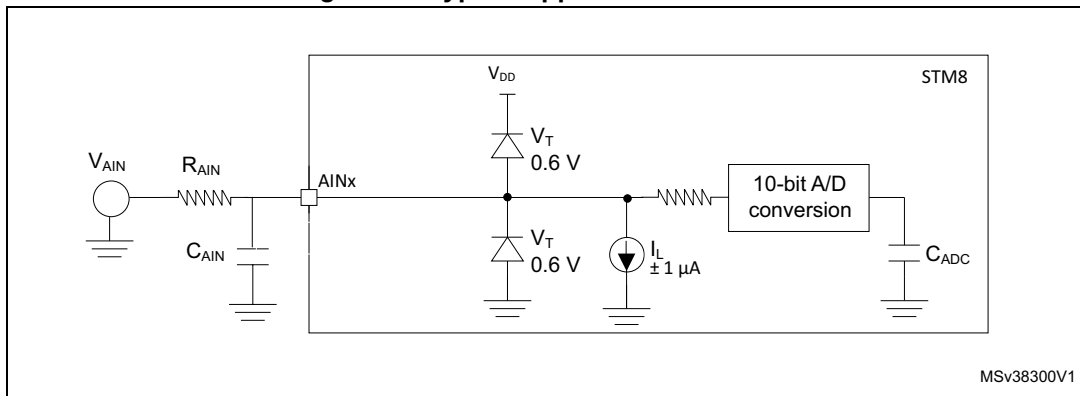
2. ADC accuracy vs. negative injection current: Injecting negative current on any of the analog input pins should be avoided as this significantly reduces the accuracy of the conversion being performed on another analog input. It is recommended to add a Schottky diode (pin to ground) to standard analog pins which may potentially inject negative current. Any positive injection current within the limits specified for $I_{INJ(PIN)}$ and $\Sigma I_{INJ(PIN)}$ in [Section 10.3.6](#) does not affect the ADC accuracy.

Figure 45. ADC accuracy characteristics



1. Example of an actual transfer curve
 2. The ideal transfer curve
 3. End point correlation line
- E_T = Total unadjusted error: maximum deviation between the actual and the ideal transfer curves.
 E_O = Offset error: deviation between the first actual transition and the first ideal one.
 E_G = Gain error: deviation between the last ideal transition and the last actual one.
 E_D = Differential linearity error: maximum deviation between actual steps and the ideal one.
 E_L = Integral linearity error: maximum deviation between any actual transition and the end point correlation line.

Figure 46. Typical application with ADC



1. Legend: R_{AIN} = external resistance, C_{AIN} = capacitors, C_{smp} = internal sample and hold capacitor.

10.3.12 EMC characteristics

Susceptibility tests are performed on a sample basis during product characterization.

Functional EMS (electromagnetic susceptibility)

While executing a simple application (toggling 2 LEDs through I/O ports), the product is stressed by two electromagnetic events until a failure occurs (indicated by the LEDs).

- **ESD:** Electrostatic discharge (positive and negative) is applied on all pins of the device until a functional disturbance occurs. This test conforms with the IEC 61000-4-2 standard.
- **FTB:** A burst of fast transient voltage (positive and negative) is applied to V_{DD} and V_{SS} through a 100 pF capacitor, until a functional disturbance occurs. This test conforms with the IEC 61000-4-4 standard.

A device reset allows normal operations to be resumed. The test results are given in the table below based on the EMS levels and classes defined in application note AN1709 (EMC design guide for STM microcontrollers).

Designing hardened software to avoid noise problems

EMC characterization and optimization are performed at component level with a typical application environment and simplified MCU software. It should be noted that good EMC performance is highly dependent on the user application and the software in particular.

Therefore it is recommended that the user applies EMC software optimization and prequalification tests in relation with the EMC level requested for his application.

Software recommendations

The software flowchart must include the management of runaway conditions such as:

- Corrupted program counter
- Unexpected reset
- Critical data corruption (control registers...)

Prequalification trials

Most of the common failures (unexpected reset and program counter corruption) can be recovered by applying a low state on the NRST pin or the oscillator pins for 1 second.

To complete these trials, ESD stress can be applied directly on the device, over the range of specification values. When unexpected behavior is detected, the software can be hardened to prevent unrecoverable errors occurring. See application note AN1015 (Software techniques for improving microcontroller EMC performance).

Table 47. EMS data

Symbol	Parameter	Conditions	Level/class
V_{FESD}	Voltage limits to be applied on any I/O pin to induce a functional disturbance	$V_{DD} = 5\text{ V}$, $T_A = 25\text{ }^{\circ}\text{C}$, $f_{MASTER} = 16\text{ MHz}$ (HSI clock), Conforms to IEC 1000-4-2	2/B ⁽¹⁾
V_{EFTB}	Fast transient voltage burst limits to be applied through 100 pF on V_{DD} and V_{SS} pins to induce a functional disturbance	$V_{DD} = 5\text{ V}$, $T_A = 25\text{ }^{\circ}\text{C}$, $f_{MASTER} = 16\text{ MHz}$ (HSI clock), Conforms to IEC 1000-4-4	4/A ⁽¹⁾

1. Data obtained with HSI clock configuration, after applying the hardware recommendations described in AN2860 (EMC guidelines for STM8S microcontrollers).

Electromagnetic interference (EMI)

Based on a simple application running on the product (toggling 2 LEDs through the I/O ports), the product is monitored in terms of emission. This emission test is in line with the norm IEC 61967-2 which specifies the board and the loading of each pin.

Table 48. EMI data

Symbol	Parameter	Conditions				Unit
		General conditions	Monitored frequency band	Max f _{HSE} /f _{CPU} ⁽¹⁾		
				8 MHz/ 8 MHz	8 MHz/ 16 MHz	
S _{EMI}	Peak level	V _{DD} = 5 V, T _A = 25 °C, LQFP48 package. Conforming to IEC 61967-2	0.1 MHz to 30 MHz	13	14	dBμV
			30 MHz to 130 MHz	23	19	
			130 MHz to 1 GHz	-4.0	-4.0	
	EMI level	EMI level	2.0	1.5	-	

1. Data based on characterization results, not tested in production.

Absolute maximum ratings (electrical sensitivity)

Based on two different tests (ESD, DLU and LU) using specific measurement methods, the product is stressed to determine its performance in terms of electrical sensitivity. For more details, refer to the application note AN1181.

Electrostatic discharge (ESD)

Electrostatic discharges (a positive then a negative pulse separated by 1 second) are applied to the pins of each sample according to each pin combination. The sample size depends on the number of supply pins in the device (3 parts x (n+1) supply pin). One model can be simulated: Human body model. This test conforms to the JESD22-A114A/A115A standard. For more details, refer to the application note AN1181.

Table 49. ESD absolute maximum ratings

Symbol	Ratings	Conditions	Class	Maximum value ⁽¹⁾	Unit
$V_{ESD(HBM)}$	Electrostatic discharge voltage (Human body model)	$T_A = 25\text{ }^{\circ}\text{C}$, conforming to JESD22-A114	A	2000	V
$V_{ESD(CDM)}$	Electrostatic discharge voltage (Charge device model)	$T_A = 25\text{ }^{\circ}\text{C}$, conforming to SD22-C101	IV	1000	

1. Data based on characterization results, not tested in production

Static latch-up

Two complementary static tests are required on 10 parts to assess the latch-up performance.

- A supply overvoltage (applied to each power supply pin), and
- A current injection (applied to each input, output and configurable I/O pin) are performed on each sample.

This test conforms to the EIA/JESD 78 IC latch-up standard. For more details, refer to the application note AN1181.

Table 50. Electrical sensitivities

Symbol	Parameter	Conditions	Class ⁽¹⁾
LU	Static latch-up class	$T_A = 25\text{ °C}$	A
		$T_A = 85\text{ °C}$	
		$T_A = 125\text{ °C}$	

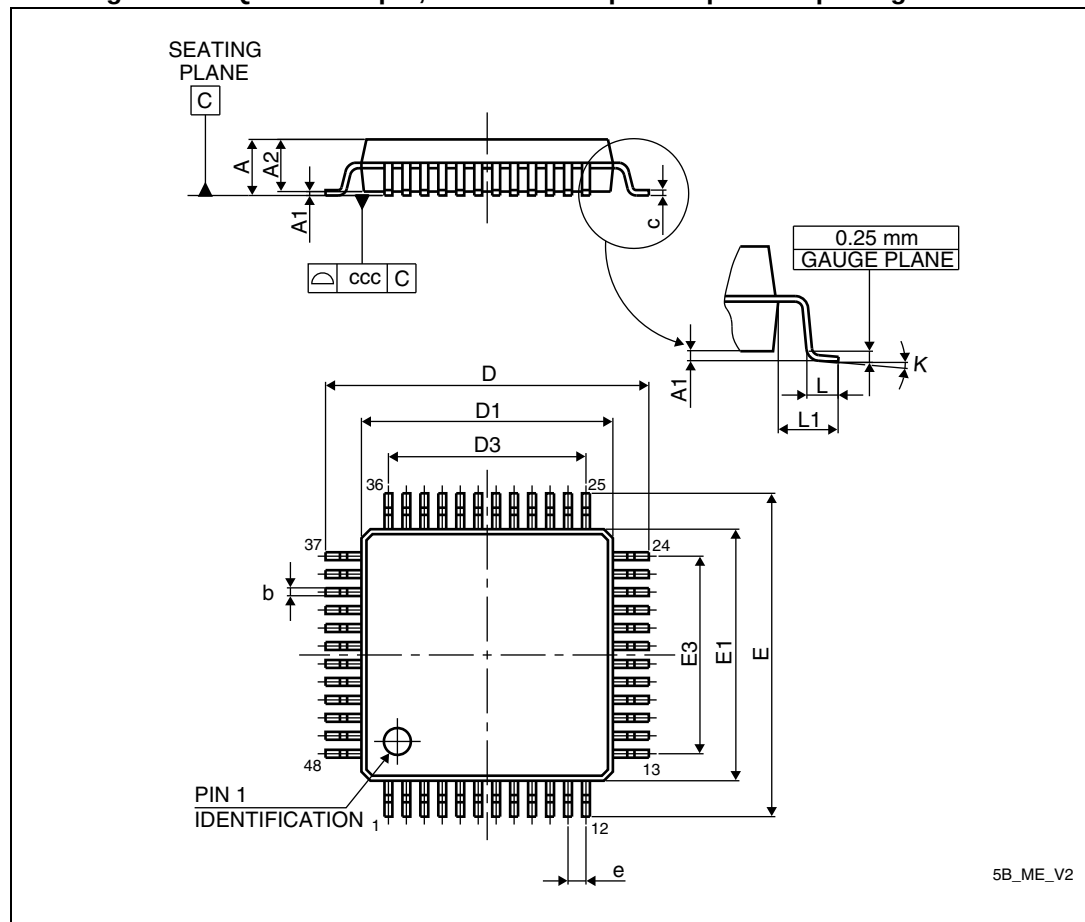
1. Class description: A Class is an STMicroelectronics internal specification. All its limits are higher than the JEDEC specifications, that means when a device belongs to class A it exceeds the JEDEC standard. B class strictly covers all the JEDEC criteria (international standard).

11 Package information

In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK® packages, depending on their level of environmental compliance. ECOPACK® specifications, grade definitions and product status are available at: www.st.com. ECOPACK® is an ST trademark.

11.1 LQFP48 package information

Figure 47. LQFP48 - 48-pin, 7 x 7 mm low-profile quad flat package outline



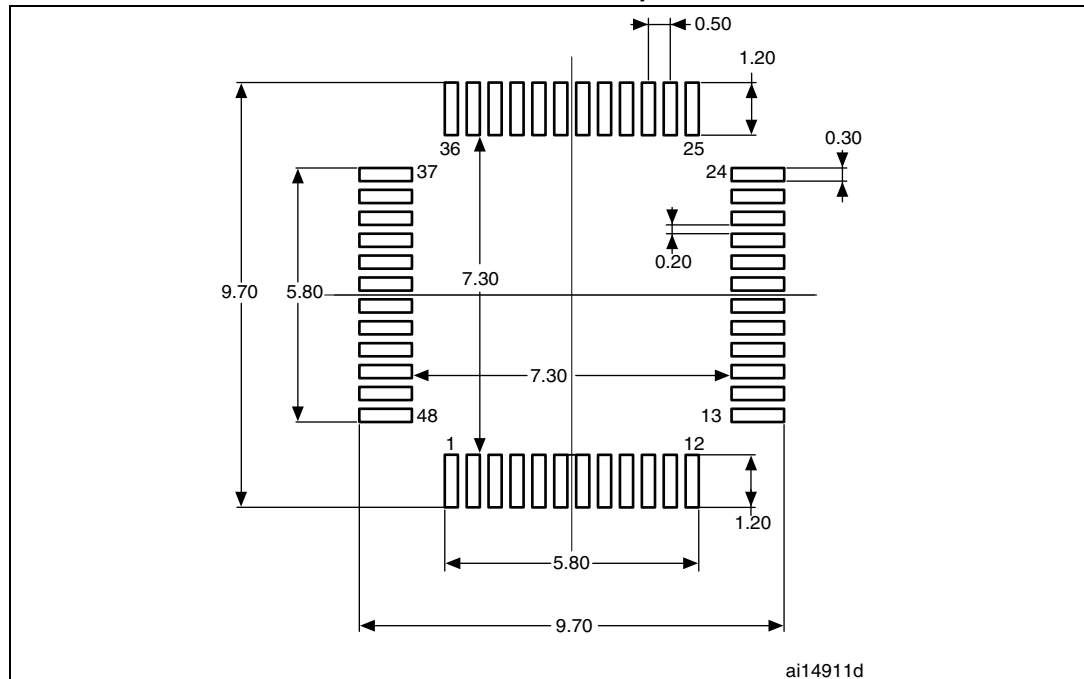
1. Drawing is not to scale.

Table 51. LQFP48 - 48-pin, 7 x 7 mm low-profile quad flat package mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	-	-	1.600	-	-	0.0630
A1	0.050	-	0.150	0.0020	-	0.0059
A2	1.350	1.400	1.450	0.0531	0.0551	0.0571
b	0.170	0.220	0.270	0.0067	0.0087	0.0106
c	0.090	-	0.200	0.0035	-	0.0079
D	8.800	9.000	9.200	0.3465	0.3543	0.3622
D1	6.800	7.000	7.200	0.2677	0.2756	0.2835
D3	-	5.500	-	-	0.2165	-
E	8.800	9.000	9.200	0.3465	0.3543	0.3622
E1	6.800	7.000	7.200	0.2677	0.2756	0.2835
E3	-	5.500	-	-	0.2165	-
e	-	0.500	-	-	0.0197	-
L	0.450	0.600	0.750	0.0177	0.0236	0.0295
L1	-	1.000	-	-	0.0394	-
k	0°	3.5°	7°	0°	3.5°	7°
ccc	-	-	0.080	-	-	0.0031

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 48. LQFP48 - 48-pin, 7 x 7 mm low-profile quad flat package recommended footprint

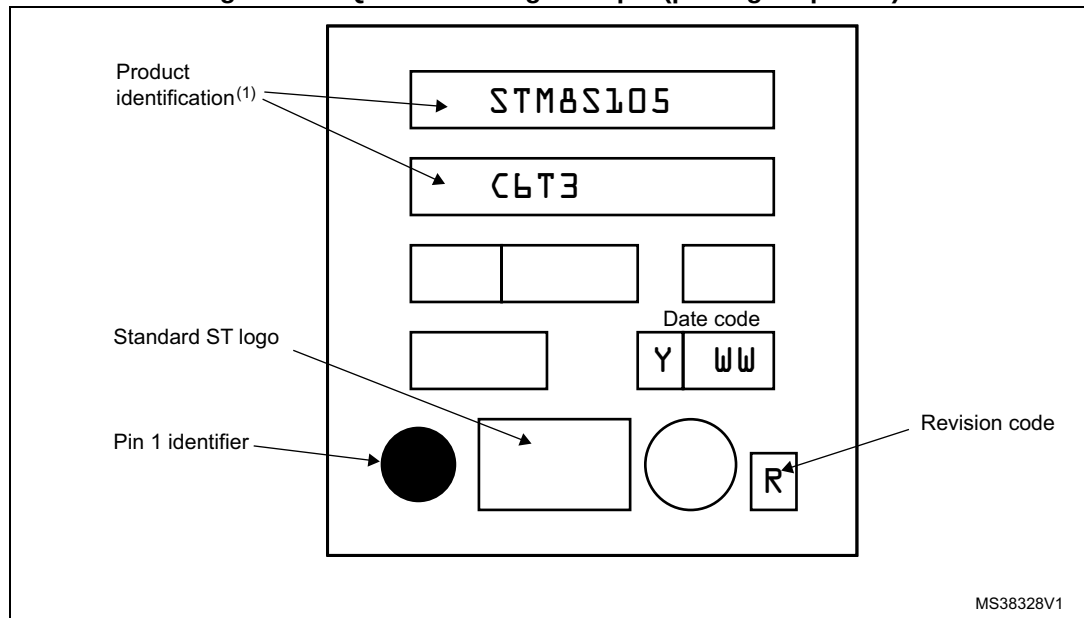


1. Dimensions are expressed in millimeters.

Device marking

The following figure gives an example of topside marking orientation versus pin 1 identifier location.

Figure 49. LQFP48 marking example (package top view)

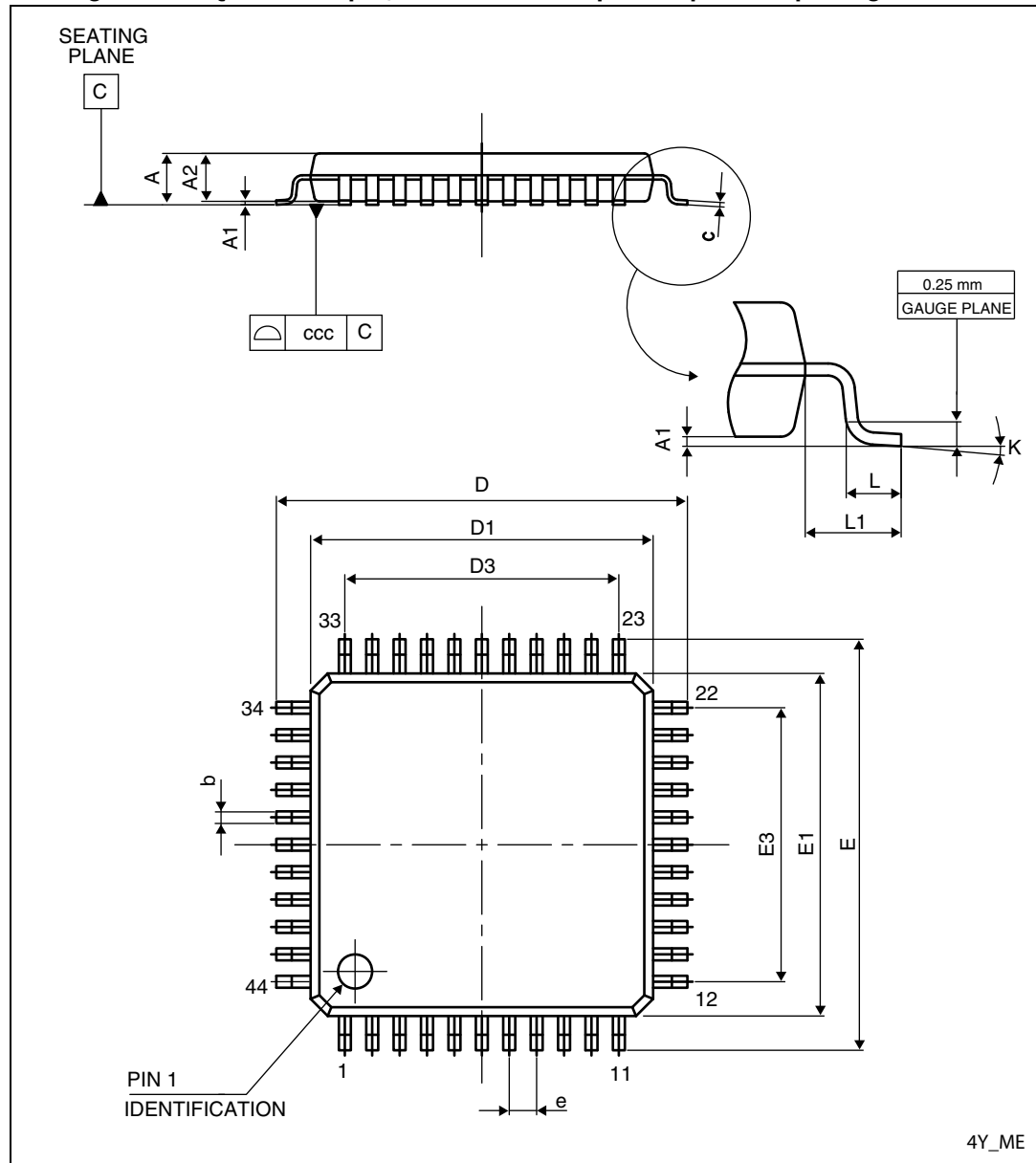


1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not yet ready to be used in production and any consequences deriving from such

usage will not be at ST charge. In no event, ST will be liable for any customer usage of these engineering samples in production. ST Quality has to be contacted prior to any decision to use these Engineering Samples to run qualification activity.

11.2 LQFP44 package information

Figure 50. LQFP44 - 44-pin, 10 x 10 mm low-profile quad flat package outline



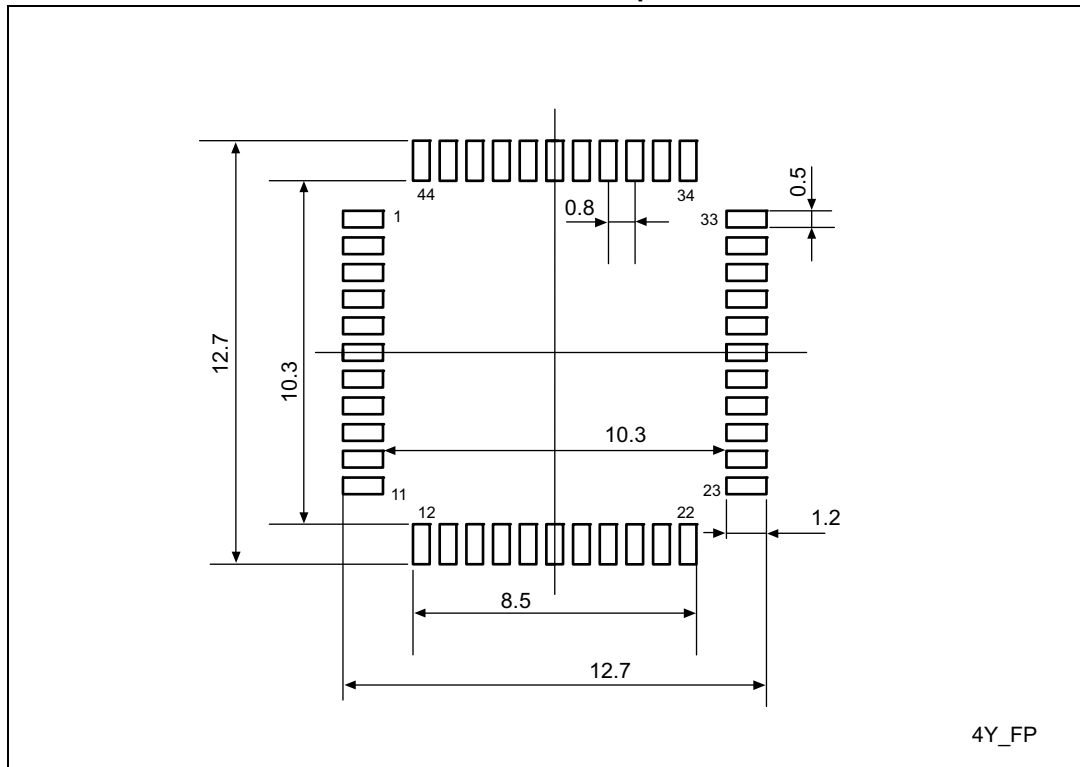
1. Drawing is not to scale.

Table 52. LQFP44 - 44-pin, 10 x 10 mm low-profile quad flat package mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	-	-	1.600	-	-	0.0630
A1	0.050	-	0.150	0.0020	-	0.0059
A2	1.350	1.400	1.450	0.0531	0.0551	0.0571
b	0.300	0.370	0.450	0.0118	0.0146	0.0177
c	0.090	-	0.200	0.0035	-	0.0079
D	11.800	12.000	12.200	0.4646	0.4724	0.4803
D1	9.800	10.000	10.200	0.3858	0.3937	0.4016
D3	-	8.000	-	-	0.3150	-
E	11.800	12.000	12.200	0.4646	0.4724	0.4803
E1	9.800	10.000	10.200	0.3858	0.3937	0.4016
E3	-	8.000	-	-	0.3150	-
e	-	0.800	-	-	0.0315	-
L	0.450	0.600	0.750	0.0177	0.0236	0.0295
L1	-	1.000	-	-	0.0394	-
k	0°	3.5°	7°	0°	3.5°	7°
ccc	-	-	0.100	-	-	0.0039

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 51. LQFP44 - 44-pin, 10 x 10 mm low-profile quad flat package recommended footprint

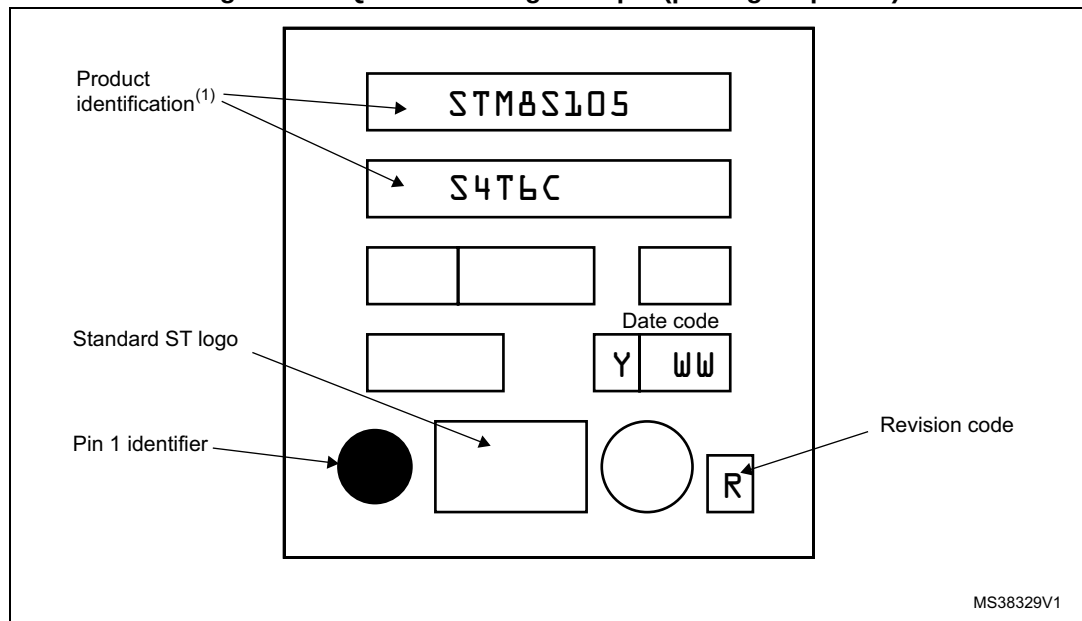


1. Dimensions are expressed in millimeters.

Device marking

The following figure gives an example of topside marking orientation versus pin 1 identifier location.

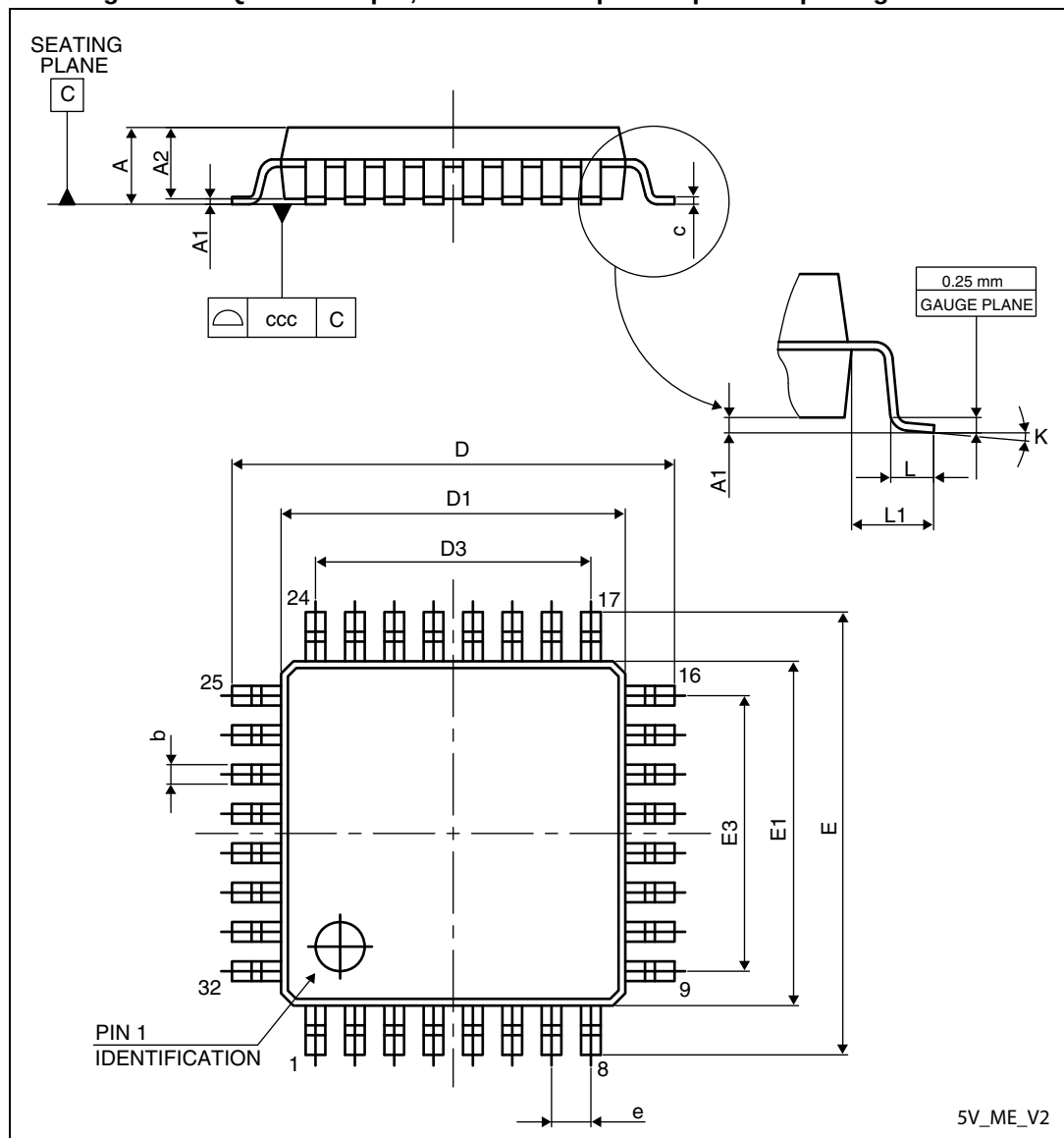
Figure 52. LQFP44 marking example (package top view)



1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not yet ready to be used in production and any consequences deriving from such usage will not be at ST charge. In no event, ST will be liable for any customer usage of these engineering samples in production. ST Quality has to be contacted prior to any decision to use these Engineering Samples to run qualification activity.

11.3 LQFP32 package information

Figure 53. LQFP32 - 32-pin, 7 x 7 mm low-profile quad flat package outline



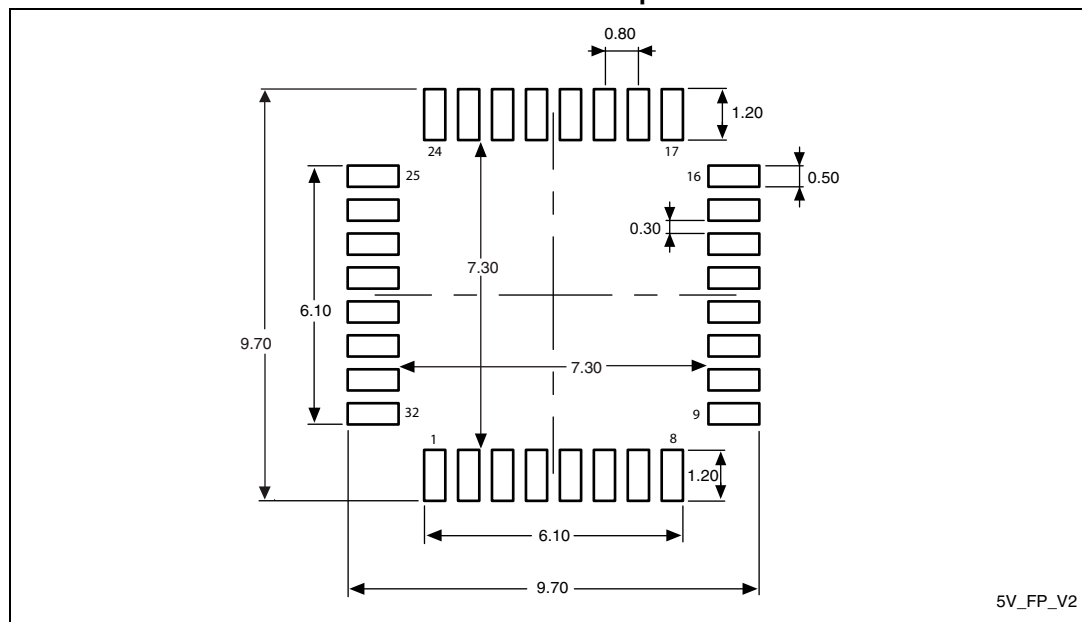
1. Drawing is not to scale.

**Table 53. LQFP32 - 32-pin, 7 x 7 mm low-profile quad flat package
mechanical data**

Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	-	-	1.600	-	-	0.0630
A1	0.050	-	0.150	0.0020	-	0.0059
A2	1.350	1.400	1.450	0.0531	0.0551	0.0571
b	0.300	0.370	0.450	0.0118	0.0146	0.0177
c	0.090	-	0.200	0.0035	-	0.0079
D	8.800	9.000	9.200	0.3465	0.3543	0.3622
D1	6.800	7.000	7.200	0.2677	0.2756	0.2835
D3	-	5.600	-	-	0.2205	-
E	8.800	9.000	9.200	0.3465	0.3543	0.3622
E1	6.800	7.000	7.200	0.2677	0.2756	0.2835
E3	-	5.600	-	-	0.2205	-
e	-	0.800	-	-	0.0315	-
L	0.450	0.600	0.750	0.0177	0.0236	0.0295
L1	-	1.000	-	-	0.0394	-
k	0°	3.5°	7°	0°	3.5°	7°
ccc	-	-	0.100	-	-	0.0039

1. Values in inches are converted from mm and rounded to 4 decimal digits.

Figure 54. LQFP32 - 32-pin, 7 x 7 mm low-profile quad flat package recommended footprint

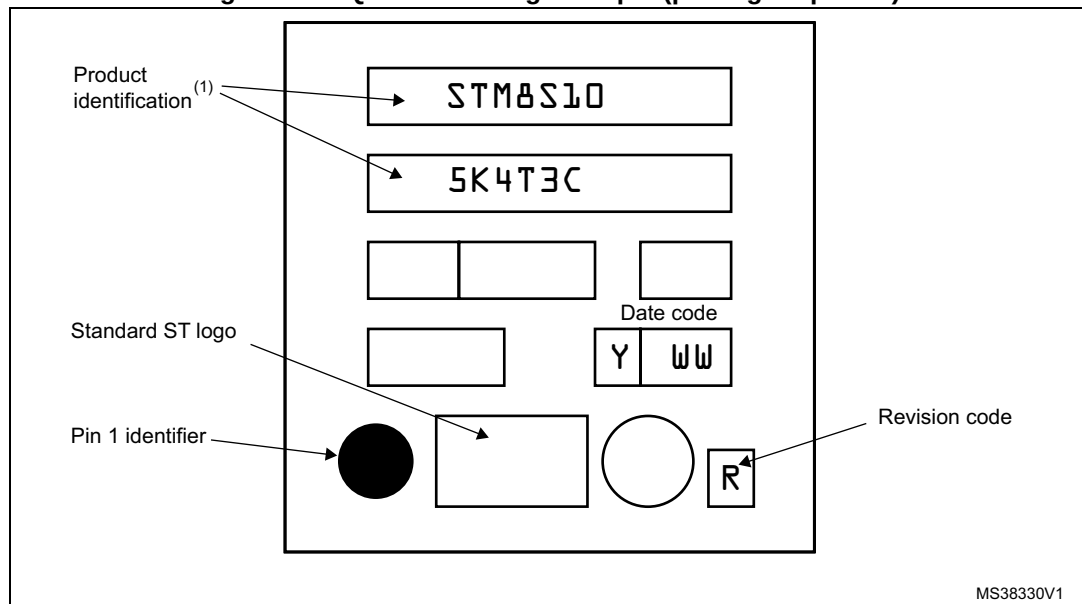


1. Dimensions are expressed in millimeters.

Device marking

The following figure gives an example of topside marking orientation versus pin 1 identifier location.

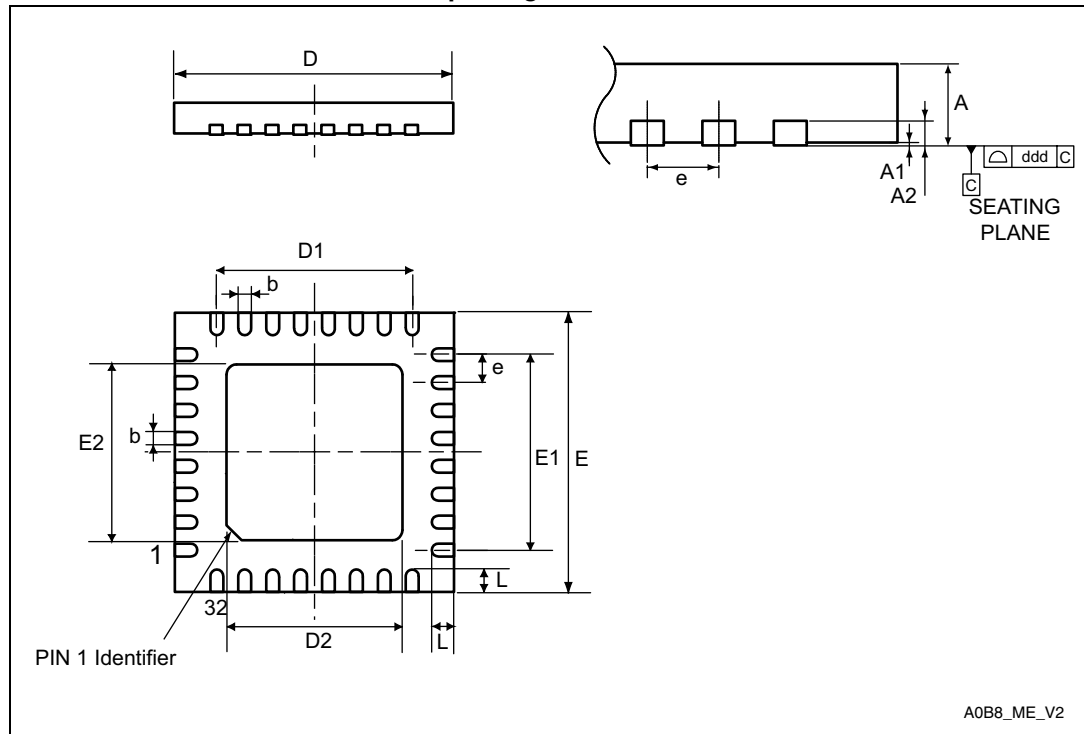
Figure 55. LQFP32 marking example (package top view)



1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not yet ready to be used in production and any consequences deriving from such usage will not be at ST charge. In no event, ST will be liable for any customer usage of these engineering samples in production. ST Quality has to be contacted prior to any decision to use these Engineering samples to run qualification activity.

11.4 UFQFPN32 package information

Figure 56. UFQFPN32 - 32-pin, 5x5 mm, 0.5 mm pitch ultra thin fine pitch quad flat package outline

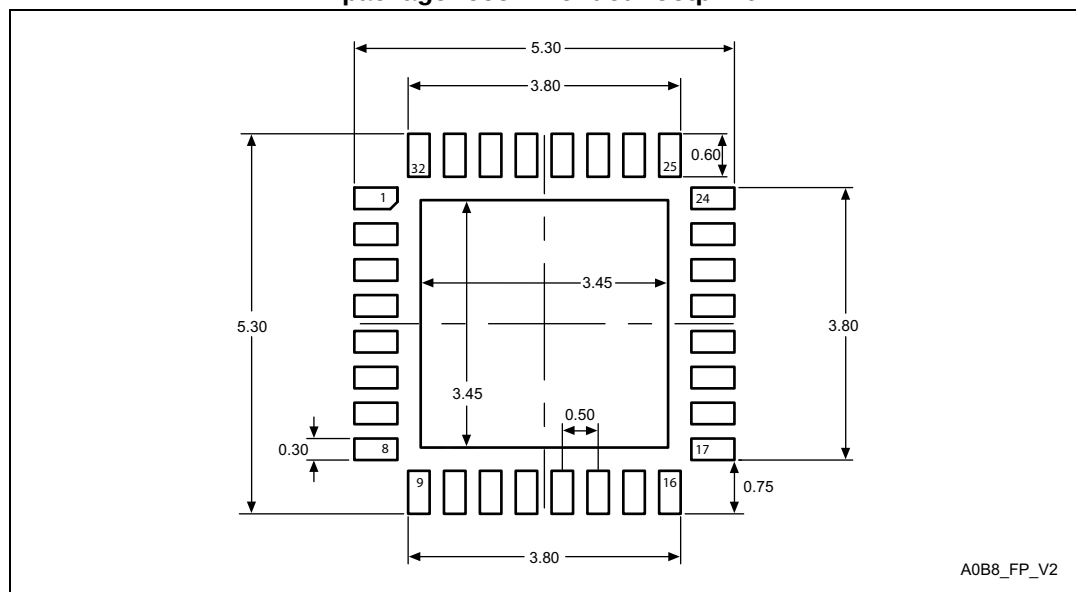


1. Drawing is not to scale.
2. All leads/pads should be soldered to the PCB to improve the lead/pad solder joint life.
3. There is an exposed die pad on the underside of the UFQFPN package. It is recommended to connect and solder this backside pad to PCB ground.
4. Dimensions are in millimeters.

Table 54. UFQFPN32 - 32-pin, 5x5 mm, 0.5 mm pitch ultra thin fine pitch quad flat package mechanical data

Symbol	millimeters			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	0.500	0.550	0.600	0.0197	0.0217	0.0236
A1	0.000	0.020	0.050	0.0000	0.0008	0.0020
A3	-	0.152	-	-	0.0060	-
b	0.180	0.230	0.280	0.0071	0.0091	0.0110
D	4.900	5.000	5.100	0.1929	0.1969	0.2008
D1	3.400	3.500	3.600	0.1339	0.1378	0.1417
D2	3.400	3.500	3.600	0.1339	0.1378	0.1417
E	4.900	5.000	5.100	0.1929	0.1969	0.2008
E1	3.400	3.500	3.600	0.1339	0.1378	0.1417
E2	3.400	3.500	3.600	0.1339	0.1378	0.1417
e	-	0.500	-	-	0.0197	-
L	0.300	0.400	0.500	0.0118	0.0157	0.0197
ddd	-	-	0.080	-	-	0.0031

1. Values in inches are converted from mm and rounded to 4 decimal digits.

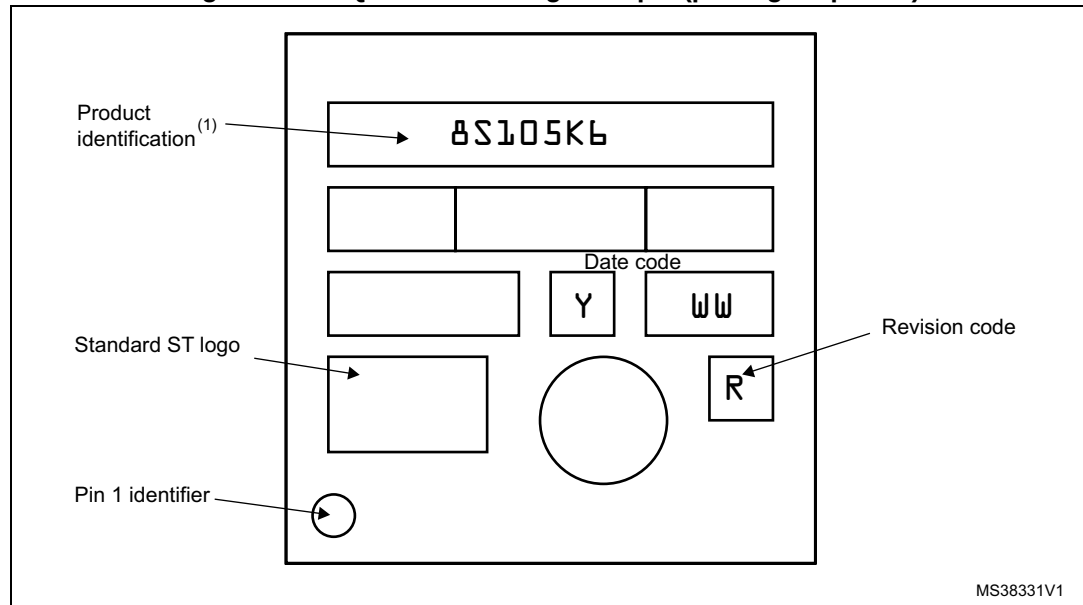
Figure 57. UFQFPN32 - 32-pin, 5 x5 mm, 0.5 mm pitch ultra thin fine pitch quad flat package recommended footprint

1. Dimensions are expressed in millimeters.

Device marking

The following figure gives an example of topside marking orientation versus pin 1 identifier location.

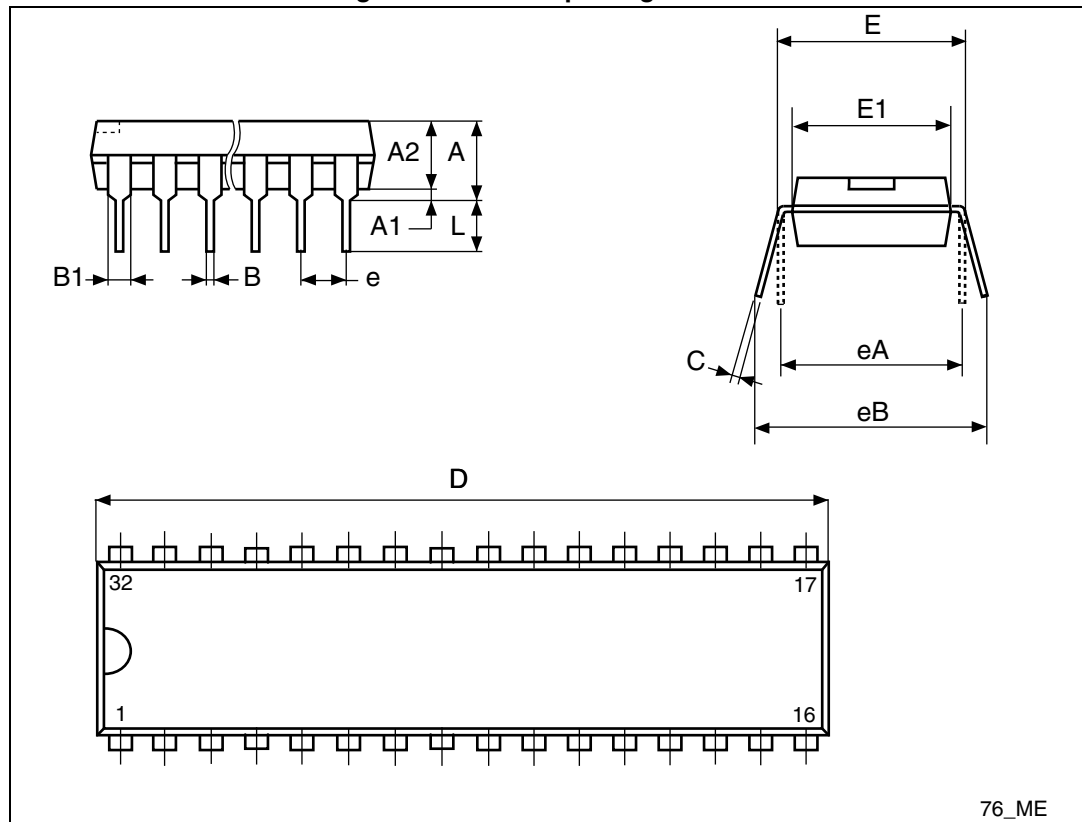
Figure 58. UFQFPN32 marking example (package top view)



1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not yet ready to be used in production and any consequences deriving from such usage will not be at ST charge. In no event, ST will be liable for any customer usage of these engineering samples in production. ST Quality has to be contacted prior to any decision to use these Engineering samples to run qualification activity.

11.5 SDIP32 package information

Figure 59. SDIP32 package outline



76_ME

Table 55. SDIP32 package mechanical data

Dim.	mm			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
A	3.556	3.759	5.080	0.1400	0.1480	0.2000
A1	0.508	-	-	0.0200	-	-
A2	3.048	3.556	4.572	0.1200	0.1400	0.1800
B	0.356	0.457	0.584	0.0140	0.0180	0.0230
B1	0.762	1.016	1.397	0.0300	0.0400	0.0550
C	0.203	0.254	0.356	0.0079	0.0100	0.0140
D	27.430	27.940	28.450	1.0799	1.1000	1.1201
E	9.906	10.410	11.050	0.3900	0.4098	0.4350
E1	7.620	8.890	9.398	0.3000	0.3500	0.3700
e	-	1.778	-	-	0.0700	-
eA	-	10.160	-	-	0.4000	-

Table 55. SDIP32 package mechanical data (continued)

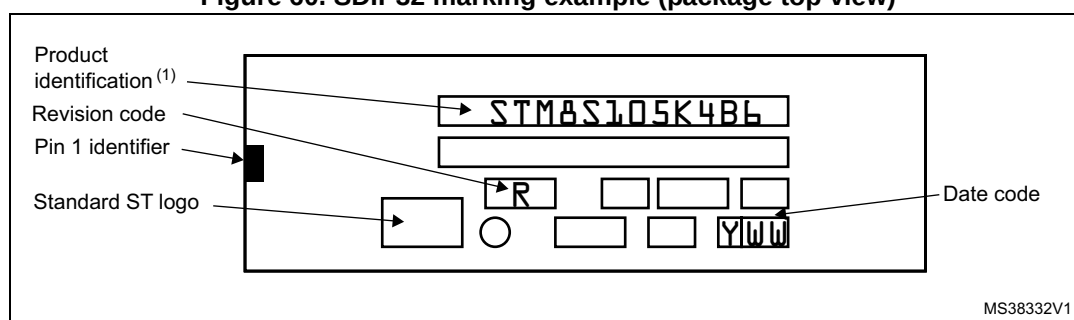
Dim.	mm			inches ⁽¹⁾		
	Min	Typ	Max	Min	Typ	Max
eB	-	-	12.700	-	-	0.5000
L	2.540	3.048	3.810	0.1000	0.1200	0.1500

1. Values in inches are converted from mm and rounded to 4 decimal digits

Device marking

The following figure gives an example of topside marking orientation versus pin 1 identifier location.

Figure 60. SDIP32 marking example (package top view)



1. Parts marked as "ES", "E" or accompanied by an Engineering Sample notification letter, are not yet qualified and therefore not yet ready to be used in production and any consequences deriving from such usage will not be at ST charge. In no event, ST will be liable for any customer usage of these engineering samples in production. ST Quality has to be contacted prior to any decision to use these Engineering samples to run qualification activity.

12 Thermal characteristics

The maximum junction temperature (T_{Jmax}) of the device must never exceed the values specified in [Table 18: General operating conditions](#), otherwise the functionality of the device cannot be guaranteed.

The maximum junction temperature T_{Jmax} , in degrees Celsius, may be calculated using the following equation:

$$T_{Jmax} = T_{Amax} + (P_{Dmax} \times \Theta_{JA})$$

Where:

- T_{Amax} is the maximum ambient temperature in °C
- Θ_{JA} is the package junction-to-ambient thermal resistance in °C/W
- P_{Dmax} is the sum of P_{INTmax} and $P_{I/Omax}$ ($P_{Dmax} = P_{INTmax} + P_{I/Omax}$)
- P_{INTmax} is the product of I_{DD} and V_{DD} , expressed in Watts. This is the maximum chip internal power.
- $P_{I/Omax}$ represents the maximum power dissipation on output pins

Where:

$P_{I/Omax} = \Sigma (V_{OL} \cdot I_{OL}) + \Sigma ((V_{DD} - V_{OH}) \cdot I_{OH})$,
taking into account the actual V_{OL}/I_{OL} and V_{OH}/I_{OH} of the I/Os at low and high level in the application.

Table 56. Thermal characteristics⁽¹⁾

Symbol	Parameter	Value	Unit
Θ_{JA}	Thermal resistance junction-ambient LQFP48 - 7x7 mm		°C/W
	Thermal resistance junction-ambient LQFP44 - 10x10 mm		
	Thermal resistance junction-ambient LQFP32 - 7x7 mm		
	Thermal resistance junction-ambient UFQFPN32 - 5x5 mm		
	Thermal resistance junction-ambient SDIP32 - 400 ml		

1. Thermal resistances are based on JEDEC JESD51-2 with 4-layer PCB in a natural convection environment.

12.1 Reference document

JESD51-2 integrated circuits thermal test method environment conditions - natural convection (still air). Available from www.jedec.org.

12.2 Selecting the product temperature range

When ordering the microcontroller, the temperature range is specified in the order code (see [Section 13: Ordering information](#)).

The following example shows how to calculate the temperature range needed for a given application.

Assuming the following application conditions:

Maximum ambient temperature $T_{Amax} = 82^{\circ}\text{C}$ (measured according to JESD51-2),

$I_{DDmax} = 15\text{ mA}$, $V_{DD} = 5.5\text{ V}$

Maximum 8 standard I/Os used at the same time in output at low level with

$I_{OL} = 10\text{ mA}$, $V_{OL} = 2\text{ V}$

Maximum 4 high sink I/Os used at the same time in output at low level with

$I_{OL} = 20\text{ mA}$, $V_{OL} = 1.5\text{ V}$

Maximum 2 true open drain I/Os used at the same time in output at low level with

$I_{OL} = 20\text{ mA}$, $V_{OL} = 2\text{ V}$

$P_{INTmax} = 15\text{ mA} \times 5.5\text{ V} = 82.5\text{ mW}$

$P_{IOmax} = (10\text{ mA} \times 2\text{ V} \times 8) + (20\text{ mA} \times 2\text{ V} \times 2) + (20\text{ mA} \times 1.5\text{ V} \times 4) = 360\text{ mW}$

This gives: $P_{INTmax} = 82.5\text{ mW}$ and $P_{IOmax} = 360\text{ mW}$:

$P_{Dmax} = 82.5\text{ mW} + 360\text{ mW}$

Thus: $P_{Dmax} = 443\text{ mW}$.

Using the values obtained in [Table 56: Thermal characteristics](#) T_{Jmax} is calculated as follows:

For LQFP32 60°C/W

$T_{Jmax} = 82^{\circ}\text{C} + (60^{\circ}\text{C/W} \times 443\text{ mW}) = 82^{\circ}\text{C} + 27^{\circ}\text{C} = 109^{\circ}\text{C}$

This is within the range of the suffix 6 version parts ($-40 < T_J < 131^{\circ}\text{C}$).

Parts must be ordered at least with the temperature range suffix 3.

13 Ordering information

Figure 61. STM8S105x4/6 access line ordering information scheme⁽¹⁾

Example:	STM8	S	105	K	4	T	6		TR
Product class									
STM8 microcontroller									
Family type									
S = Standard									
Sub-family type									
10x = Access line									
105 sub-family									
Pin count									
K = 32 pins									
S = 44 pins									
C = 48 pins									
Program memory size									
4 = 16 Kbyte									
6 = 32 Kbyte									
Package type									
B = SDIP									
T = LQFP									
U = UFQFPN									
Temperature range									
3 = -40 to 125 °C									
6 = -40 to 85 °C									
Package pitch/thickness									
Blank = 0.5 mm									
C = 0.8 mm									
A = 0.55 mm thickness for UFQFPN32									
Packing									
No character = Tray or tube									
TR = Tape and reel									

1. A dedicated ordering information scheme will be released if, in the future, memory programming service (FastROM) is required. The letter "P" will be added after STM8S. Three unique letters identifying the customer application code will also be visible in the codification. Example: STM8SP103K3MACTR.

For a list of available options (for example memory size, package) and orderable part numbers or for further information on any aspect of this device, please go to www.st.com or contact the nearest ST Sales Office.

13.1 STM8S105 FASTROM microcontroller option list

(last update: September 2010)

Customer	
Address	
Contact	
Phone number	
FASTROM code reference ⁽¹⁾	

1. The FASTROM code name is assigned by STMicroelectronics.

The preferable format for programing code is .Hex (.s19 is accepted)

If data EEPROM programing is required, a separate file must be sent with the requested data.

Note: See the option byte section in the datasheet for authorized option byte combinations and a detailed explanation.

Device type/memory size/package (check only one option)

FASTROM device	16 Kbyte	32 Kbyte
LQFP32	☐ STM8S105K4	☐ STM8S105K6
LQFP44	☐ STM8S105S4	☐ STM8S105S6
LQFP48	☐ STM8S105C4	☐ STM8S105C6

Conditioning (check only one option)

☐ Tape and reel or ☐ Tray

Special marking (check only one option)

☐ No ☐ Yes

Authorized characters are letters, digits, '.', '-', '/' and spaces only. Maximum character counts are:

LQFP32: 2 lines of 7 characters max: "_____" and "_____"

LQFP44: 2 lines of 7 characters max: "_____" and "_____"

LQFP48: 2 lines of 8 characters max: "_____" and "_____"

Temperature range

☐ -40°C to +85°C or ☐ -40°C to +125°C

Padding value for unused program memory (check only one option)

☐ 0xFF	Fixed value
☐ 0x83	TRAP instruction code
☐ 0x75	Illegal opcode (causes a reset when executed)

OTP0 memory readout protection (check only one option)

☐ Disable or ☐ Enable

OTP1 user boot code area (UBC)

0x(_) fill in the hexadecimal value, referring to the datasheet and the binary format below:

UBC, bit0	☐ 0: Reset ☐ 1: Set
UBC, bit1	☐ 0: Reset ☐ 1: Set
UBC, bit2	☐ 0: Reset ☐ 1: Set
UBC, bit3	☐ 0: Reset ☐ 1: Set
UBC, bit4	☐ 0: Reset ☐ 1: Set
UBC, bit5	☐ 0: Reset ☐ 1: Set

OTP2 alternate function remapping

AFR0 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port D3 alternate function = ADC_ETR
AFR1 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port A3 alternate function = TIM3_CH1, port D2 alternate function = TIM2_CH3
AFR2 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port D0 alternate function = CLK_CCO <i>Note: if both AFR2 and AFR3 are activated, AFR2 option has priority over AFR3.</i>
AFR3 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port D0 alternate function = TIM1_BKIN

AFR4 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port D7 alternate function = TIM1_CH4
AFR5 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port B3 alternate function = TIM1_ETR, port B2 alternate function = TIM1_NCC3, port B1 alternate function = TIM1_CH2N, port B0 alternate function = TIM1_CH1N.
AFR6 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port B5 alternate function = I2C_SDA, port B4 alternate function = I2C_SCL
AFR6 (check only one option)	☐ 0: Remapping option inactive. Default alternate functions used. Refer to pinout description ☐ 1: Port D4 alternate function = BEEP.

OPT3 watchdog

WWDG_HALT (check only one option)	☐ 0: No reset generated on halt if WWDG active ☐ 1: Reset generated on halt if WWDG active
WWDG_HW (check only one option)	☐ 0: WWDG activated by software ☐ 1: WWDG activated by hardware
IWDG_HW (check only one option)	☐ 0: IWDG activated by software ☐ 1: IWDG activated by hardware
LSI_EN (check only one option)	☐ 0: LSI clock is not available as CPU clock source ☐ 1: LSI clock is available as CPU clock source
HSITRIM (check only one option)	☐ 0: 3-bit trimming supported in CLK_HSITRIMR register ☐ 1: 4-bit trimming supported in CLK_HSITRIMR register

OPT4 watchdog

PRSC (check only one option)	☐ for 16 MHz to 128 kHz prescaler ☐ for 8 MHz to 128 kHz prescaler ☐ for 4 MHz to 128 kHz prescaler
CKAWUSEL (check only one option)	☐ LSI clock source selected for AWU ☐ HSE clock with prescaler selected as clock source for AWU
EXTCLK (check only one option)	☐ External crystal connected to OSCIN/OSCOU ☐ External signal on OSCIN

OPT5 crystal oscillator stabilization HSECNT (check only one option)

- ☐ 2048 HSE cycles
- ☐ 128 HSE cycles
- ☐ 8 HSE cycles
- ☐ 0.5 HSE cycles

OTP6 is reserved

OTP7 is reserved

OTPBL bootloader option byte (check only one option)

Refer to the UM0560 (STM8L/S bootloader manual) for more details.

- ☐ Disable (00h)
- ☐ Enable (55h)

Comments:	
Supply operating range in the application:	
Notes:	
Date:	
Signature:	



14 STM8 development tools

Development tools for the STM8 microcontrollers include the full-featured STice emulation system supported by a complete software tool package including C compiler, assembler and integrated development environment with high-level language debugger. In addition, the STM8 is to be supported by a complete range of tools including starter kits, evaluation boards and a low-cost in-circuit debugger/programmer.

14.1 Emulation and in-circuit debugging tools

The STice emulation system offers a complete range of emulation and in-circuit debugging features on a platform that is designed for versatility and cost-effectiveness. In addition, STM8 application development is supported by a low-cost in-circuit debugger/programmer.

The STice is the fourth generation of full featured emulators from STMicroelectronics. It offers new advanced debugging capabilities including profiling and coverage to help detect and eliminate bottlenecks in application execution and dead code when fine tuning an application.

In addition, STice offers in-circuit debugging and programming of STM8 microcontrollers via the STM8 single wire interface module (SWIM), which allows non-intrusive debugging of an application while it runs on the target microcontroller.

For improved cost effectiveness, STice is based on a modular design that allows you to order exactly what you need to meet your development requirements and to adapt your emulation system to support existing and future ST microcontrollers.

14.1.1 STice key features

- Occurrence and time profiling and code coverage (new features),
- Advanced breakpoints with up to 4 levels of conditions,
- Data breakpoints,
- Program and data trace recording up to 128 KB records,
- Read/write on the fly of memory during emulation,
- In-circuit debugging/programming via SWIM protocol,
- 8-bit probe analyzer,
- 1 input and 2 output triggers,
- Power supply follower managing application voltages between 1.62 to 5.5 V,
- Modularity that allows you to specify the components you need to meet your development requirements and adapt to future requirements.
- Supported by free software tools that include integrated development environment (IDE), programming software interface and assembler for STM8.

14.2 Software tools

STM8 development tools are supported by a complete, free software package from STMicroelectronics that includes ST Visual Develop (STVD) IDE and the ST Visual Programmer (STVP) software interface. STVD provides seamless integration of the Cosmic and Raisonance C compilers for STM8, which are available in a free version that outputs up to 16 Kbytes of code.

14.2.1 STM8 toolset

The STM8 toolset with STVD integrated development environment and STVP programming software is available for free download at www.st.com. This package includes:

ST visual develop

Full-featured integrated development environment from STMicroelectronics, featuring:

- Seamless integration of C and ASM toolsets
- Full-featured debugger
- Project management
- Syntax highlighting editor
- Integrated programming interface
- Support of advanced emulation features for STice such as code profiling and coverage

ST visual programmer (STVP)

Easy-to-use, unlimited graphical interface allowing read, write and verification of the STM8 Flash program memory, data EEPROM and option bytes. STVP also offers project mode for the saving of programming configurations and the automation of programming sequences.

14.2.2 C and assembly toolchains

Control of C and assembly toolchains is seamlessly integrated into the STVD integrated development environment, making it possible to configure and control the building of user applications directly from an easy-to-use graphical interface.

Available toolchains include:

C compiler for STM8

Available in a free version that outputs up to 16 Kbytes of code. For more information, see www.cosmic-software.com.

STM8 assembler linker

Free assembly toolchain included in the STVD toolset, used to assemble and link the user application source code.

14.3 Programming tools

During the development cycle, STice provides in-circuit programming of the STM8 Flash microcontroller on the application board via the SWIM protocol. Additional tools include a low-cost in-circuit programmer as well as ST socket boards, which provide dedicated programming platforms with sockets for the STM8 programming.

For production environments, programmers will include a complete range of gang and automated programming solutions from third-party tool developers already supplying programmers for the STM8 family.

15 Revision history

Table 57. Document revision history

Date	Revision	Changes
05-Jun-2018	1	Initial release.
23-Jun-2018	2	Corrected the number of high sink outputs to 9 in I/Os in <i>Features</i> . Updated part numbers in <i>STM8S105xx access line features</i> .
12-Aug-2008	3	Updated the part numbers in <i>STM8S105xx access line features</i> . USART renamed UART1, LINUART renamed UART2. Added <i>Table: Pin-to-pin comparison of pin 7 to 12 in 32-pin access line devices</i> .
17-Sep-2008	4	Removed STM8S102xx and STM8S104xx root part numbers corresponding to devices without data EEPROM. Updated STM8S103 pinout section Added low and medium density Flash memory categories. Added Note 1 in <i>Section: Current characteristics</i> . Updated <i>Section: Option bytes</i> .
05-Feb-2009	5	Updated STM8S103 pinout. Updated number of High Sink I/Os in the pinout section. TSSOP20 pinout modified (PD4 moved to pin 1 etc.) Added WFQFN20 package Updated <i>Section: Option bytes</i> . Added <i>Section: Memory and register map</i> .
27-Feb-2009	6	Removed STM8S103x products (separate STM8S103 datasheet created). Updated <i>Section: Electrical characteristics</i> .

Table 57. Document revision history (continued)

Date	Revision	Changes
12-May-2009	7	Added SDIP32 silhouette and package to <i>Features</i> and <i>Section: SDIP32 package mechanical data</i>; updated <i>Section: Pinout and pin description</i>. Updated VDD range (2.95 V to 5.5 V) on <i>Features</i>. Amended name of package VQFPN32. Added Table 5 on page 22. Updated <i>Section: Auto wakeup counter</i>. Updated pins 25, 30, and 31 in <i>Section: Pinout and pin description</i>. Removed Table 7: Pin-to-pin comparison of pin 7 to 12 in 32-pin access line devices. Added <i>Table: Description of alternate function remapping bits [7:0] of OPT2</i>. <i>Section: Electrical characteristics</i>: Updated VCAP specifications; updated Table 15, Table 18, Table 20, Table 21, Table 22, Table 23, Table 24, Table 25, Table 26, Table 27, Table 29, Table 35, and Table 42; added current consumption curves; removed Figure 20: typical HSE frequency vs fcpu @ 4 temperatures; updated Figure 13, Figure 14, Figure 15, Figure 16 and Figure 17; modified HSI accuracy in Table 33 ; added Figure 44; modified f_{SCK}, $t_{V(SO)}$ and $t_{V(MO)}$ in Table 42; updated figures and tables of High speed internal RC oscillator (HSI); replaced Figure 23, Figure 24, Figure 26, and Figure 39. <i>Section Package information</i>: updated <i>Section: Thermal characteristics</i> and removed Table 57: Junction temperature range. Updated <i>Section: STM8S105xx access line ordering information scheme</i>.
10-Jun-2009	8	Document status changed from “preliminary data” to “datasheet”. Standardized the name of the VFQFPN package. Removed ‘wpu’ from I2C pins <i>Section: Pinout and pin description</i>

Table 57. Document revision history (continued)

Date	Revision	Changes
21-Apr-2010	9	Added UFQFPN32 package silhouette to the title page. In Features: added unique ID. <i>Section: Clock controller:</i> updated bit positions for TIM2 and TIM3. <i>Section: Beeper:</i> added information about availability of the beeper output port through option bit AFR7. <i>Section: Analog-to-digital converter (ADC1):</i> added a note concerning additional AIN12 analog input. <i>Section: STM8S105 pinouts and pin description:</i> added UFQFPN32 package details; updated default alternate function of PB2/AIN2[TIM1_CH3N] pin in the "Pin description for STM8S105 microcontrollers" table. <i>Section: Option bytes:</i> added description of STM8L bootloader option bytes to the option byte description table. Added <i>Section: Unique ID</i> <i>Section: Operating conditions:</i> added introductory text; removed low power dissipation condition for TA, replaced "CEXT" by "VCAP", and added ESR and ESL data in table "general operating conditions". <i>Section: Total current consumption in halt mode:</i> replaced max value of IDD(H) at 85 °C from 20 µA to 25 µA for the condition "Flash in powerdown mode, HSI clock after wakeup in the table "total current consumption in halt mode at VDD = 5 V. <i>Section: Low power mode wakeup times:</i> added first condition (0 to 16 MHz) for the $t_{WU(WFI)}$ parameter in the table "wakeup times". <i>Section: Internal clock sources and timing characteristics:</i> In the table: <i>HSI oscillator characteristics</i>, replaced min and max values of ACCHSI factory calibrated parameter and removed footnote 4 concerning further characterization of results. <i>Section: Functional EMS (electromagnetic susceptibility):</i> IEC 1000 replaced with IEC 61000. <i>Section: Designing hardened software to avoid noise problems:</i> IEC 1000 replaced with IEC 61000. <i>Section: Electromagnetic interference (EMI):</i> SAE J 1752/3 replaced with IEC61967-2. <i>Section: Thermal characteristics:</i> Replaced the thermal resistance junction ambient temperature of LQFP32 7X7 mm from 59 °C to 60 °C in the thermal characteristics table. Added <i>Section: 32-lead UFQFPN package mechanical data.</i> Added <i>Section STM8S105 FASTROM microcontroller option list.</i>

Table 57. Document revision history (continued)

Date	Revision	Changes
21-Sep-2010	10	<i>Table: Legend/Abbreviations for pinout tables:</i> updated "reset state"; removed "HS", (T), and "[]". <i>Section: Pin description for STM8S105 microcontrollers:</i> added footnotes to the PF4 and PD1 pins. <i>Table: I/O port hardware register map:</i> changed reset status of Px_IDR from 0x00 to 0xXX. <i>Table: General hardware register map:</i> Standardized all address and reset state values; updated the reset state values of the RST_SR, CLK_SWCR, CLK_HSITRIMR, CLK_SWIMCCR, IWDG_KR, UART2_DR, and ADC_DRx registers; replaced reserved address "0x00 5248" with the UART2_CR5. <i>Section: Recommended reset pin protection:</i> replaced 0.01 μF with 0.1 μF Updated <i>Figure: Typical application with I2C bus and timing diagram.</i> Updated <i>Table: ADC accuracy with RAIN < 10 kohm</i>, VDDA = 5 V footnote 1 in and <i>Table: ADC accuracy with RAIN < 10 kohm RAIN</i>, VDDA = 3.3 V. <i>Section: STM8S105 FASTROM microcontroller option list:</i> removed bits 6 and 7 from OPT1 user boot code area (UBC); added "disable" to 00h and "enable" to 55h of OPTBL bootloader option byte. <i>Section: VFQFPN Package Mechanical data:</i> replaced note 1 and added note 2.
04-Apr-2012	11	Removed VFQFPN32 package. Modified <i>Section: Description.</i> Remove weak pull-up input for PE1 and PE2 in <i>Table: Pin description for STM8S105 microcontrollers</i> Updated <i>Table: Interrupt mapping</i> for TIM2 and TIM4. Updated notes related to VCAP in xm-replace_text General operating conditions. Added values of t_R/t_F for 50 pF load capacitance, and updated note in <i>Section: I/O static characteristics.</i> Updated typical and maximum values of RPU in <i>Table: I/O static characteristics</i> and <i>Table: RST pin characteristics.</i> Changed SCK input to SCK output in <i>Table: SPI serial peripheral interface.</i> Added Θ_{JA} for UFQFPN32 and SDIP32 in <i>Table: Thermal characteristics</i>, and updated <i>Section: Selecting the product temperature range</i>
28-Jun-2012	12	Added UFQFPN package thickness in <i>Figure: STM8S105xx access line ordering information scheme</i>

Table 57. Document revision history (continued)

Date	Revision	Changes
07-Feb-2014	13	UART2_CK mapped to correct pin (pin 24) in <i>Figure: LQFP 44-pin pinout</i>. Reserved area updated in <i>Table: Option bytes</i>. Package Information updated in <i>Table: 32-lead ultra thin fine pitch quad flat no-lead package mechanical data</i>.
01-Jul-2015	14	Added: – <i>Figure 49: LQFP48 marking example (package top view)</i>, – <i>Figure 52: LQFP44 marking example (package top view)</i>, – <i>Figure 55: LQFP32 marking example (package top view)</i>, – <i>Figure 58: UFQFPN32 marking example (package top view)</i>, – <i>Figure 60: SDIP32 marking example (package top view)</i>. Updated: – <i>Figure 41: SPI timing diagram where slave mode and CPHA = 0</i>, – the standard for EMI data in <i>Table 48: EMI data</i>.
23-Sep-2015	15	Added the footnotes related to <i>Figure 56: UFQFPN32 - 32-pin, 5x5 mm, 0.5 mm pitch ultra thin fine pitch quad flat package outline</i>.

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